Memory–Output Tradeoffs for Merging Dispersed Rotations in Downstream Quantum Compilers

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Time-dependent schedules split the angle of each rotation in a commuting layer across many rounds, and randomized compiling re-signs the pieces; when the per-round coefficients vary freely, no single instruction reveals the angle. We ask what a compiler that sees only the resulting instruction stream must pay to merge the pieces into one rotation per term. For a contiguous layer of m independent rotations on an accuracy-ϵ grid, we prove that any compiler that is correct on every such stream and reads it in forward passes must spend about $m \log_2(1/\epsilon)$ bits. They are counted as output committed while reading the earlier rounds plus state carried into the last round on each pass. A one-pass compiler that emits one aggregated rotation per term attains this to leading order, and so does a multipass compiler that merges a different block of terms on each pass. A one-pass stream optimizer with a much smaller buffer must emit unmerged output early. Memory and early output are not always interchangeable. Suppose k of the m terms, chosen at random, are split over two rounds, and the compiler emits all their first-round rotations before the second round, where they are applied. That prefix then needs at least $\log_2 \binom{m}{k}$ extra bits if each record must address its term, and a smaller amount, still linear in k for fixed $k/m < 1$, if it need only implement the rotations. A prefix buffered until the next round is read need not pay extra. Independently of the bound, on a four-qubit benchmark that repeats one diagonal layer, under one disclosed synthesis and surface-code model, Qiskit's highest preset optimization level, given the pre-lowered circuit, needs 385–1,753 times the logical T states of our compiler's output, which merges the repetitions but not the decomposition of each controlled phase; the ratio grows with the number of repetitions. TKET's PauliSimp, given the same input, also merges across that decomposition and needs about half as many as ours.

I. INTRODUCTION

Two standard steps of a fault-tolerant software stack can split rotation angles. A time-dependent schedule of a commuting Hamiltonian, or a block of consecutive commuting layers with independent per-term parameters [1–5], applies an accumulated phase over several contiguous rounds, and when the per-round coefficients vary freely no single round reveals the total angle of a phase term. Standard QAOA and VQE layers, and generic Trotter steps, are separated by non-commuting layers and do not by themselves form such blocks. Randomized compiling then dresses each round with random Pauli gates, so that coherent errors average into stochastic Pauli noise [6, 7]; for a Z-type rotation, conjugation by a random Pauli flips the sign of its angle. Both steps are correct and useful, and together they leave the total angle spread over many re-signed instructions (Fig. 1a).

Whether those pieces are merged before gate synthesis matters for fault-tolerant cost. Synthesis approximates each rotation by a Clifford+T word; at accuracy ϵ a generic angle costs about $3 \log_2(1/\epsilon)$ T gates, roughly 30 at $\epsilon = 10^{-3}$ [8], and every T gate consumes a distilled magic state [9, 10]. A term left in r pieces is synthesized r times. Merging commuting rotations is accordingly a standard optimization: phase folding and phase-polynomial methods [11–15], Pauli-gadget and ZX rewriting [16–18] all exploit it, and fault-tolerant resource estimates exploit commutation in a related way, replacing many equal-angle commuting rotations by a few rotations applied to their Hamming weight [19, 20].

These methods assume an optimizer that sees, or can rebuild, a global description of the layer. Some compilation instead runs downstream of the steps that dispersed it. A transpiler service may receive a circuit already lowered to standard gates, for instance serialized as OpenQASM [21], rather than the operators from which it was synthesized, and qubit remapping after routing already shapes what finally reaches the device [22]. A vendor pass post-processes an instruction stream that has already been emitted, and the classical control electronics process the instruction stream in real time, tracking Pauli gates in a frame to relax the timing of syndrome measurement and decoding [23]. We ask what a compiler in this position must pay to merge the pieces: how much information it must hold between rounds, and what it costs to commit output instead.

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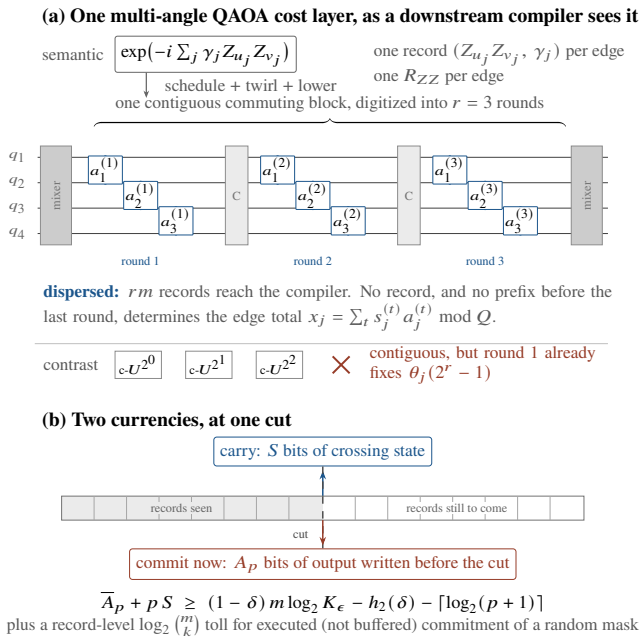


Figure 1. Dispersal and the two ways to undo it. (a) A diagonal cost layer digitized under a schedule that varies each grid-rounded edge coefficient freely across r contiguous rounds (a lowering we construct for a multi-angle QAOA cost layer) is optionally sign-randomized and reaches a downstream compiler as r rounds of grid residues; no record and no prefix that ends before the last round determines an edge total x_j. The bounds count $\mathbb{F}_2$-independent edge terms, $|V|-1$ of them for a connected graph. A standard QAOA layer, in which every round rescales one weight vector, is not such a stream. A controlled-power ladder (bottom) is also contiguous, but its first round already fixes the total when the powers are lowered exactly, so nothing is dispersed. Boxes marked C are Clifford records that permute the generators; the mixer layers bound the block. (b) At the boundary before the last round, the information about earlier rounds either crosses as compiler state or has already been committed as output. The $\log_2 \binom{m}{k}$ toll printed in (b) is the record-level one; commitment that need only implement the rotations pays a smaller toll, and buffered commitment pays none.

Our first result answers this question for a commuting layer of m independent Z-type rotations whose angles lie on an ϵ-calibrated grid and arrive split over r rounds (Fig. 1b). Any compiler that is correct on every such stream, makes at most p forward scans, and fails with probability at most $\delta < 1/2$ obeys

$$\overline{A}_p + pS \;\geq\; (1-\delta)\, m \log_2 K_\epsilon - h_2(\delta) - \lceil \log_2(p+1) \rceil \quad (1)$$

(Theorem 5). Here $K_\epsilon = \Theta(1/\epsilon)$ is the number of angles of the grid, which is also the largest number of rotations about a common axis that are pairwise 4ϵ-separated (Lemma 5); $K_\epsilon = 785$ at $\epsilon = 10^{-3}$. $\overline{A}_p$ is the expected number of output bits committed while the earlier rounds are being scanned, summed over passes, and S caps the state carried across a crossing between the earlier rounds and the last. Of the $2p-1$ crossings that p forward scans make, only the p forward ones enter: a snapshot taken when the scan rewinds is determined by the preceding forward snapshot and the last round. S charges every channel through which a compiler restarted at a crossing could resume—RAM, caches, random seeds, any copy of the input it keeps, and any intermediate representation it builds—so rebuilding a global phase table is allowed but paid for. The final output alone, encoded either relative to the known family or as a self-contained circuit, also needs $(1-\delta)m \log_2 K_\epsilon - h_2(\delta)$ bits. At $\delta = 0$ a single-pass compiler that keeps the running sums and emits one aggregated rotation per term, the compact-output corner, attains Eq. (1) to leading order. For p passes, with the stream re-delivered by its source at the start of each pass as in standard multipass streaming, a block construction that merges a different block of terms on each pass also attains it to leading order, as long as p is small compared with $m/\log m$; a compiler given the layer in aggregated form needs only $O(\log m \cdot \log(1/\epsilon))$ bits of state (Propositions 10 and 11).

Per term the price is $\log_2 K_\epsilon \approx 9.6$ bits at $\epsilon = 10^{-3}$, and it grows linearly with m. What Eq. (1) fixes is where those bits must live, and this is where dispersal enters. The inequality would hold even if the last round were known in advance; what dispersal adds is that output written before the last round arrives cannot yet contain the merged angles. A one-pass compiler whose state is small compared with $m \log_2 K_\epsilon$, such as a stream optimizer with a buffer much smaller than the layer's $m \log_2 K_\epsilon$ bits, must instead commit about that much unmerged output before the last round arrives. A deferred compiler, which commits nothing early, must carry all of it across the crossings. The proof is a standard multipass streaming argument [24–26]: once the last round is fixed, the merged angles are an injective function of the sum of the earlier rounds, and p forward scans simulate a $(2p-1)$-message protocol, of which only the p messages into the last round carry information about the earlier rounds. A closely related classical setting, streaming with a read-only input and a write-only output [27], likewise separates what an algorithm holds from what it has already written. What is specific to compilation is the fooling family and the accounting. When the grid-rounded coefficients vary freely from round to round, time-dependent schedules and layerwise-parameterized blocks emit exactly such fooling streams: the rounds of the schedule play the role of the shares, so no cryptographic secret sharing is added. Randomized compiling of such a stream, in compile-time or frame-tracked form, keeps it a fooling stream; it does not create the dispersal, but in frame-tracked form it adds the Pauli frame to the state that must cross (Propositions 4 and 5).

Equation (1) charges committed output and carried state at par. Our second result exhibits a setting in which they are not interchangeable, and shows what removes the gap. Take a two-round stream in which only k of the m terms are dispersed, the k drawn uniformly at random so that one compiler and one output code must

serve every pattern, and suppose the compiler writes the first-round records of all k dispersed terms before the boundary and that this prefix is applied there, before the next round is read—as in a control stack that streams instructions to the device. If each record of the prefix must address its term, as when the stack forwards one addressed record per term, the prefix needs on average at least $\log_2 \binom{m}{k}$ bits beyond its payload, about $h_2(k/m)/(k/m)$ bits per committed term, while a deferred compiler carries the same payload as state with only $O(\log m)$ bits of overhead (Proposition 7). That addressing requirement is an assumption about the compiler (Definition 18), and given it the bound is a source-coding count. Write-once output does not imply the requirement, since a write-once compiler that buffers its prefix pays no toll, and neither does the applied unitary, because a term whose first-round residue is zero acts as the identity. If the prefix need only implement the first-round rotations, a smaller toll remains, still linear in k: for fixed $\eta = k/m < 1$ it grows at the asymptotic rate $h_2(\eta(1-1/Q))/\eta - h_2(1/Q)$ bits per committed term as $m \to \infty$, measured against a deferred compiler that stores the sign-folded residues (Corollary 5). On the $\epsilon = 10^{-3}$ grid the two bounds nearly coincide, asymptotically 1.99 against 2 bits per term at $k = m/2$; they separate only on coarse grids, 0.92 against 2 at $Q = 3$. This unitary-level bound is proved but not measured. Conversely, a compiler that keeps the same content as state, or that writes it irrevocably but applies it only after reading on, can learn the pattern from the stream that follows and need pay nothing extra. The extra cost therefore requires the prefix to be executed at the cut; irrevocability alone does not create it. It resembles the advice costs of online algorithms [28, 29], the distinction between executed and merely written output parallels relaxed-irrevocability models of online computation [30], and it separates two kinds of commitment that Eq. (1) does not distinguish: output that is executed at the cut and output that is merely written.

A software model of the two-round stream with $Q = 3$, whose state at the cut between rounds is serialized and checked by restarting from it, makes both results concrete (Section IV C). As the dispersed fraction of a layer varies, the state of our deferred compiler grows at a slope within 5% of $\log_2 3{+}1$ bits per dispersed term at $m = 32$, 64, and 128 (Fig. 2). That is the cost of storing each residue and its frame bit separately, one bit above the floor of $\log_2 3$ proven for a deterministic one-pass zero-error compiler (Proposition 6); folding the frame bit into the residue would remove that bit. Under a preregistered protocol, an explicit encoder that ranks the dispersal pattern combinadically attains $\log_2 \binom{m}{k}/k$ to within byte rounding (at most 0.06 bits per term over the preregistered analysis cells), flat over a 16-fold range of m, and the buffered control, whose prefix is the deferred snapshot itself, pays zero at every dispersed fraction (Fig. 3). These are constructive checks of a source-coding bound, not tests that could have failed: they show that the record-level toll is attained and that buffering removes it. For external toolchains we report behavior (Section VIII B). On a four-qubit benchmark that repeats one commuting diagonal layer between Hadamard layers, Qiskit's optimization level 3 and staq's rotation folding return circuits that grow linearly with the number of repetitions, while our semantic reference returns a constant 42 gates and TKET's PauliSimp a small form of 52–73 gates. Tools that rebuild a global representation are not defeated by dispersal itself: on the matched flat representations of Section VIII B, including the masked-share stream, PyZX and TKET's PauliSimp (in its recommended native configuration) reach one rotation per active term. The bounds price that rebuild in state; they do not forbid it.

Finally, we propagate the rotation counts of three pipelines, each output carrying an exact symbolic or a dense unitary equivalence certificate, through checked Clifford+T synthesis and two surface-code profiles under a common error budget (Section X). On the four-qubit benchmark at two sizes, each across 48 settings of total error, synthesis allocation, code profile, and factory count, the output of Qiskit's optimization level 3, applied to the pre-lowered circuit as is TKET's PauliSimp, while our compiler receives the unlowered one, keeps 4,792 and 23,992 rotations, about 12 per repetition, where our merged reference keeps 22. The reference merges the repetitions but keeps the three phase terms into which each controlled phase decomposes; a full phase-polynomial merge of this layer would keep 10. It needs 385–501 times the logical T states of the merged reference at the smaller size and 1,367–1,753 times at the larger, and 538–5,573 times the spacetime volume overall. The ratios grow with the number of repetitions: Qiskit's T count grows about in proportion to it, while the merged reference's changes only through its 22 aggregate angles. PauliSimp, which rebuilds the phase structure before synthesis and also merges across that decomposition, needs 0.44–0.54 times our T states. The inputs differ between the pipelines, so the ratios measure what is at stake when the repetitions reach synthesis unmerged, not the quality of Qiskit's optimizer. This benchmark repeats a layer rather than dispersing it, so these numbers follow from one disclosed model and from the measured outputs, not from Eq. (1), which counts bits rather than gates.

The bounds concern compilers that see the stream after scheduling, randomization, and lowering, and that read it in forward scans. They need the rounds of a layer to be contiguous, up to Clifford records that permute the generators, and their coefficients to vary freely from round to round (Lemma 6, Remark 10). Non-commuting interleavings, as in a generic Trotter step, lie outside them, as do reverse scans and uncharged random access to the input. A compiler that sees the layer before scheduling and lowering escapes them; for one that holds the λ-bit seed of a pseudorandom twirl, the separate m-bit lower bound on a tracked Pauli frame falls to λ bits (Remark 9). The record-level extra cost of executed commitment is attained in our own software model; two preregistered attempts to detect it through third-party toolchains failed,

so whether it is visible at a third-party interface remains open. Read the other way, these limits are a design rule: where the interface and latency budget allow, randomize and lower late, and keep phase semantics until the coefficients have been merged.

Sections II and III define the operational setup and the budget-explicit compiler model with its cut accounting; the exact-semantics results for fixed-width circuits appear there as a corollary. Section IV builds the accuracy-calibrated packing, the dispersed streams and their physical sources, the dispersal–memory tradeoff, and the commitment toll. Sections V and VI prove the output and frontier lower bounds and their matching constructions. Sections VII to IX describe the reference compiler and report the controlled measurements, external baselines, and application workloads, and Section X propagates rotation counts into fault-tolerant resources. Sections XI and XII discuss related work, scope, and limitations.

II. OPERATIONAL SETUP

We define the operational pipeline in four stages: *semantic input*, *basis lowering*, *recovery*, and *synthesis/resource estimation*.

Definition 1 (Semantic input)**.** *A semantic input is a circuit description C_{sem} in a representation that preserves generator–coefficient structure. For a commuting diagonal phase layer with m independent generators $P_1, \ldots, P_m$ and coefficients $\theta_1, \ldots, \theta_m$, the semantic input has $O(m)$ records of the form (P_j, θ_j).*

Definition 2 (Basis lowering)**.** *Let $\mathcal{L}_B(\cdot)$ denote exact lowering into a fixed target basis B. The lowered circuit $\mathcal{L}_B(C_{\mathrm{sem}})$ is a flat instruction stream over B implementing the same unitary up to global phase. Lowering preserves unitary semantics but need not preserve the generator–coefficient structure.*

Definition 3 (Recovery)**.** *A recovery procedure $\mathcal{R}$ takes either a semantic input or a lowered circuit $\mathcal{L}_B(C_{\mathrm{sem}})$ and produces an output circuit C_{out} such that $d_{\mathrm{proj}}(U(C_{\mathrm{out}}), U(C_{\mathrm{sem}})) \le \epsilon$ under the projective operator norm d_{proj}. The recovery is* semantic-first *if the aggregate structure remains available before basis lowering; it is* materialize-first *if recovery begins only from the lowered stream. A materialize-first procedure may still reconstruct a global phase-polynomial, Pauli, or ZX representation; the serialized state needed for that reconstruction is charged by Definition 9.*

Definition 4 (Synthesis and resource estimation)**.** *A fault-tolerant synthesis procedure maps C_{out} to a discrete-gate circuit over a fault-tolerant gate set (e.g., Clifford+T) subject to a total error budget $\epsilon_{\mathrm{total}}$. A quantum resource estimator (QRE) then reports logical T count, magic-state demand, physical qubits, cycles, and spacetime volume under specified architecture assumptions.*

The central question is: for a fixed target unitary U and fixed ϵ, how do the compiler resources required by $\mathcal{R}$ depend on whether $\mathcal{R}$ receives C_{sem} or $\mathcal{L}_B(C_{\mathrm{sem}})$?

III. APPROXIMATE RECOVERABILITY FRAMEWORK

A. Approximate Distance and Recoverability

Definition 5 (Projective operator norm distance)**.** *For unitaries U, V on n qubits,*

$$d_{\mathrm{proj}}(U, V) = \inf_{\phi \in \mathbb{R}} \|U - e^{i\phi} V\|_{\mathrm{op}}. \tag{2}$$

The infimum is attained because the circle $\{e^{i\phi}\}$ is compact and $\|U - e^{i\phi}V\|_{\mathrm{op}}$ is continuous in ϕ.

The projective distance directly eliminates global phase and serves as the primary correctness criterion below; see (2).

Lemma 1 (Unitary-channel diamond comparison)**.** *For the unitary channels $\mathcal{U}(X) = UXU^\dagger$ and $\mathcal{V}(X) = VXV^\dagger$, define the* unnormalized *diamond distance $d_\diamond(\mathcal{U}, \mathcal{V}) := \|\mathcal{U} - \mathcal{V}\|_\diamond \in [0, 2]$. Then*

$$d_{\mathrm{proj}}(U, V) \le d_\diamond(\mathcal{U}, \mathcal{V}) \le 2 d_{\mathrm{proj}}(U, V).$$

If the normalized convention $\widetilde{d}_\diamond := \frac{1}{2}\|\mathcal{U} - \mathcal{V}\|_\diamond$ is used, the corresponding explicit relation is $\frac{1}{2} d_{\mathrm{proj}} \le \widetilde{d}_\diamond \le d_{\mathrm{proj}}$.

Proof. Let $\Theta \in [0, 2\pi]$ be the length of a shortest closed arc containing the spectrum of $U^\dagger V$. Centering that arc by a global phase puts every eigenvalue within angle $\Theta/2$ of 1; conversely, for every phase the rotated spectrum still spans an arc of length Θ, so some eigenvalue lies at angle at least $\Theta/2$ from 1. Hence

$$d_{\mathrm{proj}}(U, V) = 2 \sin(\Theta/4).$$

The numerical-range formula for two unitary channels [31] gives

$$d_\diamond(\mathcal{U}, \mathcal{V}) = \begin{cases} 2\sin(\Theta/2), & 0 \le \Theta \le \pi, \\ 2, & \pi \le \Theta \le 2\pi. \end{cases}$$

For $\Theta \le \pi$, the ratio is $2\cos(\Theta/4) \in [\sqrt{2}, 2]$; for $\Theta \ge \pi$, one has $d_{\mathrm{proj}} \in [\sqrt{2}, 2]$ and $d_\diamond = 2$. The two inequalities follow. □

Consequently, a projective packing with separation s is also a diamond packing with separation at least s. Every lower bound below therefore holds with the *same numerical accuracy threshold* when correctness is instead required as $d_\diamond(\mathcal{U}(C_{\mathrm{out}}), \mathcal{U}) \le \epsilon$. Under the normalized convention, the same argument uses packing separation $s/2$ and correctness threshold $\epsilon/2$. Conversely, a projective-error upper bound ϵ implies diamond error at most 2ϵ; the exact constructive upper bounds used here have zero error in either metric. No diamond packing conclusion is inferred from the upper inequality alone.

Definition 6 (ϵ-approximate recoverability)**.** *Fix $\epsilon > 0$. Given a target unitary U presented as a lowered circuit $\mathcal{L}_B(C_{\mathrm{sem}})$, the ϵ-approximate recoverability problem asks for an output C_{out} with $d_{\mathrm{proj}}(U(C_{\mathrm{out}}), U) \le \epsilon$, using compiler resources charged by the model below.*

B. Budget-Explicit Compiler Model

The exact corollary of Section III uses a fixed-width finite-state transducer model, defined in Remark 5; the growing-width approximate results use the following randomized, random-access model. The two models are not nested: the transducer model does not charge head positions, whereas the model below does.

Definition 7 (Sparse formal-parameter serialization)**.** *Fix formal angle symbols $\theta_1, \dots, \theta_M$. A parameter is stored in normalized sparse form*

$$\lambda = \sum_{\ell=1}^{s} c_\ell \theta_{j_\ell} + \frac{a}{d}\pi,$$
$$j_1 < \cdots < j_s, \quad c_\ell, a \in \mathbb{Z}, \quad d \ge 1,$$

with zero coefficients omitted and $\gcd(a, d) = 1$. Let $\mathsf{u}(z)$ and $\mathsf{s}(z)$ be fixed self-delimiting codes for nonnegative and signed integers. The executable parameter serialization is

$$\begin{aligned} \mathsf{ser}_{\mathrm{par}}(\lambda) = \mathsf{u}(s) \prod_{\ell=1}^{s} \big(\mathsf{u}(j_\ell)\mathsf{s}(c_\ell)\big) \\ \cdot\, \mathsf{s}(a)\mathsf{u}(d), \\ \mathsf{prec}(\lambda) := |\mathsf{ser}_{\mathrm{par}}(\lambda)|. \end{aligned} \tag{3}$$

Thus one nonzero formal coefficient costs $O(\log M)$ rather than $\Omega(M)$ bits. Qubit and generator locations are not part of this parameter payload; they are serialized by the complete-token codec below.

Definition 8 (Width-aware complete-token codec)**.** *Fix a self-delimiting integer code u, a signed variant s, a two-bit Pauli code p, and a reserved record terminator d. A token τ with gate-type ID g, arity r, ordered operand qubits $q_1, \dots, q_r$, sparse Pauli support $\{(z_\ell, \sigma_\ell)\}_{\ell=1}^{s}$, and k parameter slots is encoded as*

$$\begin{aligned} \mathsf{ser}_{\mathrm{tok}}(\tau) = \mathsf{u}(g)\mathsf{u}(r) \prod_{i=1}^{r} \mathsf{u}(q_i)\ \mathsf{u}(s) \prod_{\ell=1}^{s} \big(\mathsf{u}(z_\ell)\mathsf{p}(\sigma_\ell)\big) \\ \cdot\, \mathsf{u}(k) \prod_{a=1}^{k} \big(\mathsf{u}(\mathrm{id}_a)\mathsf{ser}_{\mathrm{par}}(\lambda_a)\big)\mathsf{d}. \end{aligned}$$

The zero values of r, s, or k are explicitly encoded, so the record is executable and uniquely parseable without out-of-band arity, support, or delimiter information. Update, IR-node, and output records use the same field order, with an opcode specifying which fields are active.

At growing width n, an arbitrary qubit or generator choice alone requires $\lceil \log_2 n \rceil$ bits in any injective codec; the concrete codec above uses $O(\log n)$ bits per listed address and $O(s \log n)$ bits for an s-sparse Pauli support. Consequently there is no width-independent "structural cost per token". The literal bit strings $\mathsf{ser}_{\mathrm{tok}}$ are charged wherever a token is live or emitted. Only when $n = n_0$ and a complete-token precision bound q are both fixed is the token set $\Gamma_{B,n_0}^{\le q}$ finite; the fixed-width exact-semantics statements abbreviate it as $\Gamma_B(q)$.

Definition 9 (Randomized dynamic-pass compiler and serialized cut budget)**.** *A compiler $\mathcal{A}$ may use random-access RAM, stacks, hash tables, caches, DAG/graph IRs, and intermediate tapes; it may also select a finite number $p = p(I, \rho)$ of scans or passes dynamically from the input I and a finite random seed/tape ρ. The external input is append-only on its first exposure; a new forward pass is redelivered by the input source from the start of the stream and is accounted for by the pass count (Definition 22). Any other access to a consumed prefix—a copy kept for a later scan through a file, memory mapping, spool, or cache—is live compiler storage and is charged below. The final output is write-once. At every execution cut c, serialize a complete restart snapshot in five disjoint fields:*

- $\mathsf{ser}_{\mathrm{control}}(c)$*: format version, pass and finite-control IDs, dynamic pass count and termination/scheduling state, all input/output heads and emitted-prefix lengths, end flags, framing lengths, and the random seed/tape with its cursor;*
- $\mathsf{ser}_{\mathrm{store}}(c)$*: the persistent and ephemeral RAM, stacks, hash-table keys, bucket layout, cache contents and metadata, with parameter payloads and IR objects replaced by location tags;*
- $\mathsf{ser}_{\mathrm{window}}(c)$*: the ordered live read-window token structures, including the structural fields of Definition 8 but excluding parameter payloads;*
- $\mathsf{ser}_{\mathrm{parameter}}(c)$*: the location-tagged concatenation of every live parameter payload in control, store, window, and IR, each encoded by* (3)*;*
- $\mathsf{ser}_{\mathrm{IR}}(c)$*: every other live intermediate tape/DAG/table object and every rereadable consumed-input block that can affect future execution, serialized by node opcode, the structural fields of Definition 8, edge endpoints, multiplicity, and object boundaries, but with parameter payloads omitted.*

The corresponding bit-level budget is

$$\boxed{\begin{aligned} B_{\mathrm{cut}}(c) = B_{\mathrm{control}}(c) + B_{\mathrm{store}}(c) \\ + B_{\mathrm{window}}(c) + B_{\mathrm{parameter}}(c) + B_{\mathrm{IR}}(c). \end{aligned}} \tag{4}$$

where each summand is the literal bit length of the named serialization. Parameter payloads are the formal angle parameters of live tokens and IR objects; raw integer data that a compiler computes and keeps, such as residues or partial sums held in a packed RAM word, belong to

$\mathsf{ser}_{\mathrm{store}}$ *and are charged at their literal bit length. Every upper bound below uses this convention, and no lower bound depends on it, because the lower bounds apply to any self-delimiting serialization whose control field records the emitted-prefix length. For a fixed run,* B_{cut} *is the maximum over its cuts. For randomized compilers we use the worst-input expected budget*

$$\overline{B}_{\mathrm{cut}}(\mathcal{A},\mathcal{I}) := \max_{I\in\mathcal{I}} \mathbb{E}_{\rho}\big[\max_{c} B_{\mathrm{cut}}(c; I, \rho)\big]. \tag{5}$$

No theorem below substitutes a single field, such as $\max_c B_{\mathrm{parameter}}(c)$ *or* $\max_c B_{\mathrm{IR}}(c)$*, for the total* B_{cut}*. The final output is write-once, is a codeword of a fixed prefix-free output code, and is charged separately by the output codes below.*

Remark 1 (Model scope and escape channels)**.** *Equation* (4) *charges all restart information, not only the finite-control state. In particular, a ZX graph, phase-polynomial table, spilled update log, hash table, cache, random seed, or materialized intermediate tape is serialized in the indicated field; formal angle parameters are serialized in* $B_{\mathrm{parameter}}$ *and computed integer data in* B_{store}*, so every coefficient is charged exactly once. Internal rescanning of retained data is allowed, but the pass schedule, heads, and any retained copy of old input are part of the snapshot. A pass re-delivered by the source is accounted by the pass count rather than by the snapshot, so a bound read off a single cut, such as Theorem 2, requires the stream to be delivered once; the transcript bounds of Section VI cover re-delivered passes. The five fields are disjoint by construction, so parameter bits are not counted again as store, window, or IR structure. A compiler may escape a RAM-only workspace lower bound by materializing a large IR, but it then pays the same bits in* B_{cut}*.*

Definition 10 (Three noninterchangeable output measures)**.** *Fix two prefix-free, streaming binary codes. The self-contained code* E_{self} *serializes a target-basis circuit using gate opcodes, all qubit/generator addresses, record delimiters, and sparse parameters using Definition 8; it assumes no family dictionary. For a public packing-family dictionary* $\mathcal{F}_m = (m, P_1, \ldots, P_m, K_{\mathrm{pack}}, \Delta, B)$*, the conditional code* $E_{\mathrm{rel}}(\cdot \mid \mathcal{F}_m)$ *has four modes: (i) an aggregate mode consisting of a self-delimiting header, which declares a block length* b*, followed by the* m *aggregate digits packed in base* K_{pack} *in consecutive blocks of* b *digits, each written least significant bit first as a* $\lceil b \log_2 K_{\mathrm{pack}} \rceil$*-bit integer, a final block of* $b' < b$ *digits taking* $\lceil b' \log_2 K_{\mathrm{pack}} \rceil$ *bits; (ii) a literal mode containing complete target-basis records; (iii) an ordered-share mode whose header declares a split point and whose fixed public schedule permits generator addresses to be omitted from its residue slots, which are packed in base* Q*, while retaining round-boundary and packed-block delimiters; and (iv) a* p*-block hybrid mode whose header declares the pass blocks and passthrough coordinates and whose data section contains packed aggregate blocks and, for each round, the passthrough residues of that round packed in base* Q*, in the order in which one left-to-right scan per pass produces them, a public function of* (m, p, q, r)*. Define*

$$\begin{aligned} B_{\mathrm{rel}}(C \mid \mathcal{F}_m) &:= |E_{\mathrm{rel}}(C \mid \mathcal{F}_m)|, \\ B_{\mathrm{self}}(C) &:= |E_{\mathrm{self}}(C)|, \\ G(C) &:= \textit{number of target-basis gates in } C. \end{aligned} \tag{6}$$

For a compiler, a superscript "out" and a maximum over valid inputs denote the worst-case output value of the corresponding measure; a barred superscript "out" denotes the worst-input expectation over ρ*. These quantities are not converted into one another: conditional description bits, self-contained circuit bits, and gate count are reported and bounded separately.*

C. Executable Model–Implementation Correspondence

The formal fields above are implemented by one canonical binary codec rather than by dimension-changing estimates. The executable specification is `encoding/FORMAT.md`; its reference implementation is `encoding/codec.py`, and the only bit-accounting entry point is `instrumentation/resource_accounting.py`. A complete codec frame has the form

$$\texttt{UCCACCTO} \,\|\, \mathsf{u}(\text{version}) \,\|\, \mathsf{u}(\text{type}) \,\|\, \mathsf{u}(|z|) \,\|\, z, \tag{7}$$

where the displayed terminal zero denotes a null byte, u is the canonical base-128 unsigned varint, and z is a typed canonical payload. Signed integers use zigzag coding; rational coefficients are reduced signed numerator/positive-denominator pairs. Maps are ordered by their UTF-8 key bytes. Qubit, generator, symbol, location, node, edge-endpoint, port, pass, and head addresses are literal varints in z.

Lemma 2 (Canonical prefix-free implementation)**.** *Version-one complete frames are injective and prefix-free. Concatenations of frames are uniquely parseable without an out-of-band arity or record count. The semantic and flat decoders preserve their exact aggregate coefficient table.*

Proof. Canonical varints have a unique terminating byte, and the version, type, and payload length are therefore uniquely decoded. The declared payload length locates the end of the frame. A complete valid frame cannot be a proper prefix of another complete valid frame: the shorter declared payload ends at its frame boundary, whereas any following byte belongs to a subsequent frame. Typed payload values have distinct one-byte tags; lists and maps carry counts, maps have strictly increasing keys, and reduced rational pairs are unique. This proves injectivity and unique concatenated parsing. Finally, `semantic_signature` sums exact rational sparse parameters by the literal pair (generator address, Pauli support), so encode–decode leaves the aggregate table unchanged for both input representations. □

Formal object	Codec object/file	Included information	Measured field
semantic input	`SemanticStream`	width and address-complete aggregate records	input artifact bits
dispersed flat input	`FlatUpdateStream`	width, round count, and every additive update	input artifact bits
$\mathsf{ser}_{\mathrm{control}}$	`ControlState/control.uccbin`	pass count/index, heads, scheduling, status, framing and seed state when present	B_{control}
$\mathsf{ser}_{\mathrm{store}}$	`StoreState/store.uccbin`	aggregate tables, maps, caches, stacks and their layout, excluding tagged parameter/IR payloads	B_{store}
$\mathsf{ser}_{\mathrm{window}}$	`LiveWindow/window.uccbin`	ordered live complete-token structure and explicit parameter-slot count	B_{window}
$\mathsf{ser}_{\mathrm{parameter}}$	`ParameterLedger/parameter.uccbin`	sparse symbolic parameters under complete location tuples	$B_{\mathrm{parameter}}$
$\mathsf{ser}_{\mathrm{IR}}$	`IRField/ir.uccbin`	serialized DAG width, opcodes, operands, nodes, edges, ports, boundaries and metadata, with parameters removed	B_{IR}
committed prefix	`committed-prefix.bin`	literal immutable prefix of a codec output frame	B_{com}
external output	`ExternalToolOutput`	tool, format and literal API payload	output artifact bits
certificate input	`CertificateInput`	certificate kind, certified-artifact SHA-256 and literal sidecar	certificate artifact bits
pass count	manifest `passes.count` and row `p`	actual declared scans of the instrumented compiler	p

Table I. One-to-one map from formal serialized objects to executable artifacts. No row permits a gate, token, support, or IR-node count to stand in for bits.

Proposition 1 (File-backed accounting soundness)**.** *Suppose the version-one verifier accepts a cut manifest M. For every $f \in \{\mathrm{control}, \mathrm{store}, \mathrm{window}, \mathrm{parameter}, \mathrm{IR}, \mathrm{com}\}$, let A_f be the file named by M. Then the implementation reports*

$$\begin{aligned} B_f &= 8|A_f|_{\mathrm{byte}}, \\ B_{\mathrm{cut}} &= B_{\mathrm{control}} + B_{\mathrm{store}} + B_{\mathrm{window}} \\ &\quad + B_{\mathrm{parameter}} + B_{\mathrm{IR}}, \end{aligned} \tag{8}$$

and reports the manifest pass count as p. The reported values change when a serialized address crosses a varint-width boundary. They cannot be obtained from a node or gate count alone.

Proof. The accountant first writes the five complete frames and the literal committed prefix, then obtains every byte count by `stat`, stores the SHA-256 of each file, writes M, and invokes the verifier before returning a row. The verifier reopens all six files, recomputes sizes and digests, decodes the five restart fields, and rejects any disagreement. It then forms the sum in (8); there is no count-to-bit code path. Since an address is a literal canonical varint, for example 127 and 128 occupy different address lengths. Conversely, two one-node DAGs may contain those different addresses and hence have unequal file lengths despite equal node count. □

Remark 2 (Observable boundary versus restart cut)**.** *The masked-share tracer serializes the five declared cut fields after every update. The verifier reopens and decodes every recorded field slice, checks its size and digest, and recomputes the cut sum; its maximum is therefore a file-backed measurement of the declared B_{cut} coordinate. The W3 campaign does not claim that the Python process is destroyed and restarted between events. External compilers do not expose their private in-process DAGs, caches, or tapes. Their matched-matrix and natural-factorial rows consequently report a file-backed* API-boundary *snapshot and are diagnostic only; they are not promoted to measurements of the maximum private B_{cut}. Peak RSS remains a separate byte dimension and is not substituted for any field in* (4)*.*

Remark 3 (Self-contained and conditional descriptions)**.** *Both input-stream objects declare either* `self_contained` *or* `conditional`*. The former forbids an external dictionary. The latter requires the SHA-256 of the exact public family dictionary, and its manifest records the dictionary's separate file size and whether it is charged at the cut. Thus a conditional description cannot silently be reported as a self-contained circuit code.*

Remark 4 (Physical scope of the bounded-memory streaming model)**.** *The charged streaming model of Definition 9—bounded live state, explicit pass accounting, append-only first exposure—is the operating regime of a real-time fault-tolerant control stack, not a convenience abstraction. Between rounds, the classical electronics adjacent to the quantum device holds decoder state, Pauli-frame records, and instruction buffers under hard latency and buffer budgets: Pauli corrections can be tracked in a classical frame instead of being applied, which relaxes the timing constraints on syndrome measurement and decoding [23], but they must be resolved before the next non-Clifford gate [32], and surface-code compilers are built as streaming pipelines that target real-time operation [33]. Distributed and modular architectures motivate the same scope from the interconnect side: a cross-module interaction splits the* support *of a gate across modules while its coefficient stays within one module, so modularity bounds the state that crosses a partition without by itself dispersing any coefficient. These architectures therefore ground the model's memory and pass constraints only; no*

claim is made that modular partitioning instantiates the dispersed streams of Section IV C, whose instances come from scheduling and twirling alone.

D. Exact-Semantics Corollary from Prior Work

We record the exact-semantics ($\epsilon = 0$) results as corollaries; the full transducer proofs are restated in Section A 3. The witness family is fixed-width.

Remark 5 (Separate fixed-width and growing-width encoding regimes)**.** *The exact corollary in this subsection fixes $n = n_0$, a precision bound q, a deterministic constant pass count p_0, and one finite alphabet $\Gamma^{\le q}_{B,n_0}$; its unit is therefore a token over that fixed alphabet. A* finite-window, finite-burst transducer *is, in this regime, a deterministic machine whose pass i reads the tape written by pass $i-1$ (pass 1 reads the lowered input) and writes a new tape; each pass is a subsequential transducer over $\Gamma^{\le q}_{B,n_0}$ whose state consists of finite control, a persistent store, and a window of at most w tokens, emits at most μ tokens per step, and has transitions that do not depend on head positions.* Constant budget *means that p_0, w, μ, and the set of reachable states are independent of the repetition count r; head positions and intermediate tapes are not part of this budget. No such constant-cost token assumption is made in the growing-width results. There $n = m + 1$ (or $n \ge m$), the pass count may be $p = p(m, I, \rho)$, and every input, output, RAM, IR record, and head is charged in bits by Definition 8 and* (4)*.*

Definition 11 (Fixed-width witness family)**.** *Let H_Λ be a fixed invertible boundary layer (a Hadamard layer in the experiments) and let $D = \prod_{j=1}^{m_0} \exp(-i\theta_j P_j/2)$ be a fixed commuting diagonal layer with Pauli-Z characters $P_j = Z^{a_j}$, $a_j \in \mathbb{F}_2^n$, and angles θ_j. The witness at repetition count r is $C_r = H_\Lambda D^r H_\Lambda^{-1}$, with exact lowering $\mathcal{L}_B(C_r) = \Pi M^r \Sigma$ for fixed token strings Π, M, Σ.*

Two instances are certified and reused here. The *rational CP witness* ($n = 4$; ten rz/cp terms; $\mathrm{rank}_{\mathbb{F}_2}(A) = 4$) satisfies finite-range noncollision with exact period $T = 131040$, certified by the self-contained rational angle, single-term order, relative-phase, and LCM derivation in Section A 2. The *irrational independent-Pauli witness* ($n = 4$; supports $1100, 0110, 0011, 1000$; $\theta_1/\pi = \sqrt{2}/11$) satisfies unbounded noncollision by the following uniqueness lemma.

Lemma 3 (Coefficient uniqueness for independent characters)**.** *If $a_1, \dots, a_{m_0} \in \mathbb{F}_2^n$ are linearly independent and two diagonal layers with coefficient vectors α, α' satisfy $D(\alpha) = e^{i\varphi} D(\alpha')$, then $\alpha_j \equiv \alpha'_j \pmod{2\pi}$ for every j. In particular, if some θ_j/π is irrational, the powers D^r are pairwise distinct up to global phase for all $r \ge 1$.*

Proof. Independence makes $z \mapsto (\chi_{a_1}(z), \dots, \chi_{a_{m_0}}(z))$ surjective onto $\{\pm 1\}^{m_0}$. Comparing the phase constraint at two sign patterns that differ only in coordinate j gives $2(\alpha_j - \alpha'_j) \equiv 0 \pmod{4\pi}$. For the second claim, $(r - r')\theta_j \in 2\pi\mathbb{Z}$ with θ_j/π irrational forces $r = r'$. □

Assumption 1 (Finite-range noncollision)**.** *On a range $1 \le r \le R$, the powers $D^1, \dots, D^R$ are pairwise distinct up to global phase.*

Assumption 2 (Unbounded noncollision)**.** *The powers D^r, $r \ge 1$, are pairwise distinct up to global phase.*

Corollary 1 (Exact-semantics final-output lower bound)**.** *Assume exact lowering ($\epsilon = 0$), a fixed-width witness $C_r = H_\Lambda D^r H_\Lambda^{-1}$ with $\mathcal{L}_B(C_r) = \Pi M^r \Sigma$, and unbounded noncollision (Assumption 2). Let $\mathcal{A}$ be a uniform deterministic finite-window, finite-burst transducer with constant budget in the sense of Remark 5 (constant p, w, μ and a finite reachable state set; head positions are not read) that is correct for every $r \ge 1$. Then there exist constants $h \ge 1$, $D \ge 1$ (depending on $\mathcal{A}$ and the witness but not r) such that for every $r \ge h + D$,*

$$|T_p(r)| \ge \frac{r - h - D + 1}{D} = \Omega_{\mathcal{A}}(r). \tag{9}$$

Proof. This is Theorem 6, proved in full in Section A 3. Constant budget gives one finite charged alphabet; per-tape pumpability propagation gives pumpability of the final output; noncollision prevents collapse of distinct pump residues. □

Corollary 2 (Finite-range tail form)**.** *Let $\mathcal{A}$ be as in Corollary 1 but correct only for $1 \le r \le R$, under finite-range noncollision (Assumption 1) with $N(R) = R$. Then there are constants h, D, given by the pumpability data of $\mathcal{A}$'s final pair family, with $|T_p(r)| \ge (r - h - D + 1)/D$ for $h + D \le r \le R$ (Corollary 10). For the rational CP witness, whose exact period is $T = 131040$, this applies for $R \le T$.*

Corollary 3 (Counting toll for short outputs)**.** *If finite-range noncollision supplies $N(R)$ distinct targets, then every deterministic exact compiler satisfies (Corollary 11)*

$$\max_{1 \le r \le R} B_{\mathrm{self}}(T_p(r)) \ge \lceil \log_2 N(R) \rceil .$$

The same statement holds for any single fixed prefix-free output code, in particular for a family-relative code with one public dictionary used on the entire range.

The irrational independent-Pauli witness instantiates Corollary 1 for all r via Lemma 3; the rational CP witness instantiates Corollary 2 on ranges $R \le T$. These exact-semantics results concern $\epsilon = 0$ and fixed width $n = 4$; the results below generalize to growing m and expose the dependence on $\epsilon > 0$.

IV. PACKING CONSTRUCTION

A. Growing-Width Family

Definition 12 (Packing family)**.** *For each m and accuracy scale ϵ, a packing family $\mathcal{F}_{m,\epsilon} = \{U_x : x \in \mathcal{X}_{m,\epsilon}\}$*

has pairwise projective separation greater than 2ϵ, a semantic description of $O(m)$ addressed records, and a lowered streaming description whose aggregate information may be dispersed across updates. The accuracy-dependent constructions below have $|\mathcal{X}_{m,\epsilon}| = \bar{K}_\epsilon^m$ for the no-wrap codebook of Proposition 3 and K_ϵ^m for the cyclic grid of Corollary 8, with $\bar{K}_\epsilon, K_\epsilon = \Theta(1/\epsilon)$. The two alphabets differ by about a factor of 2 and are kept apart notationally throughout.

Every growing-width family in this paper—the codebook that carries the separation, the streams that carry the lower bounds, the two-round device that is executed in Section VIII D, and the schedules of Section IV C—is one scheme under four parameters. We state it once and list the specializations in Table II.

Definition 13 (Phase-accumulation family)**.** *Let $P_1, \dots, P_m$ be $\mathbb{F}_2$-independent Pauli-Z characters on $n \geq m$ qubits (i.e. $P_j = Z^{a_j}$ with $\operatorname{rank}_{\mathbb{F}_2}(a_1, \dots, a_m) = m$), and write $R_{P_j}(\theta) = \exp(-i\theta P_j/2)$ and $[K]_0 = \{0, \dots, K-1\}$. A family is fixed by a step $\Delta > 0$, an aggregate alphabet $[K]_0$, a number of rounds $r \geq 1$, and a sign set $\Sigma \subseteq \{\pm 1\}$. The* semantic input *is the addressed vector $x \in [K]_0^m$, realizing*

$$U_x = \exp\Big(-\tfrac{i\Delta}{2} \textstyle\sum_j x_j P_j\Big) = \prod_{j=1}^m R_{P_j}(\Delta x_j). \tag{10}$$

The flat input *is the round-major record sequence $\mathsf{Update}(t, j, a_j^{(t)}, s_j^{(t)})$ with residues $a_j^{(t)}$ and signs $s_j^{(t)} \in \Sigma$, realizing*

$$W = \prod_{t=1}^{r} \prod_{j=1}^{m} R_{P_j}\big(\Delta\, s_j^{(t)} a_j^{(t)}\big), \tag{11}$$

whose aggregate is $x_j = \sum_t s_j^{(t)} a_j^{(t)}$. Two arithmetic regimes are used, and no result mixes them:

- no-wrap*: residues and aggregate live in $\mathbb{Z}$ and $(K-1)\Delta \leq \pi$, so distinct aggregates are distinct angles;*
- cyclic*: $\Delta = 2\pi/Q$ for a modulus $Q \geq 3$, residues lie in $\mathbb{Z}_Q$, the aggregate is taken* mod Q*, and the aggregate alphabet is all of $\mathbb{Z}_Q$ ($K = Q$) unless stated otherwise.*

Since the P_j commute, $W = U_x$ up to global phase in either regime (Proposition 2).

Proposition 2 (Flat inputs aggregate projectively)**.** *Every valid flat input of Definition 13 implements U_x up to global phase, with no promise about any individual round.*

Proof. The P_j commute, so $W = \prod_j R_{P_j}(\Delta \sum_t s_j^{(t)} a_j^{(t)})$. In the no-wrap regime the exponent is Δx_j by definition. In the cyclic regime write $\sum_t s_j^{(t)} a_j^{(t)} = x_j + Qk_j$ for integers k_j; since $Q\Delta = 2\pi$, $R_{P_j}(\Delta x_j + 2\pi k_j) = (-1)^{k_j} R_{P_j}(\Delta x_j)$, and multiplying over j changes only a global sign. □

Def.	Specialization	Regime	r	Σ	Role
14	no-wrap packing	no-wrap	1	$\{+\}$	codebook
21	dispersed update	cyclic	≥ 2	$\{+\}$	lower bounds
16	masked share	cyclic, Q=3	2	$\{+\}$	executed
17	scheduled stream	cyclic	≥ 2	$\{\pm\}$	physical
15	balanced	no-wrap	r	$\{+\}$	diagnostic

Table II. The growing-width families are specializations of Definition 13. The balanced encoding is listed for completeness: its tokens contain x_j/r, so it proves no lower bound (Remark 7). The fixed-width witness family of Definition 11 belongs to the separate exact-semantics regime of Remark 5 and is not an instance of this scheme.

Definition 14 (K-ary no-wrap construction)**.** *The no-wrap specialization of Definition 13 with $r = 1$, $\Sigma = \{+1\}$, integer $K \geq 2$, and $(K-1)\Delta \leq \pi$. Its codebook is $\{U_x : x \in [K]_0^m\}$ as in* (10).

Lemma 4 (Packing separation)**.** *For the construction in Definition 14, all $x \neq y \in [K]_0^m$ satisfy*

$$d_{\mathrm{proj}}(U_x, U_y) \geq 2\sin(\Delta/4). \tag{12}$$

The constant is tight: pairs differing by one in a single coordinate attain it.

Proof. Let $s = x - y$ and choose j^* with $s_{j^*} \neq 0$. Since P_j commute, $U_x U_y^\dagger = \exp(-\frac{i\Delta}{2}\sum_j s_j P_j)$ is diagonal in the computational basis. Its eigenvalue at basis state z is $\lambda(z) = \exp(-\frac{i\Delta}{2}\sum_j s_j \chi_{P_j}(z))$, where $\chi_{P_j}(z) = (-1)^{a_j \cdot z} \in \{\pm 1\}$.

Because $a_1, \dots, a_m$ are $\mathbb{F}_2$-independent, the map $z \mapsto (\chi_{P_1}(z), \dots, \chi_{P_m}(z))$ is surjective onto $\{\pm 1\}^m$. Hence two basis states can be chosen whose character vectors agree everywhere except at j^*. Their eigenvalue arguments differ by Δs_{j^*} modulo 2π. The no-wrap condition gives

$$\Delta \leq |\Delta s_{j^*}| \leq (K-1)\Delta \leq \pi.$$

Now include the global-phase infimum required by Definition 5. For every $\phi \in \mathbb{R}$,

$$\|U_x - e^{i\phi} U_y\|_{\mathrm{op}} = \max_z \big|\lambda(z) e^{-i\phi} - 1\big| \geq 2\sin(\Delta/4), \tag{13}$$

because two points on the unit circle whose arguments differ by $|\Delta s_{j^*}| \in [\Delta, \pi]$ cannot both lie within circular distance less than $\Delta/2$ of 1 after any common rotation, and $|e^{i\gamma} - 1| = 2|\sin(\gamma/2)|$. Taking the infimum over ϕ preserves the bound. A one-coordinate difference of one has exactly two spectral values separated by Δ, and the centered choice of ϕ attains the lower bound. □

Proposition 3 (ϵ-tuned K-ary packing)**.** *For $0 < \epsilon \leq 1/8$, set*

$$\begin{aligned} L_\epsilon &= \left\lfloor \frac{\pi}{4\arcsin(2\epsilon)} \right\rfloor, & M_\epsilon &= 2L_\epsilon, \\ \bar{K}_\epsilon &= L_\epsilon + 1, & \bar{\Delta}_\epsilon &= \frac{2\pi}{M_\epsilon}. \end{aligned} \tag{14}$$

Then $\bar{K}_\epsilon = \Theta(1/\epsilon)$, $(\bar{K}_\epsilon - 1)\bar{\Delta}_\epsilon = \pi$ (the no-wrap condition of Definition 14 with equality), and the $\bar{K}_\epsilon^m$ targets in Definition 14 obey

$$d_{\mathrm{proj}}(U_x, U_y) \geq 2\sin(\bar{\Delta}_\epsilon/4) \geq 4\epsilon \qquad (x \neq y). \tag{15}$$

Proof. The restriction $\epsilon \leq 1/8$ gives $L_\epsilon \geq 1$. Since $\arcsin(2\epsilon) = \Theta(\epsilon)$, the asserted order of $\bar{K}_\epsilon$ follows. The equality in the no-wrap statement is direct. Finally $L_\epsilon \leq \pi/(4\arcsin(2\epsilon))$, hence $\bar{\Delta}_\epsilon/4 = \pi/(4L_\epsilon) \geq \arcsin(2\epsilon)$. Apply Lemma 4 and monotonicity of sine on this range. □

The next lemma shows that the alphabet size is intrinsic rather than designer-chosen: up to an explicit factor paid for the no-wrap property, $\bar{K}_\epsilon$ is the metric-entropy count of pairwise distinguishable Z-rotations at the working separation, an intrinsic function of the tolerance rather than a construction parameter, and the cyclic alphabet size $K_\epsilon = Q_\epsilon$ used by the dispersed streams equals that count exactly.

Lemma 5 (Synthesis-grid calibration)**.** *For $0 < \epsilon \leq 1/8$, let $N(\epsilon)$ be the maximum cardinality of a set of single-axis rotations $\{R_P(\theta) : \theta \in \mathbb{R}/2\pi\mathbb{Z}\}$ that is pairwise 4ϵ-separated, i.e. with pairwise $d_{\mathrm{proj}} \geq 4\epsilon$, and let $g_\epsilon := 4\arcsin(2\epsilon)$. Then*

$$N(\epsilon) = \left\lfloor \frac{2\pi}{g_\epsilon} \right\rfloor, \qquad \frac{1}{2\epsilon} - 1 \;<\; N(\epsilon) \;\leq\; \frac{\pi}{4\epsilon}, \tag{16}$$

and the alphabet of Proposition 3 satisfies

$$\frac{N(\epsilon)}{2} \;<\; \bar{K}_\epsilon \;<\; \frac{N(\epsilon)}{2} + \frac{3}{2}, \tag{17}$$

while the cyclic modulus of Corollary 8 satisfies

$$Q_\epsilon = K_\epsilon = N(\epsilon). \tag{18}$$

Moreover the grid step is separation-tight, $g_\epsilon \leq \bar{\Delta}_\epsilon \leq g_\epsilon(1 + O(\epsilon))$. The factor 2 in (17) is forced by the no-wrap condition: $(K-1)\Delta \leq \pi$ with $\Delta \geq g_\epsilon$ allows at most $\pi/g_\epsilon + 1$ symbols.

Proof. For a fixed non-identity Pauli P, both characters ± 1 are realized, so $R_P(\theta)R_P(\theta')^\dagger$ has eigenvalue arguments $\mp\delta/2$ with $\delta = \theta - \theta'$; projectively θ is defined modulo 2π. Writing $g \in [0, \pi]$ for the circular distance between θ and θ' in $\mathbb{R}/2\pi\mathbb{Z}$, the global-phase infimum of Definition 5 is attained at the centered rotation, giving $d_{\mathrm{proj}} = 2\sin(g/4)$, increasing in g; the computation is the one in the proof of Lemma 4. Hence pairwise 4ϵ-separation is equivalent to pairwise circular distance at least g_ϵ, because $2\sin(g_\epsilon/4) = 4\epsilon$ exactly. The maximum number of points of a circle of circumference 2π with pairwise circular distance at least g_ϵ is $\lfloor 2\pi/g_\epsilon \rfloor$: ordering the points around the circle, the gaps between consecutive points sum to 2π and each is at least g_ϵ, and equal spacing attains the bound. The two-sided bound follows from $x \leq \arcsin x \leq (\pi/2)x$ on $[0, 1]$. For (17), put $y = \pi/(2\arcsin(2\epsilon))$, so that $N(\epsilon) = \lfloor y \rfloor$ and, by (14), $\bar{K}_\epsilon = \lfloor y/2 \rfloor + 1$. Then $\bar{K}_\epsilon > y/2 \geq N(\epsilon)/2$ and $\bar{K}_\epsilon \leq y/2 + 1 < (N(\epsilon)+1)/2 + 1 = N(\epsilon)/2 + 3/2$. For (18), $Q_\epsilon = \lfloor \pi/(2\arcsin(2\epsilon)) \rfloor = \lfloor 2\pi/g_\epsilon \rfloor = N(\epsilon)$ directly from the definitions, and $K_\epsilon = Q_\epsilon$ by Corollary 8. Finally $\bar{\Delta}_\epsilon = \pi/L_\epsilon$ with $\pi/(4\arcsin(2\epsilon)) - 1 < L_\epsilon \leq \pi/(4\arcsin(2\epsilon))$ gives the step bounds. For the factor 2, a no-wrap alphabet with step $\Delta \geq g_\epsilon$ has $(K-1)g_\epsilon \leq \pi$. □

Remark 6 (Clifford+T synthesis calibration of the grid)**.** *The packing grid is chosen by the target accuracy; it is not itself a Ross–Selinger synthesis lattice. Ancilla-free Clifford+T approximation of an arbitrary z-rotation to within ϵ up to a phase, which is exactly approximation in d_{proj} [8, Def. 9.1, Cor. 9.5], is solved by [8, Alg. 9.9], which runs the fixed-phase algorithm [8, Alg. 7.6] and its up-to-phase variant [8, Alg. 9.8] and keeps the cheaper circuit. Its T-count is $3\log_2(1/\epsilon) + O(\log\log(1/\epsilon))$ in the typical case [8, Abstract; Conj. 8.10] and at most $c + 4\log_2(1/\epsilon)$ for a constant $c \approx 10$ [8, §8.3]; given a factoring oracle the returned T-count is optimal per instance, and without one it is near-optimal under a number-theoretic hypothesis [8, Prop. 9.10; Hyp. 8.3]. Generic non-Clifford angles therefore have synthesis cost $\Theta(\log(1/\epsilon))$, but special grid points such as the identity or Clifford angles can be cheaper. The fixed-total-error campaign of Section X does not assume a uniform per-angle cost: it synthesizes every distinct angle with staq's `grid_synth` implementation and records the checked result. Consequently the information term $m\log_2 K_\epsilon$ and the generic synthesis scale are both $\Theta(m\log(1/\epsilon))$, while no angle-by-angle T-count equality is claimed. For the cyclic grid used by the dispersed streams and by Theorem 5 the identification is exact, $Q_\epsilon = N(\epsilon)$ by (18), so the sharing modulus equals the metric-entropy count itself; at $\epsilon = 10^{-3}$, $Q_\epsilon = K_\epsilon = 785$. The factor 2 in (17) concerns only the no-wrap codebook $\bar{K}_\epsilon$ ($\bar{K}_\epsilon = 393$ at $\epsilon = 10^{-3}$) used by Theorem 1.*

B. Streaming Encoding

Definition 15 (Balanced streaming encoding)**.** *The no-wrap specialization of Definition 13 in which every round carries the same residue x_j/r, emitted on the round-robin schedule $\sigma(i) = ((i-1) \bmod m) + 1$ over $N = rm$ slots. Each token therefore contains x_j/r; Remark 7 records why this encoding proves no lower bound.*

Definition 16 (Two-round masked-share lowering)**.** *The cyclic specialization of Definition 13 with $Q = 3$, $\alpha := \Delta = 2\pi/3$, $r = 2$, and $\Sigma = \{+1\}$, in which the aggregate alphabet is restricted to the binary packing $K_{\mathrm{pack}} = 2$, so $x \in \{0,1\}^m$ while shares range over $\mathbb{Z}_3$. A record carries the round tag, the full generator address j, and one residue in $\mathbb{Z}_3$; it does not carry x_j, since for every fixed first-round residue both $x_j = 0$ and $x_j = 1$ admit a compatible second-round residue. Each residue has $O(1)$ parameter payload, while a worst-case complete token has $\Theta(\log m)$ structural address bits as required by Definition 8.*

Remark 7 (Roles of the two flat schedules)**.** *The balanced encoding of Definition 15 is retained only as the configured experimental input in Section VIII; because each of its tokens contains x_j/r, it is not used to prove a representation lower bound. The theorem uses the masked-share stream instead. Before the second round arrives, every first-round update has two valid binary aggregate completions, whereas the semantic representation supplies x itself. The separation below therefore charges recovery of an aggregate hidden across rounds, not a voluntary delay after a full coefficient was already visible.*

C. Physically Realized Dispersed Representations

The masked-share and r-round streams above are stated adversarially: shares are chosen so that no token determines the aggregate. This subsection replaces the adversary by three standard fault-tolerant mechanisms. Time-dependent scheduling supplies the additive decomposition, randomized compiling supplies sign symmetry, and the synthesis grid of Lemma 5 supplies the modulus. The hypotheses of the lower bounds proved in Sections V and VI below are then met by streams that a fault-tolerant stack emits anyway, not by a constructed secret sharing.

Definition 17 (Scheduled phase-accumulation stream)**.** *The cyclic specialization of Definition 13 with modulus $Q \geq 3$, step $\Delta = 2\pi/Q$, $r \geq 2$ rounds, and the full sign set $\Sigma = \{\pm 1\}$. The residue $a_j^{(t)} \in \mathbb{Z}_Q$ is a round-t physical coefficient rounded to the grid and the sign $s_j^{(t)}$ is a twirl variable ($s_j^{(t)} \equiv +1$ for untwirled streams), so that the aggregate of Definition 13 reads*

$$x_j = \sum_{t=1}^{r} s_j^{(t)} a_j^{(t)} \bmod Q. \tag{19}$$

A twirled stream presents its signs in one of two modes. In folded *mode each sign is absorbed at emission, $a \mapsto sa \bmod Q$, into the residue and its dressing Clifford records, as in compile-time randomized compiling [6, 7]. In* tracked *mode the signs accumulate in a classical Pauli-frame register carried by the control stack and are settled downstream of the emission point [23, 32].*

Lemma 6 (Clifford interleaving)**.** *Let the r rounds of a scheduled stream be separated by Clifford records that are fixed across the family (they do not depend on the residues or signs), and let D_{t-1} denote the accumulated Clifford emitted before round t, including any Clifford records before the first round. Suppose each D_{t-1} carries the generating set into itself up to sign and permutation,*

$$D_{t-1}^{\dagger} P_j D_{t-1} = \varepsilon_j^{(t)} P_{\pi_t(j)}, \qquad \varepsilon_j^{(t)} \in \{\pm 1\}, \tag{20}$$

with π_t a permutation of $[m]$. Then commuting every Clifford record to the end of the stream yields a scheduled stream of Definition 17 on the same generators, in which the round-t record of generator j contributes residue $a_j^{(t)}$ with sign $\varepsilon_j^{(t)} s_j^{(t)}$ to coordinate $\pi_t(j)$, followed by the fixed Clifford suffix D_r. Both π_t and $\varepsilon^{(t)}$ are functions of the prefix, so the normalization is prefix-measurable, and every statement about Definition 17 applies to the normalized image.

Proof. Write the emitted unitary as $D_r' L_r \cdots D_1' L_1$ with L_t the round-t layer and D_t' the Clifford record following it, so $D_t = D_t' \cdots D_1'$. Moving each Clifford leftwards replaces L_t by $D_{t-1}^{\dagger} L_t D_{t-1}$ and leaves D_r as a suffix. Since $L_t = \prod_j R_{P_j}(\Delta s_j^{(t)} a_j^{(t)})$ and conjugation is an automorphism, (20) gives $D_{t-1}^{\dagger} L_t D_{t-1} = \prod_j R_{P_{\pi_t(j)}}(\Delta \varepsilon_j^{(t)} s_j^{(t)} a_j^{(t)})$, which is a round of Definition 17 after the stated re-indexing. The Clifford records lie in the stream, so D_{t-1}, and hence π_t and $\varepsilon^{(t)}$, are computable from the prefix. Finally D_r is common to every member of the family and unitary, so it preserves projective distance and the separation of Theorem 3 is unaffected. □

Remark 8 (What the interleaving lemma does and does not cover)**.** *Condition* (20) *is a real restriction, not a formality. It holds for Pauli-frame layers ($\pi_t = \mathrm{id}$, ε the frame signs), hence for the twirl insertions of randomized compiling itself; and for routing or SWAP layers whose induced qubit relabeling permutes the generating set, such as the reflection $q \mapsto n+1-q$ of a chain, which maps the chain characters $a_j = e_j + e_{j+1}$ to a_{n-j}. It fails for a general Clifford: if D_{t-1} normalizes the diagonal group but sends some P_j to a product of generators rather than to a generator, the rounds accumulate in different bases and the coordinatewise structure of Definition 17 is lost. It also does not rescue the non-commuting interleaving of Remark 10(i): the layers separating the diagonal rounds of a generic Trotter step are R_X rotations, which are not Clifford at generic angles.*

Remark 9 (Downstream compiler scope)**.** *The bounds obtained from scheduled streams constrain compilers positioned* downstream *of the schedule, the twirl, and the lowering: the input is the emitted record stream, and the upstream context—the pre-randomization circuit, the schedule generator, and the twirl seed—is not part of it. This position is not hypothetical; it is where much practical compilation runs: (1) a cloud transpiler receiving a lowered OpenQASM program without access to the user's high-level intent [21]; (2) the real-time processing of the instruction stream inside the classical control stack [23]; (3) a vendor-side pass that post-processes an already emitted instruction stream. If the compiler can see the circuit before randomization and lowering, the theorems do not apply, and this boundary is itself informative. The fooling families range over every valid schedule and sign pattern; a compiler coupled to the upstream generator—for example, one holding the λ-bit seed that pseudorandomly generated the twirl signs—faces only 2^{λ} of the 2^m sign patterns, and the frame lower bound of Proposition 5(ii) degrades from m to λ accordingly. The theorems therefore*

quantify exactly what is forfeited by discarding upstream context before compilation: they are the price attached to the design rule randomize and lower late.

Proposition 4 (Physical instantiation)**.** *Each of the following mechanisms can emit scheduled streams of Definition 17. For the families considered here, assume that the grid-rounded coefficients are free to vary independently across the r rounds. Under this coefficient-freedom hypothesis, the resulting stream family satisfies the hypotheses of Definition 21: every aggregate $x \in \mathbb{Z}_Q^m$ is realized by some valid stream, and no individual token, and no prefix that ends before the last round, determines it.*

(a) Time-dependent Trotterization. *For a commuting time-dependent Hamiltonian $H(t) = \sum_j c_j(t)P_j$ with step τ, round t applies $R_{P_j}(2c_j(t_t)\tau)$; grid rounding gives $a_j^{(t)}$ and $s \equiv +1$.*

(b) Layerwise-parameterized phase layers. *A depth-r ansatz whose round-t commuting layer carries free coefficients $\theta_j^{(t)}$ gives $a_j^{(t)}$ by grid rounding of $\theta_j^{(t)}$.*

(c) Randomized compiling of (a) or (b). *Twirling multiplies each residue by $s_j^{(t)}$, presented in folded or tracked mode. If the twirl bits are independent and uniform, then under the canonical sharing distribution of Proposition 9 every signed residue is uniform on $\mathbb{Z}_Q$ and independent of x_j.*

In contrast, the constant schedule $a_j^{(t)} \equiv a_j$ is degenerate: its first round determines the aggregate. This is the balanced-encoding degeneracy of Remark 7.

The coefficient-freedom hypothesis carries the weight of Proposition 4; the proposition records which streams can occur, not that a given pipeline produces them. The two halves of the hypothesis play different roles. The frontier inequality of Theorem 5 needs only the earlier rounds to range over $\mathbb{Z}_Q^m$; it would hold even with a fixed last round. The reading that early output must be unmerged needs the last round to be free as well: if, given the earlier rounds, the last-round residue of each term is confined to a few grid steps, as for a schedule with small increments, a compiler can commit part of each merged angle early. The workloads of Section IX repeat identical terms within each commuting block, or separate blocks by non-commuting layers; they therefore exercise the compiler's aggregation, not this bound.

Proof. In (a) and (b) the stated coefficient-freedom hypothesis makes the round coefficients unconstrained across rounds, so every residue sequence—in particular every valid share decomposition of Definition 21—arises from some schedule, and Proposition 9 supplies realizability and token privacy verbatim. For prefixes: after any prefix that ends before the last round, a final-round residue choice reaches every aggregate in $\mathbb{Z}_Q^m$. In (c), sign normalization $a \mapsto sa$ is a bijection of $\mathbb{Z}_Q$ for either sign, so normalizing every record maps the signed family onto the unsigned family record-for-record, and the same properties transfer; the distributional clause is the one proved in Proposition 9, applied after normalization. For the constant schedule, $x_j = ra_j \bmod Q$ is a function of the first round. An exhaustive enumeration at $(m, r, Q) = (2, 2, 5)$, frozen at `data/frozen/fooling_set_m2_r2_Q5.json`, independently confirms all three behaviors: every time-varying and every signed prefix class reaches all Q^m aggregates, and each constant-schedule prefix class reaches exactly one. □

Remark 10 (Contiguity of the rounds, and Clifford interleaving)**.** *Definition 17 is round-major over* one *commuting layer: the r rounds that accumulate into x_j reach the compiler as records of a single layer. Two consequences should be stated plainly, since they delimit Proposition 4.*

(i) Non-commuting interleaving is outside the scope, and is not claimed. *In a split-operator product formula such as $H = H_{ZZ} + H_X$, consecutive diagonal rounds are separated by non-commuting layers. There the rounds of the diagonal part may not be aggregated at all: no compact output exists, the premise of the recoverability question fails, and neither the hypotheses nor the conclusions of Theorems 4 and 5 are asserted. Proposition 4(a) is a statement about layers whose r rounds are genuinely contiguous—a time-dependent or shaped schedule of one commuting layer, a commuting-block Trotterization, a contiguous diagonal ladder—and not about the interleaved structure of a generic Trotter step. The same qualification applies to (b): variational cost layers separated by mixers are aggregatable only within a block of consecutive commuting rounds, and it is such a block that (b) describes, with independent per-term parameters $\theta_j^{(t)}$. Note that for a* fully *commuting H the Trotter error vanishes and the round structure is a scheduling choice rather than an accuracy requirement; what the bounds charge is precisely that choice, made upstream and visible to the downstream compiler only as r separate records.*

(ii) Clifford interleaving is inside the scope. *Rounds separated by Clifford records normalize back to Definition 17 whenever the accumulated Clifford permutes the generating set up to sign; this is Lemma 6, and Remark 8 delimits it. Contiguity in Definition 17 should therefore be read modulo such Clifford records.*

Contiguity and coefficient freedom are separate hypotheses, and different families supply them. Contiguous diagonal rounds arise structurally in controlled-power ladders and phase estimation, where c-U^{2^k} contributes a sequence of diagonal layers with no intervening non-commuting structure (the `qpe_controlled_powers` *and* `controlled_power_ladder` *families of Section IX). Round coefficients that vary freely across rounds—what the fooling families of Proposition 4 require—are supplied by time-dependent schedules and by layerwise-parameterized ansatz blocks whose per-term coefficients vary independently, such as multi-angle QAOA [2]. A standard QAOA cost layer, in which every round rescales one weight vector*

by a common angle γ_t, is contiguous but not coefficient-free: one round fixes the aggregate up to a scalar and rounding, so its streams are not a fooling family and the bounds do not constrain such an instance. The theorems also require $\mathbb{F}_2$-independent generators (Definition 13); for the edge characters Z_uZ_v of a connected graph on $|V|$ vertices the rank is $|V|-1$, and the bounds apply with $m=|V|-1$ to the edges of a spanning tree.

Proposition 5 (Frame as cross-cut state)**.** *Let a compiler in the charged model of Definition 9 be correct within projective error $\epsilon < \sin(\pi/(2Q))$ on every valid twirled stream of Definition 17.*

(i) In both presentation modes the normalized stream is a valid dispersed stream of Definition 21, so Theorems 4 and 5 apply verbatim to the physical stream.

(ii) In tracked mode, the pending-frame classes alone force the following. If the compiler is deterministic, zero-error, and reads the stream in one forward pass, then at any inter-round cut c, with committed prefix output of length ℓ_{pre} and snapshot budget $B_{\mathrm{cut}}(c)$, the worst case over valid twirled streams obeys

$$\max\big(\ell_{\mathrm{pre}} + B_{\mathrm{cut}}(c)\big) \geq m. \tag{21}$$

If it is randomized, correct with probability at least $1-\delta$, $\delta < 1/2$, and makes at most p forward passes, then

$$\overline{A}_p + \overrightarrow{\Sigma}_p \geq (1-\delta)m - h_2(\delta) - \lceil \log_2(p+1) \rceil. \tag{22}$$

The frame register is live classical restart state and is therefore charged in the snapshot by Definition 9. For a deferred one-pass compiler with $\ell_{\mathrm{pre}} = O(\log m)$, the crossing snapshot carries $m - O(\log m)$ bits that determine the frame action on the m generators.

(iii) In folded mode the flush map is a prefix-measurable sign normalization, injective on the sign components used below; the same information appears in the signed residues, and (21) and (22) hold with the same fooling family presented in folded records.

Proof. (i) Sign normalization is a bijection of $\mathbb{Z}_Q$ for either sign, so normalizing every record maps the twirled family onto the family of Definition 21 while preserving aggregates, record boundaries, and cut positions; each normalized stream is the image of the twirled streams that differ only in their sign patterns. The cited theorems apply to the image and pull back verbatim.

(ii) Take prefixes whose residue fields emit the residue 1 exactly once on each generator and are otherwise zero, and whose sign fields realize the frame class $\sigma \in \{\pm 1\}^m$; the residue fields are identical across classes. Choose one common suffix adding a residue a_j on generator j, for instance $a_j = 0$. The completed aggregates $x_j = \sigma_j + a_j \bmod Q$ then differ between two classes in every coordinate where the frames differ, because $2 \not\equiv 0 \pmod{Q}$ for $Q \geq 3$, and all completions are valid family instances. By Theorem 3 the corresponding targets are separated by $2\sin(\pi/(2Q)) > 2\epsilon$. For a deterministic one-pass compiler the suffix computation is a function of the snapshot and the common suffix; a run that halted before c would give the same output for two classes with different targets, so every run reaches c, and the pair (prefix output, snapshot) must be injective on the 2^m frame classes; prefix-freeness of the self-delimiting snapshot followed by the committed prefix and Kraft's inequality give (21). For the multipass statement, draw σ uniformly and repeat the forward-transcript argument of Theorem 5 with the common suffix in place of V: the forward transcript determines the final output, the decoder recovers σ with error at most δ, and Fano's inequality gives $I(\sigma;T) \geq (1-\delta)m - h_2(\delta)$. The frame register is classical restart state, so its bits lie in the named fields; subtracting the deferred-output cap bounds the snapshot share.

(iii) In folded mode the sign is consumed at emission, so the crossing information consists of the signed residues ± 1 themselves, which are distinct for $Q \geq 3$; the argument of (ii) applies verbatim to the folded records. □

Corollary 4 (Physically realized lower bounds)**.** *For the streams of Proposition 4(a)–(c) with the tuned modulus Q_ϵ of Corollary 8, the hypotheses of Definitions 21 and 22 are met under the coefficient-freedom hypothesis of Proposition 4, so Theorems 4 and 5 hold verbatim on the physical stream: for $0 < \epsilon \leq 1/8$, any compiler positioned downstream of the schedule and of the twirl that is correct within projective error ϵ with probability at least $1-\delta$, $\delta < 1/2$, and makes at most p forward passes obeys $\overline{A}_p + pS \geq (1-\delta)m \log_2 K_\epsilon - h_2(\delta) - \lceil \log_2(p+1) \rceil$, and under tracked randomized compiling the frame bounds of Proposition 5(ii) hold as well. For a coarse grid with $Q = 3$, accuracy $\epsilon < 1/2$, the subfamily with binary aggregates, two rounds, and $s \equiv +1$ is the masked-share stream, so the statements of Theorem 2 apply to single-delivery compilers that are correct on it.*

Proof. The main bound is immediate from Propositions 4 and 5(i). For the $Q = 3$ clause, restrict to binary aggregates, zero intermediate rounds, and $s \equiv +1$; the remaining streams are those of Definition 16, with the cut of Definition 19 before the last round. □

A single knob interpolates between the two regimes: the fraction of coordinates that reach the compiler in dispersed rather than pre-aggregated form sets a lower bound on committed output plus crossing state that is linear in that fraction. We call that fraction the *dispersal fraction* η. The name is deliberate: η is not the twirl strength of randomized compiling (the probability with which a random Pauli is applied), and the proof below never uses the twirl—it conditions on the dispersal mask and applies Theorem 5 to the dispersed coordinates. What η measures is the fraction of phase content presented upstream in late-aggregated form, which is exactly the axis of the design rule.

Proposition 6 (Dispersal–memory Pareto bound)**.** *Fix a dispersal fraction $\eta \in [0,1]$, $Q \geq 3$, $2 \leq K \leq Q$, $0 \leq \epsilon < \sin(\pi/(2Q))$, and $0 \leq \delta < 1/2$. Let each generator independently with probability η enter the stream in dispersed twirled form (canonical sharing of Proposition 9 with uniform signs), and otherwise in aggregate-exposed form. Let $\mathcal{A}$ be correct within ϵ with probability at least $1-\delta$ on every valid stream and mask, making $p = p(m)$ forward passes in the charged model of Definition 22. Write $\overline{A}_p(T)$ and $S(T)$ for the worst-input expected quantities of Definition 22 taken over the streams with dispersal mask T. Then, in expectation over T,*

$$\begin{aligned}&\mathbb{E}_T\big[\overline{A}_p(T) + pS(T)\big] \\ &\quad \geq (1-\delta)\,\eta m \log_2 K - h_2(\delta) - \lceil \log_2(p+1) \rceil.\end{aligned} \tag{23}$$

Under tracked presentation a compiler that keeps the pending signs separately pays for them in its crossing state (Proposition 5(ii)), but they do not add to this bound: the target depends on the signs only through the signed aggregate (19)*, so a deferred compiler may fold each pending sign into the residue it stores. For a deterministic one-pass zero-error compiler and $K \in \{Q-1, Q\}$, Theorem 4 applied to the dispersed coordinates strengthens* (23) *to $\mathbb{E}_T[A_1(T) + B_{\mathrm{cut}}(T)] \geq \eta m \log_2 Q$, which is a bound on crossing state alone for a deferred compiler with $A_1 = 0$.*

Proof. Condition on the mask T. Restrict the signs to $s \equiv +1$ and fix the exposed coordinates; T and the exposed values are then public, known to both parties of the transcript argument, and are built into the transition function rather than stored as state. The targets are the dispersed family on the $|T|$ generators in T multiplied by one fixed unitary W of the exposed coordinates. The proof of Theorem 5 uses only pairwise separation of the targets, which this translation preserves, so it applies unchanged and gives $\overline{A}_p(T) + pS(T) \geq (1-\delta)|T| \log_2 K - h_2(\delta) - \lceil \log_2(p+1) \rceil$ for every fixed T. Take the expectation over the Bernoulli mask and use $\mathbb{E}|T| = \eta m$. □

Figure 2 measures (23) on the executable masked-share device: the two admissible payment channels of the bound appear as the two panels, the state channel sits within 5% of its serializer's storage cost, a factor $(\log_2 3{+}1)/\log_2 3 \approx 1.63$ above the proven zero-error floor, and the exchange between them is linear in the dispersal fraction with slope proportional to m.

The two panels also expose an asymmetry that the frontier itself does not predict. Per dispersed generator the state channel of this serializer costs about 2.6 bits—$\log_2 3 + 1 = 2.585$ for a residue and a separately stored sign, against a proven floor of $\log_2 3 \approx 1.585$—while the committed channel costs about 16: an exchange rate near 6:1 between two currencies the frontier charges at par. The next proposition isolates the part of that gap that is not an artifact of the serializer. This requires one additional hypothesis beyond Definition 9, and the hypothesis is substantive rather than technical: without it, the gap is not merely unprovable but zero, and we state that converse alongside the bound.

Definition 18 (Executed and buffered commitment)**.** *Let c be the inter-round cut of a two-round partially dispersed stream as in Proposition 6. Let $T \subseteq [m]$ be its dispersal mask: coordinates in T carry two signed shares, whereas coordinates outside T arrive in aggregate-exposed form. We fix the record format: round one carries one addressed record per coordinate in T; round two carries one record per coordinate, a dispersed update for each coordinate in T, kept even when its residue is zero, and an aggregate-exposed record of a distinguishable type for each coordinate outside T. A deferred compiler therefore learns T from the round-two record types. Suppose the compiler has irrevocably written the round-one content $W := (a_j^{(1)}, s_j^{(1)})_{j \in T}$ before c. The commitment is* executed *if the prefix O_{pre} is applied at c, before any round-two record is parsed. We distinguish two readings of what an executed prefix must fix. It is executed* at record level *if it dispatches one addressed record per dispersed generator, as a control stack that forwards the compiler's records to the device does; such a prefix determines the pair (T, W) on its own, without the suffix and without the cut snapshot. This is a modelling assumption, not a consequence of applying a unitary. It is executed* at unitary level *if, on its own and for every value of the random seed, it is only required to implement, within ϵ, the round-one action $U^{(1)} := \prod_{j \in T} R_{P_j}\big(\Delta\, s_j^{(1)} a_j^{(1)}\big)$, which fixes the vector $y \in \mathbb{Z}_Q^m$ with $y_j = s_j^{(1)} a_j^{(1)} \bmod Q$ for $j \in T$ and $y_j = 0$ otherwise, but neither T nor W. Unqualified, "executed" means record level. The commitment is* buffered *if O_{pre} is written and never rewritten, but is applied only after later records have been read, in which case it need not determine (T, W). Call $\mathcal{A}$* mask-uniform *if a single algorithm is correct within ϵ on every valid partially dispersed stream for every mask, its committed prefixes over all masks form one prefix-free set, and the action of a prefix is a function of the prefix alone, not of T.*

Proposition 7 (Commitment toll under executed commitment)**.** *Fix $Q \geq 3$, $\epsilon < \sin(\pi/(2Q))$, $r = 2$, and $1 \leq k \leq m$. Let $\mathcal{A}$ be mask-uniform and correct within ϵ on every valid two-round partially dispersed stream described in Definition 18, with a prefix-free committed serializer. Draw the dispersal mask T uniformly from the $\binom{m}{k}$ subsets of size k, and the committed content W uniformly on $\mathbb{Z}_Q^k \times \{\pm 1\}^k$ independently of T.*

(i) (Executed commitment pays the mask entropy.) *If $\mathcal{A}$ commits in record-level executed form at the inter-round cut, then*

$$\mathbb{E}_{T,W}\big[A_{\mathrm{pre}}\big] \;\geq\; \underbrace{k \log_2(2Q)}_{payload} \;+\; \underbrace{\log_2 \tbinom{m}{k}}_{toll}, \tag{24}$$

and a fortiori $\max_{T,W} A_{\mathrm{pre}}$ *obeys the same bound. A deferred compiler, holding W as one packed integer in $\mathsf{ser}_{\mathrm{store}}$, attains $A_{\mathrm{pre}} = 0$ with crossing state $S \leq k \log_2(2Q) + O(\log m)$ (Definition 22), so the identical content crosses the cut for at least $\log_2 \binom{m}{k} - O(\log m)$*

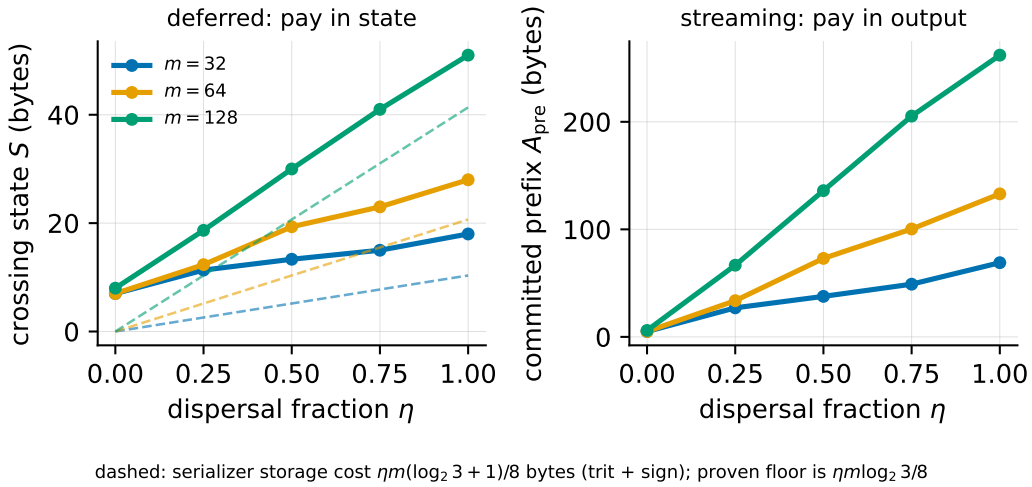


Figure 2. Measured dispersal–memory tradeoff on the executable two-round masked-share device ($Q = 3$, binary aggregates, tracked signs; every point restart-verified: the serialized snapshot alone plus the round-two suffix reproduces the exact output). Left: a deferred compact-output compiler pays the dispersal in crossing state; measured slopes 10.3/21.1/43.3 bytes per unit η at $m = 32/64/128$ match the storage cost of its serializer, one trit and one separately stored sign bit per dispersed generator, $\eta m(\log_2 3 + 1)/8$ bytes (dashed; ratios 0.99–1.05). This is not a lower bound: the proven zero-error one-pass floor is $\eta m \log_2 3/8$ bytes (Proposition 6), and the extra bit per generator is the sign, which a serializer folding it into the residue would not store. Right: a streaming compiler pays the same dispersal in committed prefix output (A_{pre} slopes 60/129/260 bytes per unit η, larger by the public addresses) while its crossing state stays flat at ≤ 8 bytes. The horizontal axis is the dispersal fraction η of Proposition 6. Frozen artifact: `data/frozen/eta_sweep_pareto.json`, generated by `PaperDraft/scripts/run_eta_sweep.py`.

fewer bits in the state channel. That difference is the toll.

(ii) (Buffered commitment pays none of it.) *For $k = \Theta(m)$ with $k \leq m - \Omega(m)$, where $\log_2 \binom{m}{k} = \Theta(m)$ exceeds the $O(\log m)$ header, the hypothesis in (i) cannot be removed. The mask-deferred compiler that writes W irrevocably in positional order and recovers T from the round-two records is admissible in Definition 9, which requires only that the final output be write-once, and is correct; it commits in buffered form, and satisfies $\mathbb{E}_{T,W}[A_{\mathrm{pre}}] \leq k \log_2(2Q) + O(\log m)$: it pays the payload and the header and no mask entropy whatever.*

The toll is therefore a property of executed *commitment, not of commitment as such. Where it is paid it is at least the entropy of the dispersal mask, up to the $O(\log m)$ header, which is at most $h_2(k/m)/(k/m)$ bits per committed generator. Among prefixes that determine (T, W), an enumerative code of the mask attains the bound up to $O(\log m)$; with one self-delimiting addressed record per generator the overhead is $O(k)$. A fixed public mask has zero entropy; within the uniform size-k ensemble above, the toll therefore vanishes at $k = m$ (and trivially at $k = 0$, which the proposition excludes).*

Proof. (i) By Definition 18 a record-level executed prefix determines (T, W), so $H(O_{\mathrm{pre}}) \geq H(T, W)$. By mask uniformity the committed prefixes form one prefix-free code over the joint source (T, W), and the expected length of a prefix-free code is at least the entropy of its source; hence $\mathbb{E}[A_{\mathrm{pre}}] = \mathbb{E}|O_{\mathrm{pre}}| \geq H(T, W)$. By construction, T and W are independent and uniform, so $H(T, W) = \log_2 \binom{m}{k} + k \log_2(2Q)$, which is (24), and a maximum dominates a mean. For the deferred compiler the round-major order is public, so its snapshot holds the k residues and signs in positional order while the round-two record types distinguish dispersed updates from aggregate-exposed coordinates and thereby disclose the mask at no charge; W is held as one packed integer of $\lceil k \log_2(2Q) \rceil$ bits in $\mathsf{ser}_{\mathrm{store}}$, and the $O(\log m)$ term is the header. Ranking T among the $\binom{m}{k}$ masks gives a prefix of length $\lceil \log_2 \binom{m}{k} \rceil + \lceil k \log_2(2Q) \rceil + O(\log m)$ that determines (T, W), which shows the bound is tight up to the header in that class.

(ii) The mask-deferred compiler never rewrites an emitted record, so it meets Definition 9, and it is correct because the round-two records supply T before the final output is written. Its prefix holds W in positional order at $k \log_2(2Q)$ bits plus a header, and reveals no function of T beyond the value of k already implied by its length, so no $\log_2 \binom{m}{k}$ term arises. There is no conflict with (i): this prefix does not determine (T, W), hence is not executable at the cut, and the compiler has bought its exemption by giving up the ability to apply what it has written. □

Remark 11 (Scope of Proposition 7, and two degeneracies)**.** *Three limits of the statement are worth naming. First, Definition 18 is an assumption about the compiler and not a consequence of Definition 9: write-once output fixes the* semantics *of an emitted record—it is never*

reinterpreted—but does not require the record to name its own targets, and clause (ii) is precisely a compiler that exploits the difference. What the definition models is the real-time control stack of Remark 4, where the prefix reaches the device before the suffix is parsed. Second, the bound is on an ensemble. For a single mask fixed in advance and known to the serializer there is nothing to pay, and the $\log_2 \binom{m}{k}$ *term is then not merely unprovable but false; mask uniformity is what makes the schedule a source rather than a constant. Third, both the payload and the toll are charged against the declared record content* $(a^{(1)}, s^{(1)})$*, not against the round-one unitary, which determines strictly less: a coordinate whose signed residue* $s_j^{(1)} a_j^{(1)}$ *is* 0 *acts as identity and is invisible, and* (s, a) *and* $(-s, -a \bmod Q)$ *induce the same rotation. The sign degeneracy affects only the payload. The zero-residue degeneracy affects the toll as well, because an invisible coordinate hides its membership in* T*: at unitary level the toll is smaller, and Corollary 5 bounds it from below. Clause (i) and Fig. 3 concern record-level commitment.*

Corollary 5 (The toll at unitary level)**.** *Under the hypotheses of Proposition 7, suppose* $\mathcal{A}$ *commits in unitary-level executed form (Definition 18). Let* $\nu \sim \mathrm{Bin}(k, 1 - 1/Q)$ *be the number of dispersed coordinates whose signed first-round residue is nonzero. Then*

$$\begin{aligned} \mathbb{E}_{T,W}\big[A_{\mathrm{pre}}\big] &\geq H(y) \\ &= \log_2 \tbinom{m}{k} + k \log_2 Q - \mathbb{E}\Big[\log_2 \tbinom{m-\nu}{k-\nu}\Big], \end{aligned} \tag{25}$$

while a deferred compiler that folds each sign into its stored residue attains $A_{\mathrm{pre}} = 0$ *with* $S \leq k \log_2 Q + O(\log m)$*, holding the folded residues as one packed integer. The unitary-level toll is therefore at least*

$$\tau_Q(m, k) := \log_2 \tbinom{m}{k} - \mathbb{E}\Big[\log_2 \tbinom{m-\nu}{k-\nu}\Big] - O(\log m), \tag{26}$$

and per committed generator, with $\eta = k/m$ *fixed and* $m \to \infty$*,*

$$\frac{\tau_Q(m, k)}{k} \;\to\; \frac{h_2\big(\eta(1 - 1/Q)\big)}{\eta} - h_2(1/Q). \tag{27}$$

The rate is attained: an executed prefix that writes ν*, the rank of* $\operatorname{supp} y$ *among the* ν*-subsets of* $[m]$*, and the nonzero values of* y *packed in base* $Q - 1$ *has* $\mathbb{E}[A_{\mathrm{pre}}] \leq H(y) + O(\log m)$*. This rate is positive for* $0 < \eta < 1$*, vanishes at* $\eta = 1$*, increases to the record-level rate* $h_2(\eta)/\eta$ *as* $Q \to \infty$*, and equals* 0.918 *bits at* $Q = 3$*,* $\eta = \frac{1}{2}$*, against* 2 *bits at record level.*

Proof. For $j \in T$ the signed residue $c_j := s_j^{(1)} a_j^{(1)} \bmod Q$ is uniform on $\mathbb{Z}_Q$, because $a_j^{(1)}$ is uniform and independent of $s_j^{(1)}$; so (T, c) is uniform on $\binom{[m]}{k} \times \mathbb{Z}_Q^k$. If $y \neq y'$, the corresponding round-one actions differ by a nonzero multiple of $\Delta = 2\pi/Q$ on some generator, and Theorem 3 with the full residue alphabet $K = Q$ separates them by $2 \sin(\pi/(2Q)) > 2\epsilon$; integer and modular residues give the same rotations up to a global phase ± 1. No prefix is within ϵ of both, so an executed prefix determines y, and mask uniformity makes the prefixes one prefix-free code; source coding gives $\mathbb{E}[A_{\mathrm{pre}}] \geq H(y)$. The map $(T, c) \mapsto (y, T)$ is injective, so $H(y) = H(T, c) - H(T \mid y)$. Given y with support of size σ, the consistent masks are the k-sets containing the support, all equally likely, so $H(T \mid y) = \mathbb{E} \log_2 \binom{m-\nu}{k-\nu}$ with $\nu = |\operatorname{supp} y| \sim \mathrm{Bin}(k, 1 - 1/Q)$; this is (25). Equivalently, writing $V := \operatorname{supp} y$, the values of y on V are uniform on $(\mathbb{Z}_Q \setminus \{0\})^V$ and independent of T given V, so $H(T \mid y) = H(T \mid V)$ and the excess of $H(y)$ over $k \log_2 Q$ is $H(T) - H(T \mid V) = I(T; V)$. The deferred compiler stores the k signed residues in the public round-major order and recovers T from the round-two record types, as in Proposition 7(i); the target depends on the signs only through the signed aggregate (19), so the folded residues suffice. Subtracting gives (26). For (27), use $I(T; V) = H(V) - H(V \mid T)$. Given T, each member of T lies in V independently with probability $1 - 1/Q$, so $H(V \mid T) = k\, h_2(1/Q)$. Each $j \in [m]$ lies in V with probability $\eta(1 - 1/Q)$, so subadditivity gives $H(V) \leq m\, h_2(\eta(1 - 1/Q))$. Conversely V is a uniformly random subset of $[m]$ given its size $\nu \sim \mathrm{Bin}(k, 1 - 1/Q)$, so $H(V) \geq \mathbb{E} \log_2 \binom{m}{\nu}$. The bound $\log_2 \binom{m}{s} \geq m\, h_2(s/m) - \log_2(m+1)$, a second-order expansion of h_2 around $\eta(1 - 1/Q)$, which lies in $(0, 1)$, and a Chernoff tail bound for ν give $H(V) \geq m\, h_2(\eta(1 - 1/Q)) - O(\log m)$. Dividing by $k = \eta m$ gives the limit, approached from below. For attainability, given ν the support is uniform over the ν-subsets of $[m]$ and the nonzero values are uniform on $(\mathbb{Z}_Q \setminus \{0\})^\nu$, so the stated prefix has expected length $H(y) - H(\nu) + O(\log m) \leq H(y) + O(\log m)$, and it implements the round-one action on its own. Strict concavity of h_2 with $h_2(0) = 0$ gives $h_2(\eta u) > \eta h_2(u)$ for $0 < \eta < 1$ and $u = 1 - 1/Q$, which is positivity, since $h_2(u) = h_2(1/Q)$. The derivative in u is $h_2'(\eta u) - h_2'(u) > 0$ because h_2' is decreasing, so the rate increases with Q, to $h_2(\eta)/\eta$ as $u \to 1$. □

The two channels are therefore *not* at par once commitment is executed rather than buffered: a record-level executed prefix must identify the dispersal mask, and even a unitary-level one must identify the part of it that its nonzero rotations reveal, while deferring—or merely buffering—obtains that same mask from the stream that follows. Because $\log_2 \binom{m}{k}/k \to h_2(\eta)/\eta$, the record-level toll, where it is paid at all, is bounded by a constant—about 2 bits per generator at $\eta = \frac{1}{2}$ and 0 at $\eta = 1$—and does not grow with m. The unitary-level toll of Corollary 5 is a lower bound measured against a deferred compiler that folds the signs, whereas the record-level toll is measured against one that stores them; at $\eta = \frac{1}{2}$ that lower bound is asymptotically 0.918 bits at $Q = 3$ and 1.99 bits on the $\epsilon = 10^{-3}$ grid ($Q_\epsilon = 785$).

Figure 3 instantiates both clauses on the same device under a preregistered protocol frozen before the run (`PaperDraft/research/E8_preregistration.md`), with every cell restart-verified. Three encoders are compared

at $m \in \{128, 512, 2048\}$ (and m up to 32768 for two of them): the deferred snapshot, the address-per-record serializer of Fig. 2, and an identity-optimal executed prefix that ranks the mask combinadically. The buffered control of Proposition 7(ii), whose prefix omits the mask and recovers it from the suffix, is the deferred snapshot written to the output, so it has no separate encoder. These runs are constructive checks of a source-coding bound, not tests that could have failed: the executed encoder's body has $\lceil \log_2(\binom{m}{k}(2Q)^k) \rceil$ bits by construction, so its excess over the deferred snapshot is $\log_2 \binom{m}{k}/k$ up to rounding, and the control's excess is zero by identity. What the runs establish is that both encoders are correct and restartable, so the bound of clause (i) is attained and buffering removes it. The measured excess matches $\log_2 \binom{m}{k}/k$ within 0.06 bits over the preregistered analysis cells ($k \geq 16$; 0.17 bits in the one excluded cell, $m = 128$, $\eta = \frac{1}{8}$) and within 0.02 bits typically; it is flat in m at fixed η (2.11/1.99/1.97 bits at $\eta = \frac{1}{2}$ across a 16× range of m); and it is exactly zero at $\eta = 1$. The comparison that matters is conceptual: the control pays the same payload into the same write-once output and rewrites nothing, so what separates it from the executed encoder is not irreversibility and not addressing but whether the prefix can be applied before the suffix is read.

This also explains the 6:1 ratio in Fig. 2. Under the identity-optimal serializer the ratio is 1.00 at $\eta = 1$ and at most 2.64 anywhere in the measured range; the remainder is codec slack, and it is the slack—not the toll—that grows with address width, from 5.4:1 at $m = 128$ to 10.8:1 at $m = 32768$. The preregistration recorded this as a prediction *against* the natural reading of Fig. 2, and we report it as such: there is no address-width law here.

Two attempts to carry the toll onto compilers we do not control were preregistered, and **both failed**; we report them rather than repair them. The first asked only whether real toolchains spend committed bytes on addresses, and its design placed every rotation on the low-index qubits, so it never varied the address width it was meant to vary (a corrected rerun, labelled non-preregistered in the frozen artifact, finds 2.95 bytes of growth per rotation over a 1000× range of register width, itself below the preregistered threshold). The second tested the mechanism directly: two circuits identical in m, in k, in rotation count and in angle multiset, differing only in whether the dispersed subset is an arithmetic progression or a random k-subset, emitted through Qiskit and TKET and compared by compressed size. Its textual control passed—the variants differ by under 0.05% of raw bytes, so the comparison is not confounded by text length—and on TKET the excess reaches 86% of $\log_2 \binom{m}{k}$ at $k = m/2$, comfortably above the seed noise. But the excess peaks at $k = m/4$ rather than $k = m/2$, and on Qiskit it is 8% of prediction and smaller than the seed-to-seed noise. Under the frozen rule this is a failure, and we make no third-party claim. Two causes are possible and the run does not separate them: a general-purpose compressor applied to a 100 kB instruction text may be too coarse an entropy estimator for a few hundred bytes of schedule information, and neither tool emits a round-major committed stream (Qiskit's OpenQASM 3 output is ordered by qubit, and TKET's lists the full first layer before the dispersed addresses), so the design premise that the output is a committed prefix was not met. The record-level toll of Proposition 7 rests on the source-coding argument and on the instrumented device of Fig. 3; whether it is measurable at a third-party API boundary remains open.

V. INFORMATION LOWER BOUNDS

We prove two lower bounds using one-way message arguments [26]. The first uses the $\bar{K}_\epsilon^m$-point no-wrap packing and Fano's inequality to obtain an explicit accuracy–description tradeoff, including randomized compilers. The second injects the 3^m possible first-round masks into the message crossing the inter-round cut. Its deterministic zero-error bound is matched by a memory-capped hybrid compiler, and a conditional-Fano extension covers randomized failure probability δ.

A. Packing Decoders

Lemma 7 (Nearest-packing decoder)**.** *Let $\epsilon > 0$ and let $\mathcal{F} = \{U_x : x \in \mathcal{X}\}$ have pairwise distance at least 4ϵ. From any output circuit C, define*

$$\widehat{x}(C) \in \arg\min_{y \in \mathcal{X}} d_{\mathrm{proj}}(U(C), U_y),$$

with a fixed tie rule. If $d_{\mathrm{proj}}(U(C), U_x) \leq \epsilon$, then $\widehat{x}(C) = x$.

Proof. The true target is at distance at most ϵ. Every other target is at distance at least $4\epsilon - \epsilon = 3\epsilon$ by the triangle inequality, so it cannot minimize. □

The following binary-coordinate version will be used for the masked-share tradeoff, where $K = 3$ denotes the sharing modulus but the aggregate alphabet remains $\{0, 1\}$.

Lemma 8 (Bit-class separation)**.** *For the binary subfamily of Definition 14, put $\epsilon_0 = \sin(\Delta/4)$. For each $i \in [m]$, partition the family into $\mathcal{F}_i^{(0)} = \{U_x : x_i = 0\}$ and $\mathcal{F}_i^{(1)} = \{U_x : x_i = 1\}$. Then the inter-class distance satisfies*

$$\mathrm{dist}\big(\mathcal{F}_i^{(0)}, \mathcal{F}_i^{(1)}\big) := \inf_{\substack{x_i=0 \\ y_i=1}} d_{\mathrm{proj}}(U_x, U_y) \;\geq\; 2\epsilon_0. \tag{28}$$

Proof. Any pair with $x_i \neq y_i$ has $x \neq y$, so Lemma 4 gives $d_{\mathrm{proj}}(U_x, U_y) \geq 2\epsilon_0$. Taking the infimum preserves the inequality. □

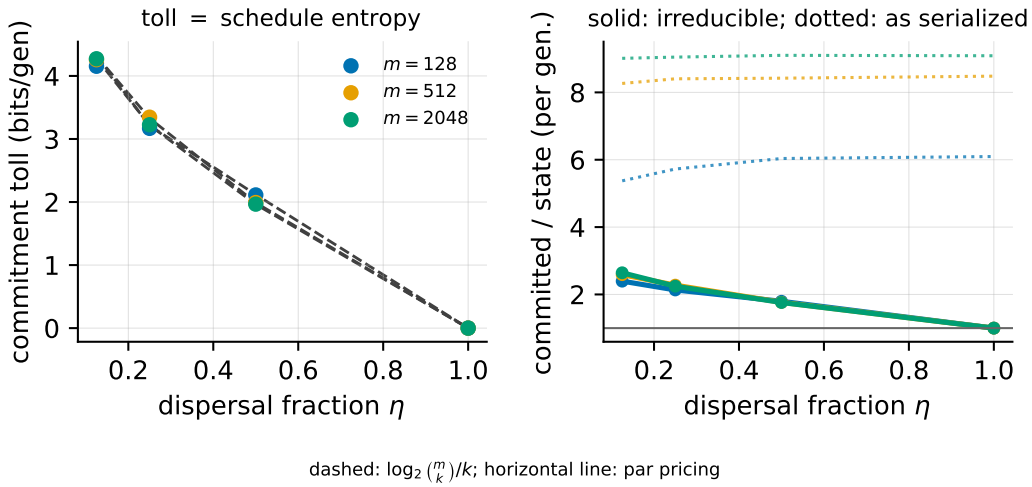


Figure 3. The record-level commitment toll attained on the two-round masked-share device under a preregistered protocol; every cell restart-verified. This is a constructive check, not a test: the executed encoder attains the source-coding bound by construction. Left: bits per committed generator paid by an executed committed prefix in excess of the deferred snapshot, against the mask entropy $\log_2 \binom{m}{k}/k$ (dashed). The toll is flat in m and vanishes at $\eta = 1$; for the buffered mask-deferred control of Proposition 7(ii), whose prefix is the deferred snapshot, it is zero at every η by identity. Right: the per-generator exchange rate between the committed and state channels. Solid: irreducible, under an identity-optimal serializer—at most 2.64, and exactly 1 when the dispersal mask is public. Dotted: the same rate as actually serialized by the address-per-record codec of Fig. 2, which is where the 6:1 comes from and which is the only part that grows with address width. Frozen artifact: `generated/e8_commit_toll.json`, generated by `PaperDraft/scripts/run_commit_toll.py`.

Lemma 9 (Representation-invariant decoder)**.** *For each $i \in [m]$ and any circuit C, define*

$$\delta_i(C) := \arg \min_{b \in \{0,1\}} \min_{y:\, y_i = b} d_{\mathrm{proj}}(U(C), U_y), \tag{29}$$

with ties broken toward 0, so that δ_i is a function of C. (Bob's local computation is unbounded, so he may exhaustively search the 2^{m-1} candidates of each class.) If $d_{\mathrm{proj}}(U(C_x), U_x) \le \epsilon < \epsilon_0$, then $\delta_i(C_x) = x_i$.

Proof. The correct class $b = x_i$ contains U_x itself, so its inner minimum is at most ϵ. Suppose the wrong class $b \neq x_i$ also attained an inner minimum $\le \epsilon$, realized by some U_y with $y_i = b$. Then the triangle inequality gives $d_{\mathrm{proj}}(U_x, U_y) \le 2\epsilon < 2\epsilon_0$, contradicting Lemma 8. Hence only the correct class attains the minimum, i.e. $\delta_i(C_x) = x_i$. □

B. Output-Description Lower Bound

Theorem 1 (Randomized ϵ–description tradeoff)**.** *Let $0 < \epsilon \le 1/8$, let $\mathcal{F}_{m,\epsilon}$ be the $\bar{K}_\epsilon$-ary family of Proposition 3, and let a possibly randomized compiler $\mathcal{A}$ output $C_{x,\rho}$ satisfying, for every x,*

$$\Pr_{\rho}[d_{\mathrm{proj}}(U(C_{x,\rho}), U_x) \le \epsilon] \ge 1 - \delta, \qquad 0 \le \delta < 1/2. \tag{30}$$

The compiler may use random-access RAM/DAG IR, hash tables, caches, and a dynamic pass count $p = p(m, x, \rho)$ as in Definition 9. Writing $h_2(\delta) = -\delta \log_2 \delta - (1-\delta)\log_2(1-\delta)$ with $0 \log 0 := 0$, each output code separately obeys

$$\begin{aligned} \max_x \mathbb{E}_\rho B_{\mathrm{rel}}(C_{x,\rho} \mid \mathcal{F}_{m,\epsilon}) &\ge (1-\delta) m \log_2 \bar{K}_\epsilon - h_2(\delta), \\ \max_x \mathbb{E}_\rho B_{\mathrm{self}}(C_{x,\rho}) &\ge (1-\delta) m \log_2 \bar{K}_\epsilon - h_2(\delta). \end{aligned} \tag{31}$$

In particular, for fixed $\delta < 1/2$ both are $\Omega(m \log(1/\epsilon))$.

At $\delta = 0$, the family-relative bound is tight to lower-order framing terms: a deterministic semantic compiler uses

$$B_{\mathrm{rel}}(C_x \mid \mathcal{F}_{m,\epsilon}) \le \lceil m \log_2 \bar{K}_\epsilon \rceil + O(\log m + \log \bar{K}_\epsilon). \tag{32}$$

Separately, for $P_j = Z_j$ and $RZ \in B$, the explicit self-contained circuit satisfies $B_{\mathrm{self}}(C_x) = O(m(\log m + \log \bar{K}_\epsilon))$ and $G(C_x) \le m$; no self-contained-bit or gate-count tightness claim is made.

Proof. Let X be uniform on $[\bar{K}_\epsilon]_0^m$ and let $C = C_{X,\rho}$. By (15), the nearest-packing decoder of Lemma 7 has error probability at most δ. Fano's inequality gives

$$H(X \mid C) \le h_2(\delta) + \delta \log_2(\bar{K}_\epsilon^m - 1),$$

and hence $I(X;C) \ge (1-\delta) m \log_2 \bar{K}_\epsilon - h_2(\delta)$. For either prefix-free code E, source coding gives $\mathbb{E}|E(C)| \ge H(C) \ge I(X;C)$. The maximum conditional expected length over x is at least this uniform average, proving each line of (31). This argument depends only on the output distribution, so random access, dynamic passes, and internal IR do not weaken it.

For (32), the public dictionary supplies $(P_1, \dots, P_m, \bar{K}_\epsilon, \bar{\Delta}_\epsilon, B)$ and aggregate mode packs the base-$\bar{K}_\epsilon$ vector into one fixed-length integer block. Its header serializes $m, \bar{K}_\epsilon$, the mode, and the block boundary. In the canonical self-contained construction, each nonzero Z_j term has an $O(\log m)$ address and an $O(\log \bar{K}_\epsilon)$ rational-π parameter under (3); there are at most m terms. □

Corollary 6 (Deterministic fixed-grid specialization)**.** *For any no-wrap (K, Δ) family of Definition 14, a deterministic compiler accurate to $\epsilon < \sin(\Delta/4)$ has, separately,*

$$B^{\text{out}}_{\text{rel}} \geq \lceil m \log_2 K \rceil , \qquad B^{\text{out}}_{\text{self}} \geq \lceil m \log_2 K \rceil .$$

In particular, the binary fixed-step family yields the linear-in-m bound used in the configured scaling panel.

Proof. Accuracy balls are disjoint by Lemma 4, so the deterministic output map is injective on K^m inputs. If ℓ_x are their lengths under either prefix-free code, Kraft's inequality gives $\sum_x 2^{-\ell_x} \leq 1$. Thus $K^m 2^{-\max_x \ell_x} \leq 1$, proving the two rounded bounds. □

Remark 12 (What the counting bound does and does not say)**.** *Equation (31) holds regardless of internal resources, but the two inequalities must not be cross-compared. The packed $\bar{K}_\epsilon$-ary block in (32) is an upper bound only for $B_{\text{rel}}(\cdot \mid \mathcal{F}_{m,\epsilon})$; it is not a self-contained target-basis circuit string. Likewise $G(C)$ is never multiplied by an implicit "bits per gate" constant. The representation-dependent question is addressed by the masked-share cut below under one fixed family-relative output code.*

C. The Masked-Share Compact-Output–Cut Tradeoff

The semantic input supplies the aggregate vector x. The flat input of Definition 16 instead supplies an arbitrary first share $u \in \mathbb{Z}_3^m$ and only later a second share v satisfying $u + v = x \pmod 3$. The cut between these rounds exposes information that is specific to the representation: a correct compiler must preserve enough about u to handle every still-valid continuation v.

Definition 19 (Masked-share cut and precommitted output)**.** *Run $\mathcal{A}$ on a masked-share stream and let c_{MS} be the arrival cut after all m first-round updates have been delivered and before any second-round update is available. Before this cut the compiler may use arbitrary RAM/DAG operations and a dynamic number of internal scans. A buffered or rereadable first-round record is included in B_{store} or B_{IR}; the only excluded prefix is one irreversibly committed to the final write-once output. Let*

$$\begin{aligned} B^{\text{rel}}_{\text{pre}}(\mathcal{A}, m \mid \mathcal{F}_m) &:= \max_{u \in \mathbb{Z}_3^m} \ell_{\text{pre}}(u), \\ B^{\text{MS}}_{\text{cut}}(\mathcal{A}, m) &:= \max_{u \in \mathbb{Z}_3^m} B_{\text{cut}}(c_{\text{MS}}(u)). \end{aligned} \tag{33}$$

Here $\ell_{\text{pre}}(u)$ is the literal bit length already committed under the prefix-monotone stream E_{rel}; it is zero if final output has not started. Thus $B^{\text{rel}}_{\text{pre}} \leq B^{\text{out}}_{\text{rel}}$. For randomized $\mathcal{A}$, barred quantities replace the maxima in (33) by $\max_u \mathbb{E}_\rho[\ell_{\text{pre}}(u, \rho)]$ and $\max_u \mathbb{E}_\rho[B_{\text{cut}}(c_{\text{MS}}(u, \rho))]$.

Theorem 2 (Representation-dependent masked-share tradeoff)**.** *Let the sharing modulus be $K_{\text{share}} = 3$, let $\alpha = 2\pi/3$, and $\epsilon_0 = \sin(\pi/6) = 1/2$. Let $\mathcal{A}$ receive the stream once: the source delivers it in a single forward pass, while internal rescans of any retained data are allowed and charged in B_{cut}. Suppose first that $\mathcal{A}$ is deterministic and correct for* every *valid two-round decomposition in Definition 16:*

$$\begin{gathered} d_{\text{proj}}(U(C(u, v)), U_{u+v \bmod 3}) \leq \epsilon < \epsilon_0, \\ u + v \bmod 3 \in \{0, 1\}^m. \end{gathered}$$

Then, under the single family-relative output code of Definition 10,

$$B^{\text{rel}}_{\text{pre}}(\mathcal{A}, m \mid \mathcal{F}_m) + B^{\text{MS}}_{\text{cut}}(\mathcal{A}, m) \geq \lceil m \log_2 3 \rceil . \tag{34}$$

Moreover, if the final output is family-relative compact,

$$B^{\text{out}}_{\text{rel}}(\mathcal{A}, m \mid \mathcal{F}_m) \leq m + c \log_2(m + 2),$$

then

$$B^{\text{MS}}_{\text{cut}}(\mathcal{A}, m) \geq \lceil m \log_2 3 \rceil - c \log_2(m + 2). \tag{35}$$

In contrast, a one-pass compiler receiving the semantic aggregate vector x streams the same aggregate-mode code with $B^{\text{out}}_{\text{rel}} = m + O(\log m)$ and $B_{\text{cut}} = O(\log m)$.

More generally, let $0 \leq \delta < 1/2$ and let $\mathcal{A}$ be randomized, receive the stream once, and satisfy the displayed accuracy condition on every valid (u, v) with probability at least $1 - \delta$. It may use arbitrary random access and any number of internal rescans of retained data. Then

$$\overline{B}^{\text{rel}}_{\text{pre}} + \overline{B}^{\text{MS}}_{\text{cut}} \geq (1 - \delta)m - h_2(\delta). \tag{36}$$

In particular, for a deferred aggregate-mode compiler with $\overline{B}^{\text{rel}}_{\text{pre}} = O(\log m)$ and fixed $\delta < 1/2$, the flat representation still requires $\overline{B}^{\text{MS}}_{\text{cut}} = \Omega(m)$. A compiler to which the source re-delivers the stream is covered instead by Theorem 5 with $Q = 3$ and $K = 2$, which bounds $\overline{A}_p + \overline{\Sigma}^{\rightarrow}_p$; a single cut does not suffice there, because a later pass can reread the first round.

Proof. Let Alice hold an arbitrary first share $u \in \mathbb{Z}_3^m$. She simulates $\mathcal{A}$ through c_{MS} and sends the already committed family-relative output prefix together with the complete five-field restart serialization of (4). All head positions, lengths, addresses, parameters, stores, windows, and materialized IR prefixes needed to resume are present by definition, including any rereadable old input. Bob, given a second share v, resumes the computation and runs its dynamically chosen remaining operations locally.

The message must distinguish every pair $u \neq u'$. To see this, choose a single continuation v coordinatewise. If $u_j = u'_j$, set $v_j = -u_j$. If $u'_j = u_j + 1 \pmod 3$, again set $v_j = -u_j$; the two aggregates are then 0 and 1. In the remaining case $u_j = u'_j + 1 \pmod 3$, set $v_j = -u'_j$; the aggregates are 1 and 0. Thus both (u, v) and (u', v) are valid, binary aggregate instances, and their aggregate vectors differ whenever $u \neq u'$.

If Alice's two messages were identical, determinism would make Bob emit the same final circuit on these two valid completions. That circuit cannot be within ϵ of both targets, because Lemma 4 gives target distance at least $2\epsilon_0 > 2\epsilon$. Hence the message map is injective on the 3^m choices of u. Serialize the restart snapshot first; its control field gives the length of the committed prefix that follows, so the complete messages are prefix-free. Kraft's inequality therefore gives maximum message length at least $\lceil m \log_2 3 \rceil$. The sum of the two separate worst-case field lengths is at least that maximum, which proves (34). For (35), fix u. Each coordinate admits two continuations v_j with $u_j + v_j \in \{0, 1\}$, so the 2^m continuations reach every binary aggregate, and correct final outputs for distinct aggregates are distinct. These 2^m codewords of the prefix-free output code all extend the committed prefix, of length $\ell_{\mathrm{pre}}(u)$, and have length at most $m + c \log_2(m+2)$, so Kraft's inequality applied to their suffixes gives $\ell_{\mathrm{pre}}(u) \leq c \log_2(m+2)$. Substituting in (34) gives (35).

For the semantic upper bound, the compiler writes the aggregate-mode header, copies the arriving bits x_j into the aggregate field (base two, so the blocks are exact), and retains only a self-delimiting index and finite control. It stores no coefficient table or intermediate graph. The sparse parameter payload is fixed and the live address/index costs $O(\log m)$, so every cut has $B_{\mathrm{cut}} = O(\log m)$ under the same serializer.

For the randomized extension, draw U uniformly from $\mathbb{Z}_3^m$ and X uniformly from $\{0,1\}^m$, independently, and give Bob $V = X - U \pmod 3$. Then X and V are independent and $H(X \mid V) = m$. Alice's message M consists of the committed prefix and the full restart snapshot after processing U. The snapshot includes the seed/tape and its cursor, RAM, caches, old-input buffers, and IR, so Bob can resume the exact randomized run upon receiving V. The binary decoder of Lemma 9 recovers X with error at most δ. Conditional Fano therefore gives

$$\begin{aligned} H(X \mid M, V) &\leq h_2(\delta) + \delta m, \\ I(X; M \mid V) &\geq (1-\delta)m - h_2(\delta). \end{aligned}$$

The two message fields are self-delimiting, so source coding lower-bounds their expected total length by this conditional information, yielding (36). This is a conditional-Fano (equivalently, distributional/Yao) argument; it allows internal rescans of retained data and does not assume sequential internal memory, but it uses that Bob's continuation reads only V, which is why the stream must be delivered once. □

Proposition 8 (Memory-capped hybrid compilers)**.** *For every $\beta \in [0,1]$, let $h = \lfloor \beta m \rfloor$. There is a deterministic zero-error compiler for the masked-share stream, using the ordered-share mode of E_{rel}, such that at the inter-round cut*

$$\begin{aligned} B_{\mathrm{cut}}^{\mathrm{MS}} &\leq h \log_2 3 + O(\log m), \\ B_{\mathrm{pre}}^{\mathrm{rel}} &\leq (m-h) \log_2 3 + O(\log m). \end{aligned} \tag{37}$$

Thus (34) *is matched to leading order for every memory fraction β under this single family-relative encoding. This statement does not assert a matching curve for B_{self} or G.*

Proof. The ordered-share header records h. During round one the compiler packs $u_1, \ldots, u_h$ into a base-three integer held in $\mathsf{ser}_{\mathrm{store}}$ (Definition 9) and commits $u_{h+1}, \ldots, u_m$ in their public generator order, packed in base three and flushed by the end of round one, before the cut. It needs only the packed block, an index, and framing state at the cut. During round two it emits the aggregates $u_j + v_j \pmod 3$ for $j \leq h$ and the separate residues v_j for $j > h$. The latter combine with the already emitted first shares; all rotations commute, and Proposition 2 gives the target up to global phase. Fixed-order packing gives the two bounds in (37); no qubit address is hidden, because the ordered-share mode declares its public schedule and split in its header. □

Corollary 7 (Compactness versus flat-share recovery)**.** *On the masked-share representation, any deterministic compiler that receives the stream once and has $B_{\mathrm{cut}}^{\mathrm{MS}} = o(m)$ must have*

$$B_{\mathrm{rel}}^{\mathrm{out}}(\mathcal{A}, m \mid \mathcal{F}_m) \geq m \log_2 3 - o(m),$$

whereas family-relative optimal output $m + O(\log m)$ requires the cut budget $\lceil m \log_2 3 \rceil - O(\log m)$ of (35)*. This is a representation tradeoff: the semantic aggregate stream attains both compact output and $O(\log m)$ cut budget.*

Proof. Use $B_{\mathrm{pre}}^{\mathrm{rel}} \leq B_{\mathrm{rel}}^{\mathrm{out}}$ in (34) for the first claim and (35) for the second, then compare with the semantic construction in Theorem 2. □

Remark 13 (Three points under one encoding)**.** *All points in* (34) *use $E_{\mathrm{rel}}(\cdot \mid \mathcal{F}_m)$. The semantic pass-through has compact relative output and logarithmic cut budget. A flat compiler that materializes the first round in a table or intermediate graph pays its full serialized structure and parameters in B_{cut}. A one-pass flat compiler may instead commit update residues before seeing their mates, but those records enlarge $B_{\mathrm{pre}}^{\mathrm{rel}}$ and hence the final relative code. The hybrids of Proposition 8 interpolate between these endpoints and attain the deterministic lower envelope to leading order for every β. No claim about B_{self} or $G(C)$ is inferred from this curve.*

Remark 14 (Relation to the exact-semantics results and model scope)**.** *The exact-semantics corollary (Corollary 1) bounds final output length under exactness and*

noncollision, via pumpability. Theorem 1 bounds two explicitly named output-description codes via counting. Theorem 2 is different: it uses the two-round completion adversary and the complete bit-level restart snapshot B_{cut}. A global ZX graph, phase-polynomial table, spilled store, large-precision parameter, addressed window, RAM/DAG, cache, hash table, internal rescan schedule, or random seed is therefore an accounted recovery channel, not an exception. Both parts of the theorem concern compilers that receive the stream once; internal rescans of retained data are allowed and charged. For passes re-delivered by the source, the transcript bound of Theorem 5 applies instead. The balanced experimental schedule is not claimed to instantiate this lower bound.

VI. UNIFIED K-ARY RANDOMIZED MULTIPASS FRONTIER

The preceding results isolate output information and the binary two-round cut. The following package aligns the packing modulus with a genuinely dispersed additive-update representation and makes the number of passes and the full serialized state cap explicit. It also separates the proved constant-factor upper envelope from the remaining model-dependent gaps.

A. Exact cyclic K-ary packing

The no-wrap construction of Definition 14 is convenient for output counting, but modular additive shares require a grid whose period is also the sharing modulus. The following formulation records exactly when wraparound creates a collision.

Definition 20 (Wrap margin)**.** *For an alphabet $[K]_0$, step $\Delta > 0$, and $K \geq 2$, define*

$$\mathrm{dist}_{2\pi}(a,0) := \min_{z\in\mathbb{Z}} |a - 2\pi z|, \tag{38}$$

$$\gamma(K,\Delta) := \min_{1\leq d\leq K-1} \mathrm{dist}_{2\pi}(d\Delta, 0). \tag{39}$$

Let $P_1,\dots,P_m$ be independent Pauli-Z characters as in Definition 14, and set

$$U_x^{(K,\Delta)} := \exp\left[-\frac{i}{2}\sum_{j=1}^{m} x_j \Delta P_j\right], \qquad x \in [K]_0^m. \tag{40}$$

Theorem 3 (Exact worst-pair separation)**.** *For every $m \geq 1$,*

$$\min_{x\neq y} d_{\mathrm{proj}}\big(U_x^{(K,\Delta)}, U_y^{(K,\Delta)}\big) = 2\sin\left(\frac{\gamma(K,\Delta)}{4}\right). \tag{41}$$

Consequently the family is projectively injective exactly when $\gamma(K,\Delta) > 0$. If $(K-1)\Delta \leq \pi$, then $\gamma(K,\Delta) = \Delta$. If $Q \geq K$ and $\Delta = 2\pi/Q$, then again $\gamma(K,\Delta) = \Delta$, even though the coefficient interval $[0,(K-1)\Delta]$ may cross the principal-argument cut at π; in particular the full residue alphabet $K = Q$ is allowed.

Proof. Fix $x \neq y$, put $s = x - y$, and choose j with $s_j \neq 0$. Independence of the characters makes their joint sign map surjective. Hence the relative unitary $U_x^{(K,\Delta)}(U_y^{(K,\Delta)})^\dagger$ has two eigenvalues obtained from sign patterns that agree off j and whose circular angular separation is $\mathrm{dist}_{2\pi}(|s_j|\Delta, 0) \geq \gamma(K,\Delta)$. After any common phase rotation, one of two circle points separated by $\eta \in [0,\pi]$ is at chordal distance at least $2\sin(\eta/4)$ from 1. This proves the lower bound in (41) after taking the global-phase infimum.

Choose $d \in \{1,\dots,K-1\}$ attaining $\gamma(K,\Delta)$ and take a pair that differs by d in one coordinate only. Its relative spectrum consists of two points with circular separation $\gamma(K,\Delta)$, so centering the shorter arc attains $2\sin(\gamma/4)$. This proves equality. Under the no-wrap hypothesis, every $d\Delta$ lies in $[\Delta,\pi]$, whose minimum is Δ. For $\Delta = 2\pi/Q$ and $1 \leq d \leq K-1 \leq Q-1$, the circular distance of $d\Delta$ from 0 is $(2\pi/Q)\min\{d, Q-d\} \geq 2\pi/Q$, with equality at $d = 1$. □

Corollary 8 (ϵ-tuned cyclic family)**.** *For $0 < \epsilon \leq 1/8$, let*

$$Q_\epsilon := \left\lfloor \frac{\pi}{2\arcsin(2\epsilon)} \right\rfloor, \qquad K_\epsilon := Q_\epsilon, \qquad \Delta_\epsilon := \frac{2\pi}{Q_\epsilon}. \tag{42}$$

Then

$$\frac{1}{4\epsilon} \leq K_\epsilon \leq \frac{\pi}{4\epsilon}, \quad \min_{x\neq y} d_{\mathrm{proj}}(U_x,U_y) = 2\sin\left(\frac{\pi}{2Q_\epsilon}\right) \geq 4\epsilon. \tag{43}$$

Thus, with $\mathcal{G}_{m,\epsilon} := \{U_x : x \in \mathbb{Z}_{Q_\epsilon}^m\}$,

$$\log_2|\mathcal{G}_{m,\epsilon}| = m\log_2 K_\epsilon = \Theta\left(m\log\frac{1}{\epsilon}\right). \tag{44}$$

The aggregate alphabet is the full residue group, so the coefficient interval $[0,(Q_\epsilon-1)2\pi/Q_\epsilon]$ wraps around the circle; Eq. (41), rather than a principal-interval argument, controls the wraparound. We keep the name K_ϵ for the alphabet size because the no-wrap codebook of Proposition 3 uses a different one, $\bar{K}_\epsilon$.

Proof. At $\epsilon = 1/8$ one has $Q_\epsilon = 6$, so $Q_\epsilon \geq 6$ throughout the range. The bound $\arcsin(2\epsilon) \leq \pi\epsilon$ and $\lfloor a\rfloor \geq a - 1$ give $K_\epsilon \geq 1/(2\epsilon) - 1 \geq 1/(4\epsilon)$; the reverse bound uses $\arcsin(2\epsilon) \geq 2\epsilon$. Since $Q_\epsilon \leq \pi/[2\arcsin(2\epsilon)]$, $\pi/(2Q_\epsilon) \geq \arcsin(2\epsilon)$. Apply Theorem 3 with $K = Q_\epsilon$. The cardinality statement is immediate. □

B. Dispersed additive-update representation

Definition 21 (r-round dispersed update stream)**.** *The cyclic specialization of Definition 13 with modulus $Q \geq 3$, aggregate alphabet $[K]_0$ with $2 \leq K \leq Q$ (the tuned*

families use the full alphabet $K = Q$), $\Delta = 2\pi/Q$, $r \geq 2$ rounds, and $\Sigma = \{+1\}$: for $x \in [K]_0^m$ the shares $a^{(1)}, \ldots, a^{(r)} \in \mathbb{Z}_Q^m$ satisfy

$$\sum_{t=1}^{r} a_j^{(t)} = x_j \pmod{Q} \qquad (j \in [m]), \tag{45}$$

the flat input is the round-major record sequence $\mathsf{Update}(t, j, a_j^{(t)})$, and the semantic input is the addressed aggregate vector x.

Proposition 9 (Semantic equivalence and token privacy)**.** *Every valid flat stream of Definition 21 implements the same U_x as the semantic input, up to global phase. Moreover, a single flat token never determines x_j: for every fixed (t, j, a) and every $b \in [K]_0$, there is a valid stream with $a_j^{(t)} = a$ and $x_j = b$. Under the canonical sharing distribution in which the first $r-1$ shares are independent uniform residues and the last is fixed by* (45)*, every individual residue is uniform on $\mathbb{Z}_Q$ and independent of x_j.*

Proof. For integer representatives there are $k_j \in \mathbb{Z}$ with $\sum_t a_j^{(t)} = x_j + Qk_j$. Commutativity and $Q\Delta = 2\pi$ give

$$\prod_t R_{P_j}(\Delta a_j^{(t)}) = R_{P_j}(\Delta x_j + 2\pi k_j) = (-1)^{k_j} R_{P_j}(\Delta x_j).$$

The product of the signs is global. For privacy, fix the named token and choose any one of the other $r-1$ residues to enforce the desired sum; the remaining residues are arbitrary. The distributional claim follows because a sum containing at least one independent uniform residue is uniform, including the last share after conditioning on x_j. □

Definition 22 (Forward-pass cut accounting)**.** *Partition the round-major stream into Alice's prefix A (rounds $1, \ldots, r-1$) and Bob's suffix B (round r). A forward p-pass compiler scans AB from left to right at most $p = p(m)$ times, so a run makes at most p forward crossings $A \to B$ and at most $p-1$ rewind crossings $B \to A$; crossings that do not occur contribute zero below. Each new pass is delivered by the input source from the start of the stream, as in the standard multipass streaming model; that re-delivery is accounted for by the pass count and the rewind crossing, not as stored state. At every crossing, the complete restart state is the five-field serialization of* (4)*; any copy of the input that the compiler itself retains, and every live IR object, are therefore included. For a run on I, let $A_p(I)$ be the number of bits irreversibly committed to the final output while the machine scans A, summed over all passes, and define*

$$\begin{aligned} \Sigma_p^{\rightarrow}(I) &:= \sum_{t=1}^{p} B_{\mathrm{cut}}(c_{A\to B,t}), \\ \Sigma_p(I) &:= \Sigma_p^{\rightarrow}(I) + \sum_{t=1}^{p-1} B_{\mathrm{cut}}(c_{B\to A,t}). \end{aligned} \tag{46}$$

Both are literal transcript lengths; the five-field serialization is self-delimiting, and its control field records the emitted-prefix length. Unbarred resource symbols take maxima over valid inputs. For randomized compilers, write

$$\overline{A}_p := \max_I \mathbb{E}_\rho A_p(I, \rho), \qquad \overline{\Sigma}_p^{(\rightarrow)} := \max_I \mathbb{E}_\rho \Sigma_p^{(\rightarrow)}(I, \rho),$$

and let $S := \max_I \mathbb{E}_\rho[\max_c B_{\mathrm{cut}}(c; I, \rho)]$ over the $2p-1$ named crossings. Then $\overline{\Sigma}_p^{\rightarrow} \leq pS$ and $\overline{\Sigma}_p \leq (2p-1)S$; the deterministic version is identical without expectations.

Theorem 4 (One-pass zero-error mask bound)**.** *Let a deterministic one-pass compiler be correct to $\epsilon < \sin(\pi/(2Q))$ on every valid stream of Definition 21 with $K \in \{Q-1, Q\}$. Then*

$$A_1 + B_{\mathrm{cut}}(c_{A\to B,1}) \geq \lceil m \log_2 Q \rceil . \tag{47}$$

Proof. For each $u \in \mathbb{Z}_Q^m$, choose a canonical prefix whose aggregate is u (put u in its first round and zero in the other prefix rounds). The committed prefix and restart state after this prefix must distinguish all Q^m choices. Indeed, for $u \neq u'$, choose one common final share v coordinatewise. For $K = Q$ take $v = 0$. For $K = Q-1$, at each coordinate at most two residues of v_j would make $u_j + v_j$ or $u'_j + v_j$ equal to the single excluded aggregate $Q-1$; since $Q \geq 3$, an allowed v_j exists. Both completions are valid, and their aggregates differ in every coordinate in which u and u' differ. By Theorem 3, one circuit cannot be within ϵ of both targets, while a deterministic one-pass suffix computation is a function of the snapshot and the suffix. A run that halted inside the prefix would give the same output for two completions with different targets, so every run reaches the crossing. Hence the map $u \mapsto \mathsf{ser}(\mathsf{s}_u) \cdot O_A(u)$, the five-field snapshot followed by the committed prefix, is injective. The snapshot serialization is self-delimiting and its control field records the emitted-prefix length, so these messages form a prefix-free set, and Kraft's inequality gives (47). □

C. Randomized multipass frontier

Put

$$L_\delta(m, K) := (1-\delta) m \log_2 K - h_2(\delta), \qquad 0 \leq \delta < \tfrac{1}{2}. \tag{48}$$

Theorem 5 (K-ary randomized multipass recoverability frontier)**.** *Let $Q \geq 3$, $2 \leq K \leq Q$, and $0 \leq \epsilon < \sin(\pi/(2Q))$. Let a possibly randomized compiler be correct within projective error ϵ, with probability at least $1-\delta$ for every $x \in [K]_0^m$ and every valid r-round dispersed representation of x. Assume that it makes at most $p = p(m)$ forward passes in the charged model of Definition 22. Then:*

$$\overline{B}_{\mathrm{rel}}^{\mathrm{out}} \geq L_\delta(m, K), \qquad \overline{B}_{\mathrm{self}}^{\mathrm{out}} \geq L_\delta(m, K), \tag{49}$$

$$\overline{A}_p + \overline{\Sigma}_p^{\rightarrow} \geq L_\delta(m, K) - \lceil \log_2(p+1) \rceil. \tag{50}$$

In particular, for the worst-input expected crossing cap S of Definition 22,

$$\boxed{\begin{aligned}&\overline{A}_p + pS\\ &\quad \geq (1-\delta)m\log_2 K - h_2(\delta) - \lceil \log_2(p+1)\rceil,\end{aligned}} \tag{51}$$

and a fortiori $\overline{A}_p + (2p-1)S$ *obeys the same bound. Here S includes RAM, parameters, cache/hash metadata, materialized input, and all IR fields from* (4)*; it is not a RAM-only bound. For the tuned choice of Corollary 8, both right-hand sides are $\Omega((1-\delta)m\log(1/\epsilon))$ for fixed $\delta < 1/2$ and $p = 2^{o(m)}$.*

Proof. For (49), take X uniform on $[K]_0^m$ and fix one valid dispersed representation for each value of X. The packing separation is greater than 2ϵ, so nearest-packing decoding recovers X from a successful output. Fano's inequality gives

$$I(X;C) \geq m\log_2 K - h_2(\delta) - \delta\log_2(K^m - 1) \geq L_\delta(m,K).$$

Expected length in either separately fixed prefix-free output code is at least this mutual information; a worst-input expectation is at least the uniform average.

For (50), independently choose a uniform prefix aggregate $U \in \mathbb{Z}_Q^m$, put it in the first prefix round, put zero in any other prefix rounds, and set the last share $V = X - U \pmod Q$. Then V is uniform and independent of X, so $H(X \mid V) = m\log_2 K$. Run the compiler and record the forward transcript

$$T := \big(c,\ \mathsf{s}_1, O_1,\ \ldots,\ \mathsf{s}_c, O_c,\ O_{\text{fin}}\big),$$

where $c \in \{0,\ldots,p\}$ is the number of forward crossings, written in $\lceil \log_2(p+1)\rceil$ bits; s_t is the five-field snapshot at the t-th crossing $A \to B$; O_t is the output written while scanning A just before that crossing; and O_{fin} is any output written in A after the last crossing (after a final rewind, or in a run that never reaches B). Given T and V, Bob reproduces the whole run. From s_t and V he simulates the scan of B, including the random tape and dynamic control that s_t serializes, and thereby obtains the Bob-side output and the rewind snapshot; the rewind snapshot is therefore a function of (s_t, V) and need not be transmitted. The control field of s_t records the emitted-prefix length, which gives $|O_t|$ once s_t is read, so s_t is placed before O_t. After the c-th crossing, O_{fin} completes a codeword of the prefix-free output code and is therefore self-delimiting. Hence T is a prefix-free code for each fixed V. It determines the final output, the output decoder recovers X with error at most δ, and conditional Fano yields

$$I(X;T \mid V) \geq L_\delta(m,K).$$

The expected length of a prefix-free code is at least this conditional information, and that length is $\lceil \log_2(p+1)\rceil + A_p + \Sigma_p^{\rightarrow}$. Maximizing over inputs proves (50). Since each of the at most p forward crossings has expected maximum budget at most S, the boxed form follows. The tuned consequence follows from Corollary 8. □

Remark 15 (What is and is not optimal)**.** *Equation* (49) *is an output-information bound and* (51) *is a transcript bound; neither converts self-contained circuit bits into gates. A forward scan alternates $2p-1$ times between the earlier rounds and the last one, as in the standard simulation of p passes by a $(2p-1)$-message protocol, but only the p messages into the last round carry information about the earlier rounds: each rewind snapshot is a function of the preceding forward snapshot and of the last round. This is why the bound charges p crossings and not $2p-1$. A random-access input oracle, uncharged rereading outside the counted forward passes, or reverse scans would be a different model; re-delivery of the whole stream at the start of each counted pass is part of this one (Definition 22). For $p = 1$ and a deterministic zero-error compiler, Theorem 4 gives the integer form $A_1 + B_{\text{cut}} \geq \lceil m\log_2 Q\rceil$ for $K \in \{Q-1, Q\}$, which for $K = Q-1$ is slightly stronger than $m\log_2 K$. For the full alphabet $K = Q$ the constructions of Section VI D attain* (50) *and* (51) *at $\delta = 0$ to leading order.*

D. Memory-capped constructions

Proposition 10 (Block hybrid)**.** *Fix $p, h \geq 1$ and let $q = \min\{m, ph\}$. There is a deterministic zero-error p-pass compiler for Definition 21 with $K = Q$ that aggregates q coordinates and passes the other update records through in the public p-block hybrid mode of E_{rel}. It obeys*

$$S \leq \left\lceil \frac{q}{p} \right\rceil \log_2 Q + O(\log(mQrp)), \tag{52}$$

$$A_p \leq (r-1)(m-q)\log_2 Q + O(r\log(mQrp)), \tag{53}$$

$$\begin{aligned}B_{\text{rel}}^{\text{out}} &\leq q\log_2 Q + r(m-q)\log_2 Q\\ &\quad + O((p+r)\log(mQrp)).\end{aligned} \tag{54}$$

In the compact-output corner $h = \lceil m/p\rceil$, this becomes

$$B_{\text{rel}}^{\text{out}} \leq m\log_2 Q + O(p\log(mQrp)), \tag{55}$$

$$S \leq \left\lceil \frac{m}{p} \right\rceil \log_2 Q + O(\log(mQrp)). \tag{56}$$

In the compact-output corner $q = m$ for any number of rounds, and for every q when $r = 2$, the construction satisfies $A_p + pS \leq m\log_2 Q + O((p+r)\log(mQrp))$, and the same bound holds for $A_p + \Sigma_p^{\rightarrow}$. Compared with (50) *and* (51) *at $\delta = 0$, whose right-hand side is $m\log_2 Q - \lceil \log_2(p+1)\rceil$, it is therefore optimal to leading order whenever $p\log(mQrp) = o(m\log Q)$; the additive terms are not claimed to be optimal, and for p comparable to m they dominate.*

Proof. Partition the first q coordinates into at most p consecutive blocks of size at most h. Each pass is re-delivered by the input source (Definition 22), so the compiler keeps no copy of the stream between passes. On pass t, keep only the residues of block t and add their first $r-1$ rounds modulo Q; when the final-round mates arrive, emit the

corresponding aggregate base-Q block. The partial sums are held as one packed integer in $\mathsf{ser}_{\mathrm{store}}$ at its literal bit length (Definition 9); the live packed block, indices, and framing give (52) at the crossings. Each block is packed separately, which costs at most one bit per block; together with the per-block framing this is the $O(p\log(mQrp))$ term in the output bound. On the first pass, emit every share of each unselected coordinate in the declared round-major order, packing the residues of each round in base Q and flushing them at the end of that round, so that the residues of round $r-1$ are written before the crossing; these rotations already have the correct product. The selected coordinates can be balanced across the p blocks, so the maximum block has size $\lceil q/p\rceil$. Their consecutive boundaries are a public function of (m,p,q), so the header stores those integers rather than p separate delimiters. The prefix contains $r-1$ residues and the whole output contains r residues for each such coordinate, giving (53) and (54). If $ph \ge m$ there are no passthrough coordinates, A_p is the header, and $p\lceil m/p\rceil \log_2 Q \le m\log_2 Q + p\log_2 Q$ gives the stated bound on $A_p + pS$ for any r. For $r=2$ and general q, $(m-q)\log_2 Q + p\lceil q/p\rceil \log_2 Q$ is at most $m\log_2 Q + p\log_2 Q$, which gives the same bound; since every forward snapshot is bounded by the same block, the bound also holds for $A_p + \Sigma_p^{\rightarrow}$. □

Proposition 11 (Semantic and randomized upper corners)**.** *Under the same public dictionary, fix a block length $b \ge 1$. A one-pass semantic compiler receiving x emits the aggregate-mode code with b-digit blocks and satisfies*

$$\begin{aligned} B_{\mathrm{rel}}^{\mathrm{out}} &\le m\log_2 K + \lceil m/b\rceil + O(\log(mQ)),\\ S &\le b\log_2 K + O(\log(mQ)), \end{aligned} \tag{57}$$

and implements the same U_x as the flat stream. With $b = \lceil \log_2(m+1)\rceil$ the state is $O(\log m\cdot\log Q)$ and the output is $m\log_2 K\,(1+o(1))$. Blocking is needed: when $K = 2^e K_{\mathrm{odd}}$ with $K_{\mathrm{odd}} > 1$, a write-once streaming writer of the single integer $\sum_j x_j K^{j-1}$ in binary, least significant bit first, reading $x_1, x_2, \dots$ in order, must hold at least $\lfloor m/2\rfloor \log_2 K_{\mathrm{odd}} - 1$ bits after half the digits. For any $0 \le \delta < 1/2$, an input-independent abort coin that runs this exact compiler with probability $1-\delta$ and otherwise emits a fixed short circuit has worst-input success at least $1-\delta$ and expected relative output

$$(1-\delta)\big(m\log_2 K + \lceil m/b\rceil\big) + O(\log(mQ)+1). \tag{58}$$

Thus the randomized family-relative output lower bound is matched to leading order, up to framing and the binary-entropy additive term. A self-contained R_Z-circuit upper bound is $O(\sum_j |\mathrm{supp}\,P_j|\log n + m\log Q)$ bits, which is $O(m(\log m + \log Q))$ for constant-weight characters on $n = \mathrm{poly}(m)$ qubits, with at most m rotations; no matching self-contained-bit or gate-count claim is made.

Proof. Buffer the incoming aggregates b at a time, held as one packed integer, and write each full buffer as one $\lceil b\log_2 K\rceil$-bit integer and a final partial block of $b' < b$ digits in $\lceil b'\log_2 K\rceil$ bits; the buffer, an index, and framing are the only live state. Each block loses less than one bit to rounding, which gives the $\lceil m/b\rceil$ term. For the single packed integer, let $P = \sum_{i\le j} x_i K^{i-1}$ be the value of the first j digits and $N = P + K^j R$ the final integer. Since $K^j R \equiv 0 \pmod{2^{ej}}$ and the higher bits of N depend on R, a correct write-once writer has written at most the low ej bits of P, so at most 2^{ej+1} distinct written prefixes occur. Two digit prefixes with the same snapshot and the same written prefix produce the same N for every R, hence $P = P'$. The K^j values of P therefore need at least $K^j/2^{ej+1} = K_{\mathrm{odd}}^j/2$ distinct snapshots, which, being self-delimiting (Definition 22), need at least $j\log_2 K_{\mathrm{odd}} - 1$ bits by Kraft's inequality; take $j = \lfloor m/2\rfloor$. Proposition 9 gives semantic equality with the flat representation. Prefix the output with the abort-coin branch tag for the randomized construction and take expectations. The self-contained construction writes each nonzero $R_{P_j}(2\pi x_j/Q)$ with its address and rational parameter. □

VII. REPRESENTATION-AWARE COMPILER

A. Bounded Supported-Pattern Recognition

Definition 23 (Supported-pattern region recognizer)**.** *The implemented recognizer accepts only configured high-level forms: (i) explicit Pauli rotations or phase-gadget records that already carry their support, (ii) local phase-ladder/Fourier spans, (iii) expanded mirrored or self-inverse composite blocks, and (iv) repeated-run or preset semantic nodes. Within one supported contiguous span it maintains the declared generators $P_1,\dots,P_k$, coefficients $\theta_1,\dots,\theta_k$, qubit support, and source records. It may extend the span while the next record belongs to the supported grammar and passes the configured commutation check; an unsupported or certified noncommuting record closes the candidate. Rejected recognition leaves the original circuit available to the conservative fallback.*

Remark 16 (Recognition, maximality, soundness, and complexity scope)**.** *The artifact does not implement a general binary-symplectic extractor for arbitrary circuit DAGs and does not claim a maximal commuting region in that sense. Its only maximality notion is the maximal contiguous span accepted by the fixed supported-pattern grammar under the chosen scan order. Records for distinct generators remain separate until aggregation. Soundness of a selected rewrite comes from the independent exact or symbolic equivalence certificate of Definition 25; failure returns the unchanged candidate. No general worst-case linear-time claim is made: the runtime is the sum of the bounded scans, signature/hash bookkeeping, certificate work, and any configured backend probes, as detailed in Appendix B.*

B. Aggregation and Certificate

Definition 24 (Aggregation of an accepted supported region)**.** *Given an accepted supported region with generators $P_1, \dots, P_k$ and coefficients $\theta_1, \dots, \theta_k$, the aggregation emits a representative circuit implementing $\exp(-i\sum_j \theta_j P_j/2)$ using at most $c_0 + c_1 k$ gates in the target basis.*

Let

$$\mathcal{C}_n = \langle H, S, S^\dagger, X, Y, Z, \mathrm{CX}, \mathrm{CZ}, \mathrm{SWAP}\rangle$$

be the Clifford gate language on addressed qubits. Fix a declared finite formal-symbol set $\mathcal{S}$ and the exact coefficient module

$$\Lambda = \mathbb{Q}\pi \oplus \bigoplus_{s\in\mathcal{S}} \mathbb{Q}\theta_s.$$

For a signed Hermitian n-qubit Pauli P and $\lambda \in \Lambda$, write $R_P(\lambda) = \exp(-i\lambda P/2)$.

Definition 25 (Exact commuting-Pauli/Clifford certificate domain)**.** *The domain $\mathcal{D}_{\mathrm{CPC}}$ consists of finite addressed circuits over $\mathcal{C}_n$, exact records $R_Z(\lambda)$ and $R_P(\lambda)$, identity and barrier records, and an arbitrary exact global phase. Every coefficient is a reduced rational π component plus a sorted sparse list of reduced rational formal-symbol components carried by the input; a floating-only parameter is not in the exact domain.*

Scan a circuit in execution order while maintaining a Clifford frame F and the invariant

$$U_{\mathrm{prefix}} = F R_{P_t}(\lambda_t)\cdots R_{P_1}(\lambda_1) \quad \text{up to global phase.} \tag{59}$$

The frame is represented by the complete signed inverse-conjugation tableau $\{F^\dagger X_j F, F^\dagger Z_j F\}_{j=1}^n$. Appending a physical rotation $R_Q(\lambda)$ appends $F^\dagger Q F$ to the Pauli list. The certificate predicate additionally requires all extracted P_j to commute pairwise.

Canonicalization moves a Pauli sign into its coefficient, discards an identity-axis rotation, adds coefficients on identical unsigned supports, reduces the rational π coefficient modulo 2 to the interval $[0, 2)$, keeps every formal-symbol coefficient exact, deletes zero records, and sorts the remaining records lexicographically by their binary (x, z) masks. Circuit and Clifford global phases are omitted. The resulting record is

$$\beta_j := (b_{j,s})_{s\in\mathcal{S}} \in \mathbb{Q}^{\mathcal{S}}, \qquad \bar{a}_j \in \mathbb{Q}\cap[0,2),$$
$$\mathrm{can}(C) = \big(n, T_F, ((x_j, z_j, \bar{a}_j, \beta_j))_{j=1}^{\ell}\big). \tag{60}$$

The symbolic checker accepts exactly when the two canonical byte records are equal.

Proposition 12 (Soundness of canonical symbolic acceptance)**.** *For $C, C' \in \mathcal{D}_{\mathrm{CPC}}$ satisfying the pairwise-commutation predicate,*

$$\mathrm{can}(C) = \mathrm{can}(C') \quad\Longrightarrow\quad U_C = e^{i\phi} U_{C'}$$

for some $\phi \in \mathbb{R}$.

Proof. Induction over the input records proves (59). For a Clifford record G, replace F by GF. For a rotation,

$$R_Q(\lambda) F = F R_{F^\dagger Q F}(\lambda),$$

which is exactly the tableau update used by the checker. Thus the extracted record implements the original circuit up to its explicitly ignored global phase.

Pairwise commutation permits deterministic reordering without changing the unitary. Sign normalization is exact because $R_{-P}(\lambda) = R_P(-\lambda)$, while coefficient reduction changes a factor only by

$$R_P(\lambda + 2k\pi) = (-1)^k R_P(\lambda).$$

All such signs multiply to a global phase; an identity-axis rotation is also a global phase. Equality of the sorted rotation records, including every formal-symbol component, therefore gives equal rotation products projectively.

Finally, equality of the signed tableaus means that $F'F^\dagger$ commutes with every Pauli generator. The Pauli representation generates the full matrix algebra, so $F'F^\dagger$ is a scalar. Hence the two Clifford frames also agree up to global phase, proving the claim. □

Remark 17 (Status and completeness boundary)**.** *Only status* `completed_valid`*, accompanied by the content-addressed canonical records and an independent certificate payload, is an acceptance. A canonical mismatch is* `completed_invalid`*; an unlisted gate or malformed address is* `unsupported`*; failure of pairwise commutation is* `predicate_error`*; and a floating coefficient without an exact sidecar, an unassigned formal symbol in the dense checker, or a dense distance inside the numerical guard band is* `numerical_inconclusive`*. Only* `completed_valid` *rows enter quality means. The proposition is a soundness statement, not a claim that every equivalent Clifford-plus-rotation circuit has the same record.*

For $n \le 6$, the audit independently forms dense matrices and checks the normalized Frobenius projective distance

$$d_{\mathrm{F,proj}}(U, V) = 2^{-n/2} \min_{|\omega|=1} \|U - \omega V\|_{\mathrm{F}}.$$

The frozen suite covers both accepted and rejected pairs at every width $1 \le n \le 6$ and changes, separately, an angle, a sign, one support bit, a qubit permutation, the Clifford frame, and a deleted or duplicated gate. The executable definition and status contract are `certificates/CERTIFICATE_SPEC.md`; the canonicalizer and checker are `certificates/canonicalize.py` and `certificates/symbolic_checker.py`.

Within the supported exact domain, the executable audit applies 56 adversarial mutations—angle, sign, support bit, qubit index/permutation, Clifford frame, deletion, and duplication—and observes 0 false accepts. It also performs 12 dense/symbolic cross-check cases for $n \le 6$ with no checker disagreement.

The status contract distinguishes `completed_valid`, `completed_invalid`, `unsupported`, `predicate_error`, and `numerical_inconclusive`; only independently certified `completed_valid` rows enter quality means.

C. Constructive Aggregation Proposition

Proposition 13 (Constructive aggregation size)**.** *For a supported semantic input accepted as a commuting diagonal phase layer with m independent phase terms, the semantic aggregation path emits an aggregate representative with at most $c_0 + c_1 m$ phase records before final lowering, for constants c_0, c_1 fixed by the representation and lowering conventions. Hence the constructive representative has $O(m)$ phase records, a number independent of the repetition count r at fixed width; the aggregated coefficients themselves can need $\Theta(\log r)$ bits each.*

Proof. By hypothesis the accepted layer is a product of m supported commuting diagonal terms. The constructive path keys terms by their diagonal generator and adds coefficients across repetitions. Repetition changes the coefficient on each key, not the number of keys. The aggregate contains at most one record per phase term, plus fixed boundary and lowering overhead. □

Proposition 13 shows that a semantic-first pipeline achieves an $O(m)$-record representative on the packing family. Under the public family dictionary it has $B_{\mathrm{rel}} = m + O(\log m)$, matching the family-relative inequality of Theorem 1; its separately reported self-contained size is $O(m \log m)$. On direct semantic input, the support mask can be streamed with $B_{\mathrm{cut}} = O(\log m)$, which is the constructive side of the separation in Theorem 2.

VIII. CONTROLLED VALIDATION

A. Matched-Representation Benchmark

Definition 26 (Matched-representation benchmark)**.** *For one sampled coefficient vector x, a matched cell fixes the target U_x, target basis, hardware model, total error budget, timeout, memory cap, and seed, and then evaluates every compiler on nine representations: (1) semantic aggregate, (2) contiguous flat, (3) round-robin flat, (4) random commuting order, (5) masked-share/update stream, (6) locally folded flat, and (7–9) QASM, TKET, and ZX bridge records. Each representation has a unique identifier, serialized payload, generation seed, and unitary certificate. A cell is numerical only after the common certificate policy returns `completed_valid`; unsupported predicates and malformed or inconclusive records remain status outcomes.*

The executable campaign uses the no-wrap target

$$U_x = \prod_{j=1}^{m} \exp\left(-\frac{i\pi x_j}{8K} Z_j Z_{j+1}\right), \qquad x_j \in \{0, \ldots, K-1\}.$$

The active support is sampled at the requested density and each active coefficient is uniform on $\{1, \ldots, K-1\}$. The frozen campaign contains 1,080 matched cells across the nine representations and five compilers. Of these, 1,065 are `completed_valid`; the remaining 15 are retained as `predicate_error`, all from the Qiskit masked-update path, and are not converted into numerical wins or losses. Every raw row records the generation parameters, sampled x, realized density, schedule, basis, timeout, memory cap, runtime, peak RSS, gates, depth, CX, serialized input and output/IR bytes, bridge sequence, and certificate kind.

The representation main effect, compiler main effect, their interaction, completion probability, and scaling exponents are computed from the complete matrix. The interaction is substantive: a compiler that reconstructs a phase polynomial or Pauli network before synthesis belongs on the semantic-capable side even when its input arrived through a flat bridge.

B. Semantic-Capable External Baselines and Bridge Loss

A separate 1,728-cell campaign freezes recommended-native and unified-target-basis configurations for the authors' exact phase-table reference, local Qiskit optimization, TKET PauliSimp, and PyZX full-reduce. It records tool version, available commit metadata, pass sequence, bridge, seed, basis, timeout, memory cap, pre/post-bridge gate/depth/CX and serialized IR, runtime, RSS, and certificate status. 1,695 cells are `completed_valid`; 33 are retained as `predicate_error`. TKET and PyZX are classified as semantic-capable, not as local baselines, whenever their configured global IR recovers the compact structure. Bridge loss is reported separately from compiler quality, and unsupported is never conflated with incorrect.

The stance of this campaign is explanatory, not competitive. TKET PauliSimp and PyZX full-reduce already implement the semantic-first principle on their global IRs; the contribution of this paper is the bound that says what any such tool must pay to do so on a dispersed stream (Corollary 4). A preregistered scan probed this on a third-party compiler the authors do not control: with predictions and decision rule committed before execution (`PaperDraft/research/E7_preregistration.md`), TKET PauliSimp was run at $m \in \{8, 16, 32, 64, 128\}$, three seeds per size, with the TKET pass sequence DecomposeBoxes, AutoRebase, RemoveRedundancies, PauliSimp, RemoveRedundancies, AutoRebase, on the four-round masked modular-share streams of the benchmark generator (`masked_share_update` with $r = 4$ and $K = 8$: shares modulo $8K = 64$ on the chain characters $Z_j Z_{j+1}$, every share nonzero and none equal to the target

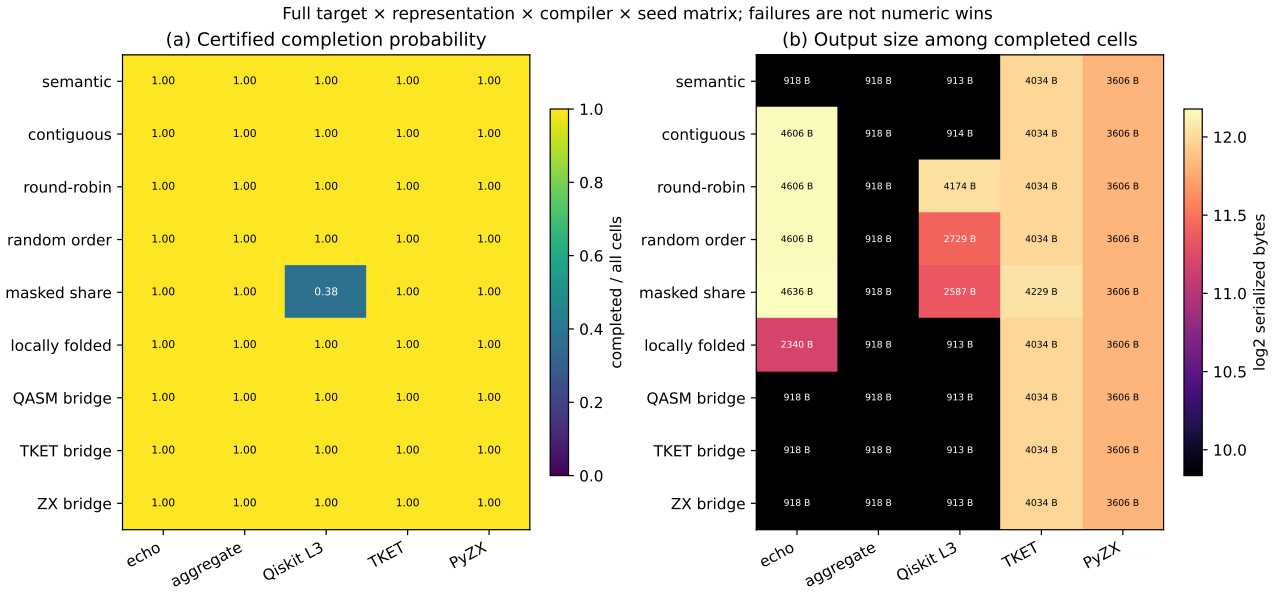


Figure 4. Complete matched-representation interaction summary. Panel (a) reports certified completion probability for every representation–compiler pair; panel (b) reports serialized output size only among completed cells. Predicate errors remain status observations and never become numerical wins.

coefficient). These are dispersed streams in the sense of Definition 21 at a fixed modulus; they are not the two-round $Q = 3$ masked-share device of Theorem 2. In every cell PauliSimp reduced the non-Clifford rotations to one per active term (508 dispersed rotations to 127 non-Clifford R_z at $m = 128$), although its output was not compact in size: at $m = 128$ it emitted 688–705 R_z gates in total and 5.3 times the input QASM text. Its process memory (mean RSS increment) grew monotonically in m, with log–log slope 1.70 over $m \in \{32, 64, 128\}$, the range selected by the preregistered noise-floor rule (1.14 over all five sizes), while the instrumented semantic-input control stayed flat (196 → 199 bytes over the same 16× range of m). The preregistered prediction was monotone growth with log–log slope in $[0.7, 1.5]$, evaluated on $m \in \{32, 64, 128\}$ when the small sizes fall below the noise floor; it was thereby rejected under its own frozen decision rule, and we report it as rejected, noting that the all-size slope would have fallen inside the interval. The frozen artifact is `data/frozen/e7_tket_state_scaling.json`. Two limits on what this scan is allowed to establish should be stated explicitly. First, the quantity measured at a third-party API boundary is a process-memory increment, which is a configured-pipeline diagnostic in the sense of Table IV: it is not promoted to a measurement of B_{cut} and is not commensurable with the S of the frontier. The scan also has no arm that runs PauliSimp on the aggregated layer, and its input text itself grows linearly with m, so it does not isolate a cost of dispersal: it describes PauliSimp's behavior on dispersed streams, while the instrumented control shows only that the authors' semantic-input compiler stays flat. It tests neither the constant nor the exponent of the frontier. Second, because the measured quantity is not B_{cut}, the scan cannot say how close this tool comes to the floor; growth faster than linear over the evaluated range is compatible with implementation structure (the pairwise commutation structure of a Pauli graph is itself superlinear in the number of terms). No claim that the authors' implementation outperforms these tools is made or implied.

Lemma 10 (Dimension-preserving instrumentation)**.** *For every file A named by an accepted version-one cut manifest (Proposition 1), its reported size is exactly* $8\,\mathrm{stat}(A)$`.st_size` *bits. Address growth is charged by the literal varint bytes; in particular* 127 *and* 128 *have different address lengths. There is no representation-independent conversion from gate count, depth,* `cx`*, token count, or IR-node count to bits.*

Proof. The first statement is Proposition 1. Gate or node count does not determine parameter numerators, denominators, qubit-address widths, edge endpoints, delimiters, or textual external payloads. Thus equal-count objects can have different artifact lengths. Conversely, family-relative dictionary codes can assign different self-contained circuits equal conditional length. Only the named artifact file supplies a bit value. □

C. Packing-Family Scaling

The complete multifactor campaign of Fig. 4 is the primary matched-representation experiment. The earlier 5-by-5 sweep below is retained as a separately keyed legacy diagnostic because it reaches the same maximum

Dimension	Recorded field	Executable serializer or definition	Permitted comparison
bits	`serialized_input_bits`	verified `SemanticStream`, `FlatUpdateStream`, or external-input artifact file	same codec/mode only
bits	`B_com_bits`	verified literal committed-output artifact/prefix file	same codec only
bits	five `B_*_bits` fields	verified control/store/window/parameter/IR files	corresponding formal field only
records/tokens	source records	exact count before serialization	never converted by a constant
IR nodes	`ir_nodes`	structural node count reported separately from `B_ir_bits`	no implicit bit or gate conversion
gates/depth/CX	structural output	Qiskit circuit operations, DAG depth, and `cx` count	like-for-like target basis

Table III. Instrumentation contract. Figures and tables compare within rows of one dimension. Cross-dimensional discussion cites the explicit codec or is stated only qualitatively.

Evidence class	Included experiments	Authorized inference
instrumented theorem-model implementation	W3 traced reference compilers and restart verifier	directly tests the serialized commitment–cut coordinates and reports theorem residuals
configured-pipeline diagnostic	matched matrix, external semantic baselines, exact-semantics witnesses, and natural workloads	compares observed tool behavior; never proves a lower bound
model-specific consequence	fixed-total-error synthesis and surface-code calculation	reports consequences under the disclosed QRE; not a hardware-independent forecast

Table IV. Formal/evidentiary separation used throughout the paper. The first row is the instrumented reference implementation of the declared resource model; configured external tools are not retroactively placed in that class.

width under a 600 s protocol and preserves its historical `predicate_error` outcomes; it is not substituted for the new matrix. This legacy experiment links the growing-m theorems to configured-pipeline behavior. Unlike the fixed-width witnesses below, which exercise the exact-semantics corollary, the packing family grows the number of independent phase terms m at fixed tolerance.

a. Instance. We instantiate Definition 14 with the chain of two-body characters $a_j = e_j + e_{j+1}$ on $n = m+1$ qubits, which are $\mathbb{F}_2$-independent for every m (certified by rerunning the exact rank verifier), using the separate experimental setting $K = 2$, $\Delta = \pi/4$. Each token is lowered as $RZ_i(\theta)\, RZ_j(\theta)\, CP_{ij}(-2\theta)$ under the convention used here. The balanced round-robin encoding (Definition 15) with dispersal $r = 4$ interleaves the terms across shared wires, so per-wire locality does not trivially reveal the aggregate. Because these balanced tokens contain x_j/r, this is a pipeline diagnostic rather than an instance of the masked-share lower bound. The sweep is $m \in \{4, 8, 16, 32, 64\}$ under the standard 600 s protocol of this section. The exact rational-π sidecar records $\Delta/\pi = 1/4$ and per-token angle $1/16$. Correctness uses the dense certificate for $n \le 6$ and the sound exact symbolic checker of Definition 25 and Proposition 12 for $n > 6$; any output outside $\mathcal{D}_{\mathrm{CPC}}$ is status-only.

b. Results. Table V reports completed output gates as a function of m for semantic UCC, `qiskit opt3`, rebased TKET PauliSimp, staq rotation folding, and the phase-polynomial reference. All five paths completed at $m = 4$; semantic UCC, `qiskit opt3`, staq rotation folding, and the phase-polynomial reference also completed at $m = 8, 16, 32, 64$, while rebased TKET PauliSimp returned `predicate_error` at those four sizes because its output fell outside the Hadamard-free symbolic-certificate predicate. Semantic UCC and the phase-polynomial reference produced the same linear resource sequence, from 25/21/8 gates/depth/`cx` at $m = 4$ to 385/321/128 at $m = 64$. Among the completed external rows, `qiskit opt3` and staq produced $17m$ and $10m$ output gates, respectively, as the rm token stream grew. Status outcomes are recorded separately and omitted from numerical curves.

The earlier two-panel figure is deliberately omitted. Its gate panel is redundant with Table V, while its family-relative bit panel was generated from a handwritten length formula and has no serialized per-point files. Under the present accounting contract such a curve may be restored only by emitting the exact conditional-code artifacts (together with the family-dictionary digest) and measuring their files.

c. Scope. The external-toolchain rows are configured-pipeline diagnostics in the sense of this section: they neither prove nor test Theorem 2, whose serialized inter-round cut budget is not observable from output gate counts, and no bit curve is inferred from these gate-count rows. The separate reference-compiler implementation below measures theorem-model cut fields directly; it is not inferred from output gates.

D. Event-Level Commitment–Cut Tracing

The instrumented suite contains echo/pass-through, full aggregation, 25%, 50%, and 75% memory-capped

m	semantic UCC	`qiskit opt3`	TKET PauliSimp (rebased)	staq folding	phase-poly ref
4	25/21/8	68/38/32	54/20/10	40/27/32	25/21/8
8	49/41/16	136/54/64	`predicate_error`	80/39/64	49/41/16
16	97/81/32	272/86/128	`predicate_error`	160/63/128	97/81/32
32	193/161/64	544/150/256	`predicate_error`	320/111/256	193/161/64
64	385/321/128	1088/278/512	`predicate_error`	640/207/512	385/321/128

Table V. Packing-family scaling at growing m (chain two-body characters, $n = m + 1$, dispersal $r = 4$, 600 s budget). Completed cells report output gates/depth/`cx`; status outcomes are recorded separately and are not scored as structural wins. No external row is claimed to be a member of the transducer class of Definition 9.

hybrids, and an IR-materializing control. Each compiler follows the formal flat-update interface and records an event after every input update and commitment. The campaign scans m, r, K, ϵ, p, S, schedule, and seed, producing 592 complete run summaries plus the compressed raw event log and content-addressed trace archive. At each named cut it records input position, pass, control/store/window/parameter and IR bytes, committed-output bytes, current output length, RSS, and wall time. The literal measured definitions are

$$B_{\mathrm{com}}(c) = 8|A_{\mathrm{com,c}}|_{\mathrm{byte}},$$
$$B_{\mathrm{cross}}(c) = 8 \sum_{f \neq \mathrm{com}} |A_{f,c}|_{\mathrm{byte}}.$$

RSS is recorded as an implementation metric and is not substituted for the serialized state cap S. The analyzer recomputes the theorem residual from the trace coordinates and retains every anomaly or failed run. All frozen points lie in the theorem-allowed region; no point was removed as an outlier.

Echo commits updates early with little retained state; full aggregation occupies the state-heavy compact-output corner; and the three capped hybrids interpolate through explicit block commitments. Their scaling fits and residuals are reported with the raw points. The measured quantities are the theorem's resources in its own bit units, but at these sizes they are dominated by a fixed serializer overhead of about 3.8 kbit, against a bound of 10–180 bits; only the increments with m are informative, and the trace is a consistency check of the accounting, not a test that could approach the bound.

E. Width-Axis Scaling

To check the constructive aggregation prediction (Proposition 13), we run a width-axis sweep for the `full_pair_cp` topology at $n = 4, 5, 6$.

The output is $O(m_{\mathrm{CP}})$ and independent of r on this finite width sweep, consistent with the constructive proposition.

F. Exact-Semantics Witness Results

These experiments exercise the exact-semantics corollary (Corollary 1), not the main resource tradeoff (Theorem 2). They verify the constructive aggregation path on the tested fixed-width witnesses and record how configured external pipelines behave predictably across the representation boundary. Full data tables are in Section C.

The rational CP HD^rH witness is the canonical fixed-width family with a constant 42-gate output. Figure 6 shows that semantic UCC remains at 42 gates while rebased TKET PauliSimp returns 52–73 gates across five scales; materialize-first baselines grow linearly. The irrational independent-Pauli witness (Fig. 7) shows the same pattern at 27-gate constant output. Key observations:

- Semantic UCC outputs a constant gate count (42 for rational, 27 for irrational) at all five tested sizes.
- Rebased TKET PauliSimp recovers a small validated form at all five scales on both witnesses, demonstrating that a generic pipeline can cross the formal boundary by reconstructing an aggregate Pauli representation.
- Public staq rotation folding and `qiskit opt3` grow linearly with input size. TKET GuidedPauliSimp returns exactly the gate count of its lowered input at every size: it applies PauliSimp only inside `CircBox` regions, which these flat inputs do not contain, so it is not an optimizing baseline here.
- PyZX `full_reduce` completes at small sizes but reaches time or correctness limits at larger scales.

These measurements exercise the constructive side of Corollary 1 and disclose the behavior of configured tools; they do not prove the resource bound for any real compiler. The rebased PauliSimp result is the classification's positive prediction in action: once a generic pass sequence reconstructs a compact Pauli representation, the output need not grow with the flat input. Conversely, staq's growing output and PyZX's status outcomes are not impossibility results—they are diagnostics of configured-pipeline behavior on this specific witness family.

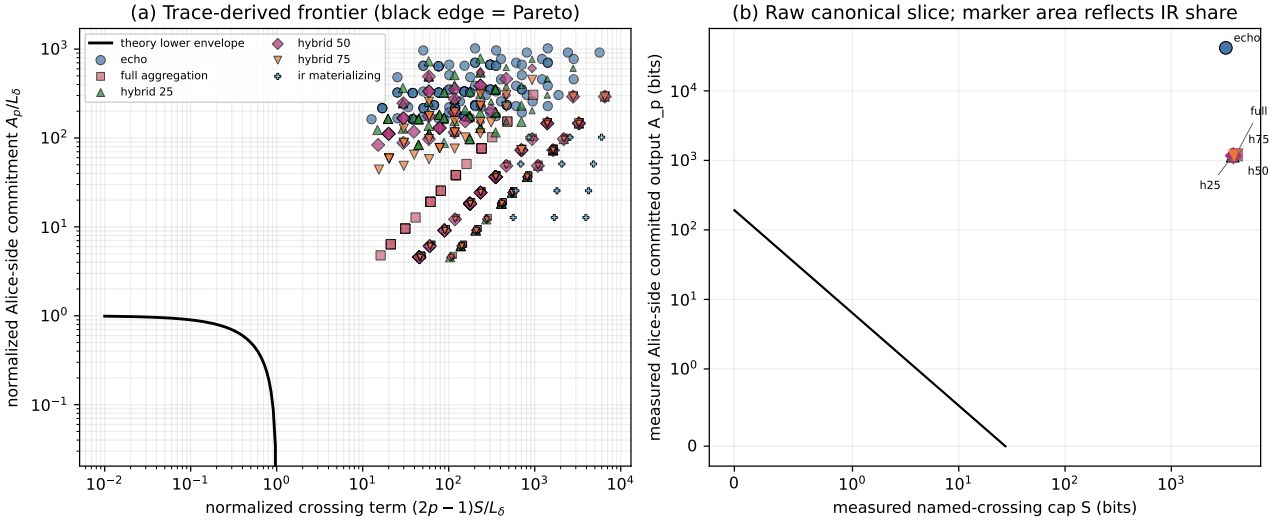


Figure 5. Trace-derived commitment–crossing-state–IR surface. The black edge marks measured Pareto points and the theoretical lower envelope is evaluated in the same serialized-bit coordinates. Marker area in the raw slice reflects IR share; RSS is recorded separately.

Width (n)	CP pairs (m_{CP})	Output Gates	CX Count
4	6	42	12
5	10	65	20
6	15	93	30

Table VI. Width-axis scaling for the semantic Fourier-layer witness. For each fixed width n, the semantic output is constant across all tested repetition counts r. Across widths, the output scales as $O(m_{\mathrm{CP}})$ where $m_{\mathrm{CP}} = n(n-1)/2$.

G. Mechanism Ablation

The ablation strengthens the mechanism interpretation. With Fourier-layer IR disabled, the compiler completes at 4k, 10k, and 20k but materializes the same large non-aggregated outputs as `qiskit opt3`; at 50k and 100k the disabled-IR rows exceed the 600 s budget (Fig. 8). In contrast, the Fourier-enabled semantic UCC path and the phase-polynomial reference recover the same 42-gate canonical circuit at every tested size. Thus the constant form is tied to semantic Fourier-layer lifting and coefficient aggregation, not merely to a favorable ordering of flat post-lowering passes.

H. Fixed-Total-Error End-to-End Synthesis and QRE

The new campaign scans $\epsilon_{\mathrm{total}} \in \{10^{-3}, 10^{-6}, 10^{-9}\}$ and assigns the same additive budget fractions to every pipeline: $0.10\epsilon_{\mathrm{total}}$ for the certified algorithmic stage, $0.45\epsilon_{\mathrm{total}}$ for rotation synthesis, and $0.45\epsilon_{\mathrm{total}}$ for logical failure. For equal allocation and N_{rot} output rotations, it chooses $p = \lceil \log_{10}(N_{\mathrm{rot}}/\epsilon_{\mathrm{synth}}) \rceil$ and $\epsilon_i = 10^{-p}$, hence $\sum_i \epsilon_i \le \epsilon_{\mathrm{synth}}$. A second measured-cost greedy allocation spends only this decimal slack; it is an upper construction, not an optimality claim. Each IEEE-serialized angle is actually synthesized by the frozen deterministic GMP build of staq `grid_synth` [8, 34]. Across 670 checked angle–precision pairs, the reported error never exceeds the allocated radius; exact multiples of $\pi/4$ use staq's checked exact branch.

The full $2\times3\times3\times2\times2\times4 = 288$ matrix crosses two target sizes, three pipelines, three total errors, two allocation policies, two QEC profiles, and 1/4/16/32 factories. All 288 rows are `completed_valid`, carry a content-addressed exact-symbolic or dense $n \le 6$ certificate, and satisfy the requested total-error bound. The exact symbolic checker is sound but deliberately incomplete: it accepts every semantic-first row and the materialize-first rows of the 4k target, does not accept the 48 materialize-first rows of the 20k target (its canonical records differ, which does not imply inequivalence), and stops with a predicate error on all 96 TKET rows. Those 144 rows carry the dense $n \le 6$ unitary certificate instead, which every row passes. The two surface-code profiles [35–37] use $p_{\mathrm{phys}} = 10^{-3}$ and 10^{-4}, respectively, with separately frozen data/factory coefficients, factory periods [10], and cycle times. They choose the least odd d meeting the disclosed union bound. These are executable sensitivity scenarios rather than vendor estimates, and follow the accounting conventions of published end-to-end estimates [38, 39].

The physically relevant point is the ordering of semantic aggregation relative to synthesis. In a fault-tolerant

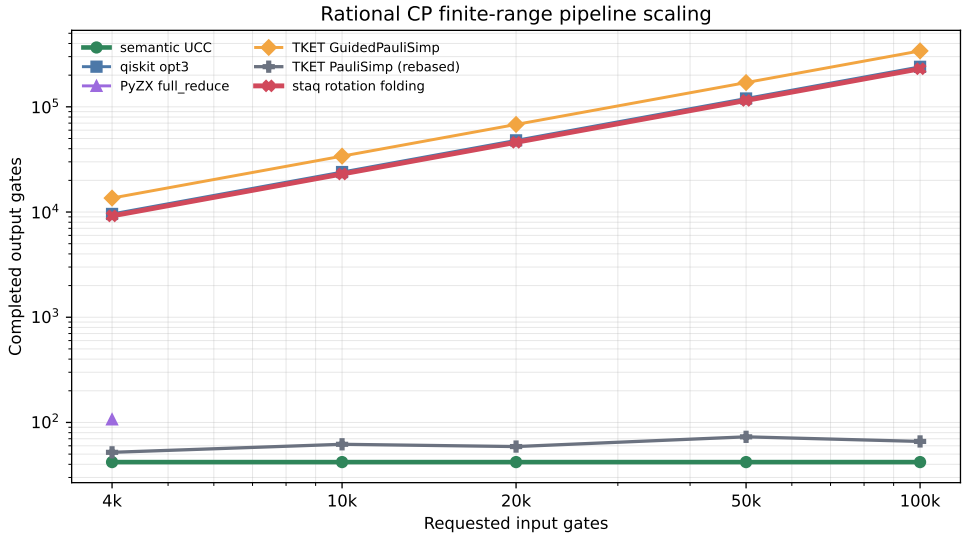


Figure 6. Fixed-width non-inverse Fourier-layer scaling on the rational HD^rH witness. Semantic UCC remains at 42 gates and rebased TKET PauliSimp returns 52–73 gates across the five scales. Public staq rotation folding and `qiskit opt3` grow with input size; TKET GuidedPauliSimp returns its lowered input unchanged in size (it acts only inside `CircBox` regions); PyZX `full_reduce` completes at 4k and reaches the time budget at larger scales. Status-only cells are omitted from numerical curves.

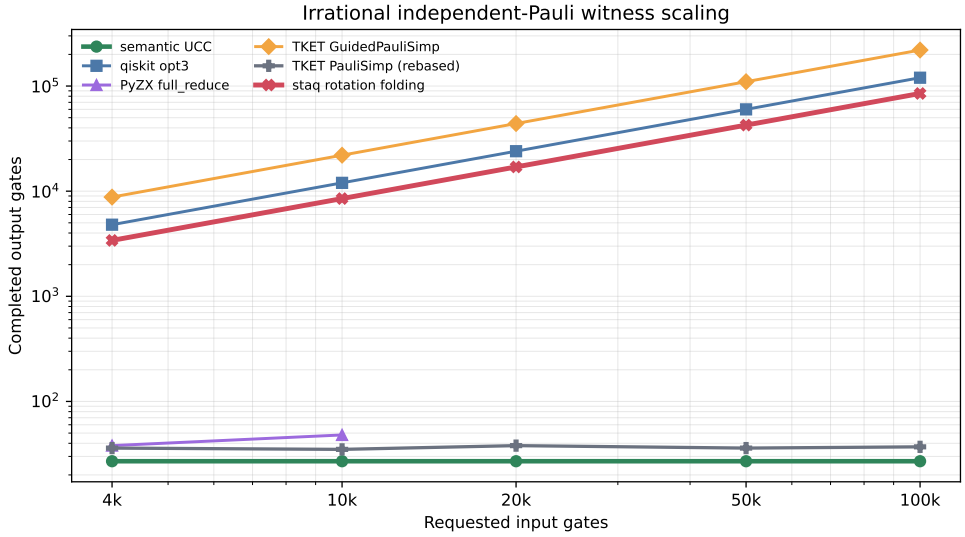


Figure 7. Configured-pipeline scaling on the irrational independent-Pauli witness. Semantic UCC remains at 27 gates. Rebased TKET PauliSimp returns a small validated form at all five scales, demonstrating that a generic pipeline can cross the formal boundary by reconstructing an aggregate Pauli representation. Public staq rotation folding and `qiskit opt3` grow with input size; TKET GuidedPauliSimp, which acts only inside `CircBox` regions, does not optimize these flat inputs. PyZX `full_reduce` completes at 4k and 10k, fails the numerical tolerance at 20k, and reaches the time budget at larger scales. Status-only cells are omitted from numerical curves.

pipeline, arbitrary rotations are eventually approximated by discrete resources such as Clifford+T sequences. If the compiler first materializes r separate rotations and only later synthesizes them, the resource estimator sees r independent approximation tasks. A resource-rich pipeline instead combines the coefficients first and presents one aggregated rotation per phase term. An FT toolchain that reconstructs the phase polynomial before synthesis collapses this multiplicity itself and should be classified on the semantic-capable side of the comparison (see Section X for the model-specific resource discussion).

Requested	semantic UCC	artifact UCC (Fourier disabled)	`qiskit` opt3	phase-poly ref
4,000	42 / 2.9 s	9,588 / 18.2 s	9,588 / 1.0 s	42 / 0.2 s
10,000	42 / 8.7 s	23,988 / 94.2 s	23,988 / 5.1 s	42 / 0.3 s
20,000	42 / 28.5 s	47,988 / 339.3 s	47,988 / 19.5 s	42 / 0.3 s
50,000	42 / 20.7 s	timeout	119,988 / 120.9 s	42 / 0.5 s
100,000	42 / 28.7 s	timeout	239,988 / 485.8 s	42 / 0.8 s

Table VII. Fourier-layer semantic-path ablation on the fixed-width HD^rH witness under the 600 s budget. Disabling the Fourier-layer IR completes at 4k–20k but materializes large outputs; 50k–100k exceed the budget. Fourier-enabled semantic UCC and the phase-polynomial reference return 42 gates at all sizes.

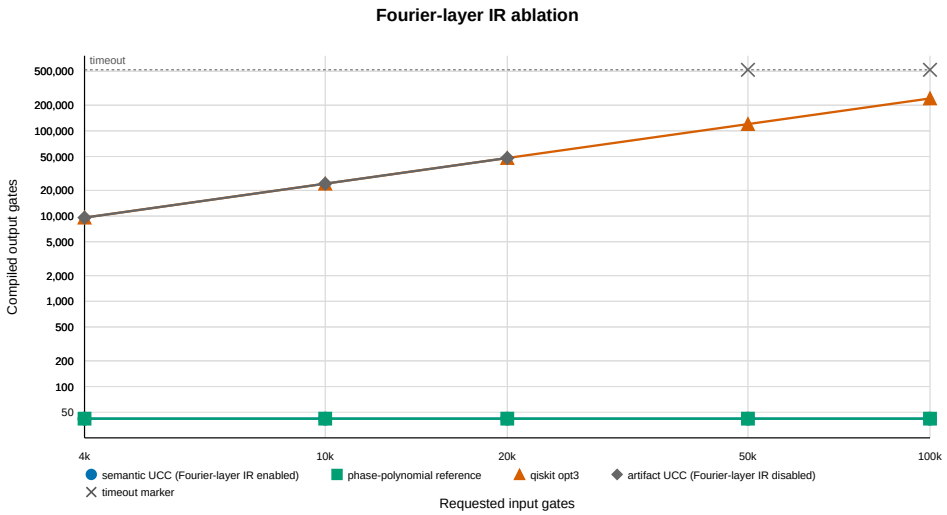


Figure 8. Mechanism ablation for the fixed-width HD^rH witness. Disabling the Fourier-layer semantic path yields completed but linearly large outputs through 20k and timeout rows at 50k and 100k, while Fourier-enabled semantic UCC and the phase-polynomial reference recover the canonical 42-gate circuit.

IX. APPLICATION WORKLOADS

A. Workload Specification

The application campaign contains seven structured families: Ising/MaxCut cost evolution [1], diagonal Hamiltonian simulation, commuting-block Trotterization [5], QFT arithmetic, QPE controlled powers, controlled-power ladders, and phase-polynomial blocks [13]. Family sizes and naming follow the low-level benchmark conventions of [40, 41]. Random Clifford+R_Z and routing-dominated [42, 43] circuits are negative controls. The separate controlled packing family (Definition 14) with growing m, fixed ϵ, is reported in Section VIII C.

B. Strong Baseline Comparison

The canonical comparison includes Qiskit `optimization_level=3` [44], TKET PauliSimp [16, 45], PyZX `full_reduce` [46], and the authors' exact phase-table reference. Each external tool has a recommended-native and a unified $\{\mathrm{CX}, R_Z, H\}$ configuration, with pass sequence, bridge, version, seed, timeout, memory cap, and certificate frozen. TKET supplies the Pauli-network/global-semantic class and PyZX the graph/global-IR class. The complete W4 interaction is Fig. 4; the independent W5 campaign and bridge-loss accounting are described in Section VIII B. Historical staq, GuidedPauliSimp, and phase-polynomial diagnostic rows remain in their original immutable campaigns and are not silently mixed into the canonical matrix. This comparison is diagnostic, not a contest: the external tools are treated as independent implementations of the semantic-first principle whose internal costs the bounds of Sections IV C and VI explain, gate-count parity is reported as parity, and no headline ranks the authors' pipeline above them.

pipeline	N_{rot}	T	physical qubits	factories	cycles	qubit-s
External semantic-capable (TKET)	29	798	258,152	16	13,317	3.44e+03
Materialize-first	23,992	2,679,110	468,968	16	57,966,528	2.72e+07
Semantic-first reference	22	1,800	305,000	16	31,675	9.66e+03

Table VIII. Predeclared 20k representative slice of the fixed-total-error campaign: $\epsilon_{\mathrm{total}} = 10^{-6}$, equal synthesis allocation, conservative $p_{\mathrm{phys}} = 10^{-3}$ scenario, and 16 factories. Each compiled output has an independent certificate, and every unique-angle staq synthesis returns `Check flag = 1`. The frozen full table additionally records code distance, data and factory qubits separately, and modeled quantum runtime.

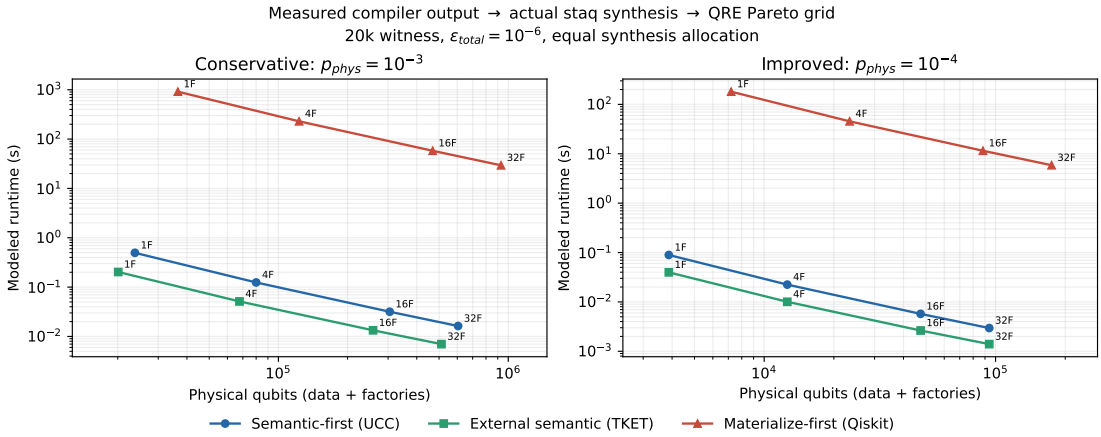


Figure 9. Physical-qubit/runtime Pareto grid at the representative 20k, $\epsilon_{\mathrm{total}} = 10^{-6}$ slice. Labels give factory counts. The external TKET path receives a materialized CX–R_Z–H bridge but reconstructs a compact phase form before synthesis, so it is classified as semantic-capable.

C. Real-Task Panel

These results support the recoverability-parity regime (Fig. 10): the semantic artifact matches `qiskit opt3`-level output quality while improving over the upstream UCC v0.4.12 pre-semantic default of Table IX by approximately 65%, 73%, and 78% gate reduction on the three instances. The selector cost in this panel is the implementation's conservative structural score $\kappa = \mathrm{gates} + \mathrm{depth} + 10 \cdot \mathrm{multi\text{-}qubit}$, with candidate-control guards; the factor 10 is a conservative lexicographic-style heuristic favoring lower two-qubit exposure, used only for portfolio selection. This explains the `REAL24-phase-est.` row (source key `phase_estimation_real`) in which translation-only has fewer total gates than the selected candidate but more `cx` gates and a higher score (957,132 versus 870,919). This quality-recovery pattern shows that semantic lifting behaves as an anti-regression mechanism on workloads beyond the minimal Fourier construction.

D. Natural-Kernel Factorial Ablation

The frozen W8 campaign crosses nine workload families, scales 1/2/4, three seeds, densities 0.5/1.0, and all-to-all/line hardware topologies. Seven are structured families: QAOA/Ising, diagonal-Hamiltonian simulation, commuting-block Trotterization, QPE controlled powers, QFT arithmetic, controlled-power ladders, and phase-polynomial blocks. Random Clifford+R_Z and routing-dominated circuits are the two negative controls. For every instance the authors' reference is evaluated on the complete 2^6 toggle matrix

$$\text{semantic lift} \times \text{aggregation} \times \text{selector} \times \text{cache/reuse} \times \text{preset recognizer} \times \text{projected-block selection}.$$

The authors' factorial contributes 20,736 cells, and the strongest matched external semantic baseline, TKET PauliSimp, contributes 324, for 21,060 cells total. Every cell is `completed_valid`, passes the independent dense projective certificate at six qubits, and carries the same fixed-total-error W7 QRE fields. Size increases repeat and densify structure rather than relaxing the certificate width.

Five structured families satisfy the predeclared criterion on their primary cells; QFT arithmetic and phase-polynomial blocks do not, and remain visible in Table X and Fig. 11. The seeds replicate less than the design suggests: the QPE controlled-power and QFT-arithmetic generators produce the same instance for every seed at both densities, and at density 1 so do the Trotter, controlled-power-ladder, and phase-polynomial generators, so the criterion's "every seed" clause is vacuous on their primary cells. Gate count is also not the fault-

Instance	Method	Gates	Depth	CX	Runtime
`REAL24-phase-est.`	translation only	159,838	121,214	67,608	0.080 s
	qiskit opt3	166,805	119,204	58,491	1.298 s
	upstream UCC v0.4.12 default	471,819	328,271	58,491	80.923 s
	semantic UCC	166,805	119,204	58,491	1.364 s
`REAL24-grover`	translation only	85,403	55,892	33,408	0.104 s
	qiskit opt3	79,029	54,163	32,544	0.552 s
	upstream UCC v0.4.12 default	287,047	152,298	32,544	1.915 s
	semantic UCC	79,029	54,163	32,544	0.556 s
`REAL24-qaoa`	translation only	36,512	2,416	23,808	0.022 s
	qiskit opt3	36,512	2,416	23,808	0.272 s
	upstream UCC v0.4.12 default	167,833	7,133	23,808	1.010 s
	semantic UCC	36,512	2,416	23,808	0.273 s

Table IX. Official real algorithm-family instances. The upstream UCC v0.4.12 row is the frozen pre-semantic default defined in the provenance map of Appendix B. Parity with `qiskit opt3` reflects conservative portfolio selection; this panel is an anti-regression check, not evidence of semantic recovery.

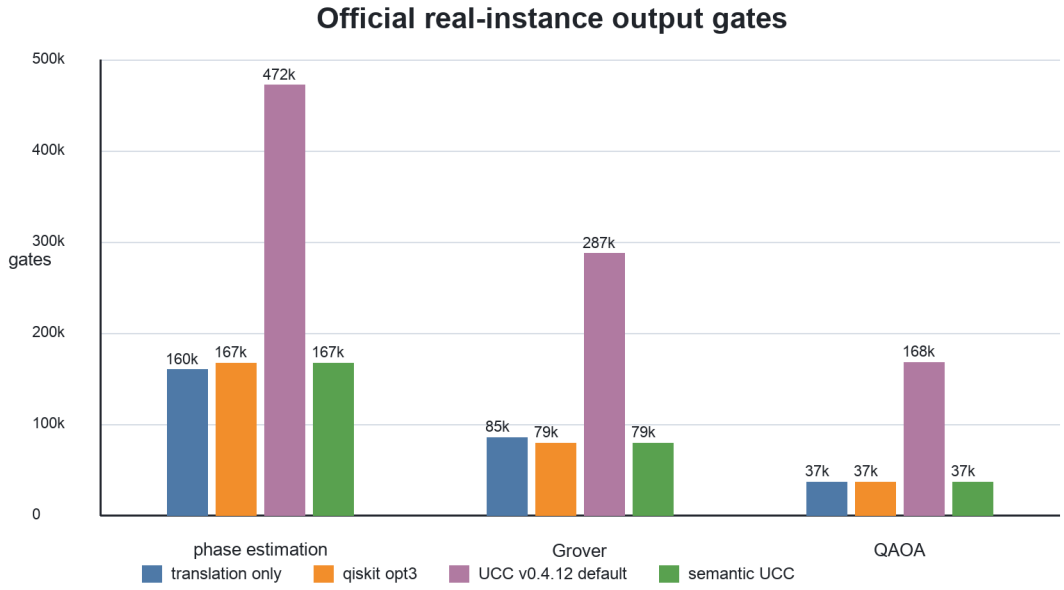


Figure 10. Total output gate counts on the three official real instances. The semantic UCC artifact matches `qiskit opt3`-level output quality while substantially improving over the upstream UCC v0.4.12 pre-semantic default.

tolerant metric: in logical T states the two compilers tie on five structured families, TKET PauliSimp is better on QAOA/Ising and on both negative controls, and the authors' path is better only on the Trotter family (`data/frozen/natural_workload_summary.csv`). The largest isolated gate-count contribution is the preset recognizer, with mean reduction 364.9 and row-level bootstrap interval [349.8,380.5]; the rows share instances across the other factors, so this interval understates the uncertainty. Aggregation and semantic lift also have positive mean effects; selector, cache/reuse, and projected-block selection have zero isolated gate effect in this controlled implementation. Both negative controls have zero all-on-minus-all-off differences in gates, depth, CX, logical T, and spacetime, hence 0 systematic quality regressions in the frozen grid. These are finite reference measurements, not empirical proofs of the streaming theorem or universal compiler superiority.

X. MODEL-SPECIFIC FAULT-TOLERANT RESOURCE CONSEQUENCES

The recoverability bounds of Sections V and VI establish an information tradeoff, but they do not by themselves convert description bits into gates or magic states. The certified pipelines of Section VIII additionally exhibit different rotation multiplicities. This section propagates those measured circuit outputs through Clifford+T synthesis and surface-code scheduling under one disclosed and versioned cost model. The resulting physical numbers are therefore a model-specific consequence of the measured

Family	external-minus-authors gates [95% CI]	strict success
controlled_power_ladder	9.0 [5.7, 12.7]	yes
diagonal_hamiltonian	38.0 [25.9, 51.8]	yes
phase_polynomial_blocks	-24.3 [-33.8, -16.3]	no
qaoa_ising	44.7 [33.7, 56.6]	yes
qft_arithmetic	8.1 [1.9, 13.8]	no
qpe_controlled_powers	-12.0 [-22.0, -3.0]	yes
trotter_commuting_blocks	33.0 [17.6, 48.2]	yes

Table X. Complete structured-family summary against the matched external semantic-capable baseline. The gate column averages all 36 matched cells of a family; the predeclared strict criterion is evaluated on the primary cells only (all-to-all topology, density 1; nine per family) and requires valid outputs, no worse gate and CX counts at every primary cell, and lower measured compiler runtime at every primary scale and seed. A family can therefore pass with a negative mean gate difference, as QPE controlled powers does. Non-winning families are retained.

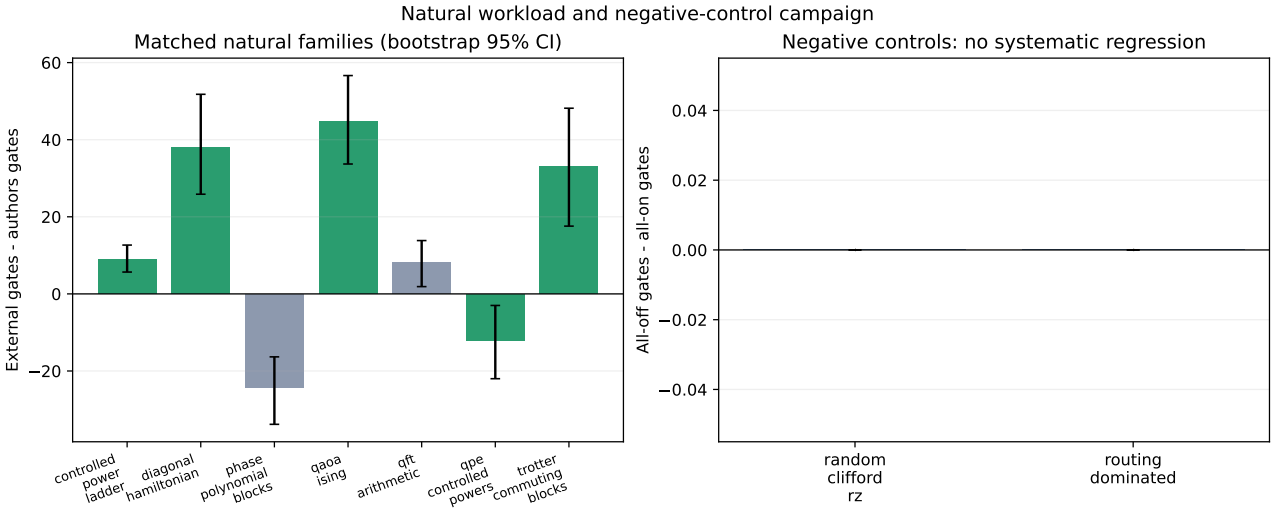


Figure 11. Matched natural-family comparison and negative controls. Error bars are paired bootstrap 95% intervals over matched cells; because the generators of several families do not depend on the seed, these intervals describe the spread over cells, not seed replication. All structured families and both controls are retained.

pipelines, not a representation-independent corollary of the bit lower bound.

The representation gap propagates into the fault-tolerant resources of the explicit model in Table VIII and Figs. 9 and 13. The materialize-first pipeline is Qiskit's preset optimization level 3 (trivial layout, no routing) applied after the witness has been lowered to the basis $\{cx, rx, ry, rz, h\}$ at optimization level 0; TKET `PauliSimp` receives the same circuit lowered to a cx–R_z–h bridge, and semantic-first receives the unlowered circuit (`scripts/run_qre_campaign.py`). Over all 96 matched target/error/allocation/QEC/factory settings, materialize-first uses 385–1,753 times as many logical T states and 538–5,573 times the spacetime volume of semantic-first: 385–501 times the T states at the 4k target and 1,367–1,753 times at the 20k target. The ratio grows with the 399 and 1,999 repetitions of the witness block because the materialize-first output keeps about 12 rotations per repetition (4,792 and 23,992 in total), so its T count grows about in proportion to the repetitions, whereas semantic-first keeps 22 rotations at both sizes and its T count changes only through their aggregate angles. No reversal occurs within the scanned error and factory grid. This is not a claim that the authors' implementation dominates other semantic tools: external TKET `PauliSimp` reconstructs the phase structure before synthesis and uses 0.443–0.536 times the semantic-first T states on this witness. Correctly crediting that recovery is part of the comparison.

These values are outputs of actual checked angle synthesis followed by the disclosed scheduling models, not universal hardware predictions. The absolute physical numbers still depend on routing, correlated-error, factory layout, feed-forward, and code-distance assumptions outside this campaign. The complete sensitivity plot and all preflight anomalies are retained in the root QRE artifact package.

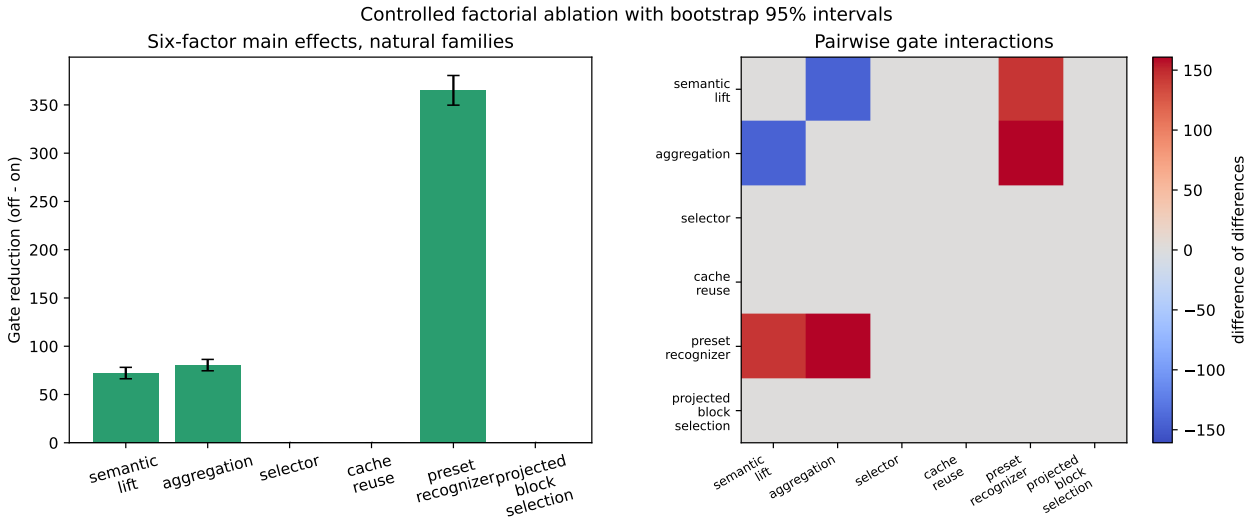


Figure 12. All six factorial gate-count main effects and pairwise interactions. Zero-effect factors are displayed rather than omitted.

XI. RELATED WORK

a. Phase-polynomial, Pauli-gadget, and ZX optimization. Commuting-rotation aggregation is a standard optimization target [11, 12, 16, 17, 45–47]. The phase-polynomial line makes the generator–coefficient structure explicit and then re-synthesizes the parity network, with or without a connectivity constraint [13, 48, 49]; the ZX line keeps a global diagrammatic normal form in which phases migrate and fuse across syntactic boundaries [50, 51]. All of these are *upper-bound* constructions: they exhibit a compiler that recovers structure on the inputs where their rewrite system applies. Our contribution is complementary and in the opposite direction: we formalize a representation-dependent recoverability boundary with an explicit ϵ–resource tradeoff that no such pass can evade without paying in the charged channels, connect it to communication complexity, and measure model-specific fault-tolerant consequences under one disclosed QRE. Where these tools do recover the aggregate—as rebased TKET `PauliSimp` does on our witnesses—the bound charges the reconstruction state rather than treating it as free. Section VIII D measures that channel for the formal-model reference compilers; the TKET API-boundary scan is reported separately as a configured-pipeline diagnostic and is not identified with S.

b. Gate synthesis and T-count accounting. The packing entropy and the generic synthesis cost share the same accuracy scale, but they are distinct statements. Solovay–Kitaev decomposition [52], exact Clifford+T synthesis [53], and the number-theoretic Ross–Selinger-type approximation algorithms [8, 54] realize generic Z-rotations with word length $\Theta(\log(1/\epsilon))$; multi-qubit T-count lower bounds sharpen the same accounting at the unitary level [55]. The metric packing fixes K_ϵ independently, and the QRE campaign then measures the actual checked synthesis cost of each distinct output angle rather than assigning one uniform repair cost to every coordinate.

c. Compiler IRs, verified compilation, and full-stack tooling. Quantum software stacks use intermediate representations to preserve information across lowering [33, 44, 45, 56–60]. Recent designs are explicit about what the IR retains: OpenQASM 3 admits timing and classical control in the instruction stream [21], while SSA- and MLIR-based representations expose dataflow so that passes can be composed and verified [61, 62]. Our budget-explicit model charges exactly this retained information—B_{IR} in Eq. (4)—rather than treating a richer IR as a free side channel, which is what makes the semantic and materialized pipelines comparable at all.

d. Routing, mapping, and benchmark suites. Architecture-directed passes reshape the stream that a downstream compiler sees [42, 43, 57], and the platform-specific compilers for shuttling-based trapped ions [63] and dynamically field-programmable neutral-atom arrays [64] show how far that reshaping can go before the stream reaches a downstream stage at all. In our model the SWAP and routing records they insert normalize away only when the accumulated Clifford permutes the declared generator set up to sign, exactly the hypothesis of Lemma 6. General routing that violates this hypothesis is outside that corollary. The routing-dominated family is used as a negative control in Section IX for precisely this reason. Benchmark suites supply the workload conventions we follow [40, 41, 59, 65].

e. Fault-tolerant resource estimation. The downstream cost of a T gate is set by distillation and by the scheduling model [9, 10, 35–37]. End-to-end estimates for concrete algorithms [38] propagate non-Clifford gate counts through distillation and layout into spacetime cost,

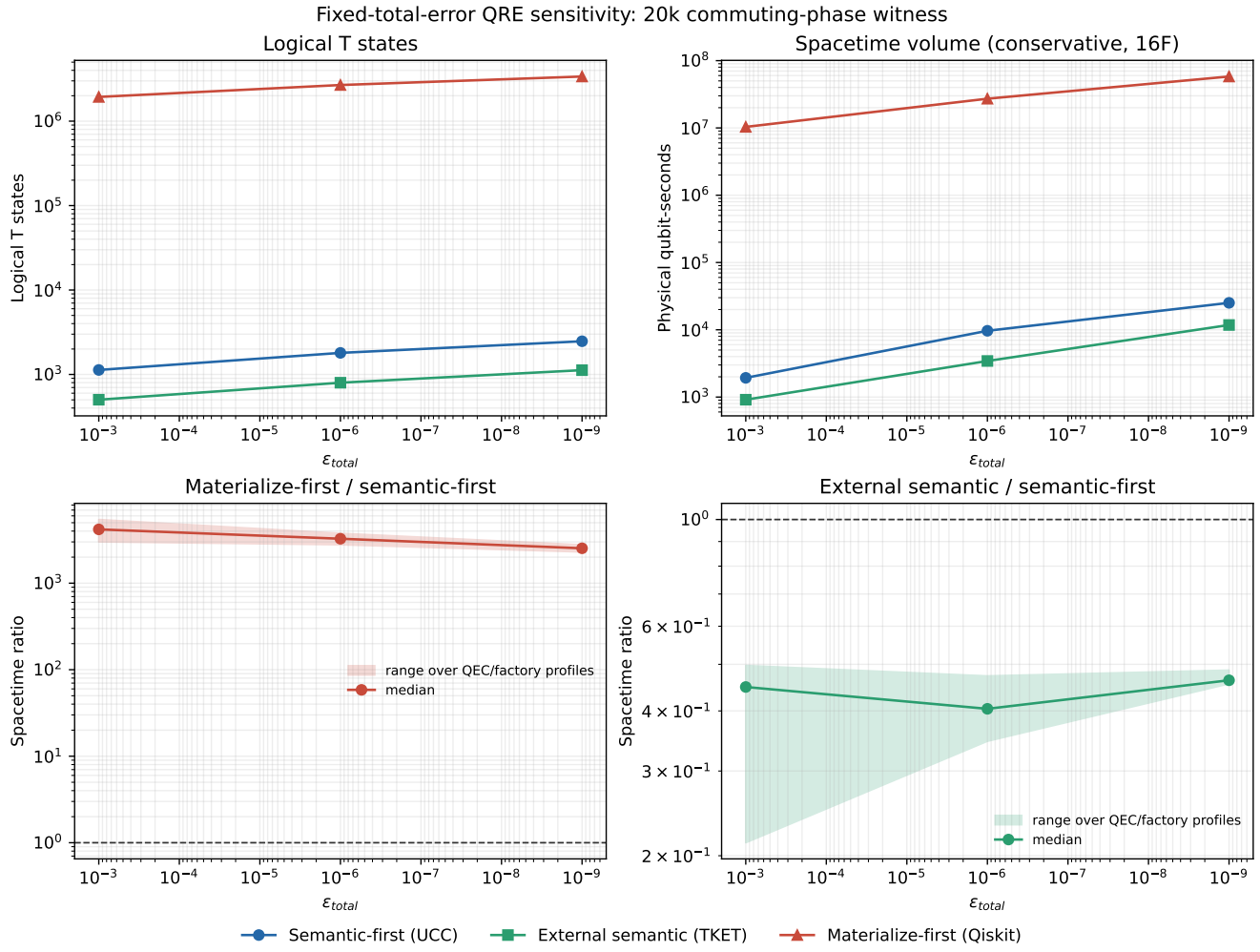


Figure 13. Sensitivity to total error, QEC profile, and factory count on the 20k witness. Top: logical T demand and the conservative-profile spacetime volume. Bottom: ratios to semantic-first, with bands spanning both QEC profiles and all four factory counts. The dashed line is equality. No materialize-first reversal occurs in the measured grid; the strong external TKET path is reported on the semantic-capable side.

and estimation frameworks that count arbitrary-angle rotations make their number and per-rotation synthesis accuracy enter through the T count per rotation [39]; this is the methodology used in Section X. The margin available to a quadratic speedup is thin enough that such factors matter [66], and resource-estimating compilers such as that of Ref. [33] implement the same accounting. Our QRE campaign is an instance of this methodology under a disclosed model, not a new estimator.

f. Noise tailoring and error mitigation. Randomized compiling and Pauli twirling [6, 7, 32] act on the dispersed streams we analyze without creating the dispersal, which comes from the schedule: their sign randomization is exactly the transformation our Lemma 6 normalizes, and in frame-tracked form it adds the Pauli frame to the crossing state (Proposition 5). The broader mitigation literature [67], including learned sparse noise models used for probabilistic error cancellation [68], shares the structural feature that matters here: the physically executed circuit is a randomized ensemble whose individual members do not display the quantities the source program wrote down.

g. Classical lower-bound methodology. The proof imports fooling-set and communication-complexity reasoning into a quantum-compilation representation question [26, 69, 70]. The pass-explicit form of the frontier follows the classical space–passes tradition for one-way and multi-pass computation over a stream [24, 71], where the object charged is the state serialized at a cut rather than total work.

XII. DISCUSSION

A. Representation as a Resource

The central message is that representation is a computational resource in quantum compilation. A compiler receiving a semantic representation that preserves generator–coefficient structure can aggregate in $O(m)$ records. The cyclic packing gives $\Omega(m \log(1/\epsilon))$ bits separately for family-relative and self-contained output codes and is compatible with an r-round modular share stream. In the stated forward-scan model its pass-explicit frontier is $\overline{A}_p + pS \geq (1-\delta)m \log_2 K - h_2(\delta) - \lceil \log_2(p+1) \rceil$ (Theorem 5). Thus a deferred flat compiler needs $S \geq (m \log_2 K)/p$ to leading order. The block hybrid attains this with compact family-relative output, to leading order whenever p is small compared with $m/\log m$, while the direct semantic vector uses only $O(\log m \cdot \log K)$ state (Propositions 10 and 11). The binary two-round theorem remains a sharper zero-error special case for single-delivery compilers.

This boundary is channel-explicit. Equation (4) serializes control, store, window structure, every live parameter, every live IR object, cache/hash layout, dynamic-pass state, and random seed. A compiler that reconstructs a global aggregate—as rebased TKET PauliSimp does on the fixed-width witnesses—may succeed, but its reconstruction table or graph is charged in B_{cut} rather than treated as a free semantic side channel.

B. Scope and Limitations

1. The results have different scopes. The unified theorem (Theorem 5) covers r-round additive shares, randomized failure, and $p = p(m)$ forward scans of the round-major stream; only the p forward crossings carry information, and the matching construction attains the bound to leading order for p small compared with $m/\log m$. Reverse scans, an uncharged random-access input oracle, or a different output code are outside that theorem. The older representation-dependent tradeoff (Theorem 2) is scoped to the two-round masked-share stream and to compilers that receive the stream once; it gives the stronger deterministic one-pass mask count. The exact-semantics corollary is scoped to fixed width, constant pass/cut budget, and certified noncollision. The growing-width theorems instead use the complete bit codec and allow random access and dynamic $p = p(m, I, \rho)$. The decoder is representation-invariant, but all syntax lengths retain their declared encoding semantics.

2. The commitment toll (Proposition 7) is the most narrowly scoped result here, and deliberately so. It is a two-sided statement about a compiler class, not about commitment in general: clause (i) needs the prefix to be *executed* at the cut and assumes that it dispatches one record per dispersed coordinate, and clause (ii) exhibits an admissible buffered compiler that pays nothing. The bound is also on an ensemble of masks compiled by one mask-uniform algorithm; for a single mask known to the serializer there is no toll to pay. Payload and toll are charged against the declared record content rather than the round-one unitary, which fixes it only up to zero residues and the $(s, a) \mapsto (-s, -a)$ degeneracy (Remark 11); at unitary level the proved lower bound on the toll is smaller, asymptotically 0.918 instead of 2 bits per committed generator at $Q = 3$, $k = m/2$, and 1.99 instead of 2 on the $\epsilon = 10^{-3}$ grid (Corollary 5); it is proved but not measured, and only the record-level toll is measured. Both clauses are measured only on the instrumented device of Fig. 3; the two preregistered attempts to observe the toll at a third-party compiler boundary failed and are reported as failures.

3. The tuned cyclic packing is stated for $0 < \epsilon \leq 1/8$ and uses the full residue alphabet, $K_\epsilon = Q_\epsilon$. Its coefficient interval wraps around the circle; Theorem 3 handles circular distance and global phase exactly. The prior no-wrap packing and the binary/$\mathbb{Z}_3$ theorem remain separate valid specializations. The configured scaling experiment uses $K = 2$, $\Delta = \pi/4$ and is not claimed to instantiate the masked-share theorem.

4. The fixed-total-error experiment performs actual Clifford+T synthesis and executable surface-code calculations under two disclosed physical/QEC profiles, but these profiles share one phenomenological model family. Its input is the measured rotation multiplicity of the three configured pipelines; the information lower bound alone does not imply a T-count or spacetime-volume ratio. Absolute physical counts still require cross-validation under an independent estimator and architecture-specific routing/factory layouts.

5. The packing-family scaling experiment (Section VIII C) links the growing-m theorems to configured-pipeline behavior. The instrumented reference compilers measure the masked-share cut fields directly (Section VIII D), but they are controlled formal-model objects, not instrumented versions of every external compiler.

6. Dense independent certification fixes the W8 natural-workload width at six qubits; the size axis repeats and densifies structure rather than increasing Hilbert-space width. Runtime and RSS are local-machine measurements. The two negative controls and three zero gate-effect ablation factors are retained as finite null results, not safety theorems.

7. The clean-room reconstruction creates an independent working directory and fresh virtual environment bound to locked project packages. It proves directory-level reconstruction of selected theorem, benchmark, and

QRE results, not binary or container independence from the host.

XIII. CONCLUSION

Consider a compiler that receives a commuting phase layer after scheduling has split it and randomized compiling has re-signed the pieces, and that reads the stream in forward passes. Provided the rounds of the layer are contiguous up to generator-permuting Clifford records, the round coefficients vary freely, and the compiler is correct with probability at least $1 - \delta$ on every such stream, its early output and the state it carries across its crossings must total at least $(1 - \delta)m \log_2 K_\epsilon - h_2(\delta)$ bits, about $\log_2(1/\epsilon)$ per independent term. At $\delta = 0$ a single-pass memory-capped compiler that emits one aggregated rotation per term meets that price to leading order. The two channels part ways when the committed first-round rotations are applied at the cut. For a randomly chosen dispersal pattern that prefix then needs, on average, at least the entropy of the pattern if it must address its targets, and a smaller but still linear amount if it need only implement the rotations; buffered output need not pay. A software model attains the record-level toll, and a buffered control pays none, in a preregistered run. Under one disclosed synthesis and surface-code model, a repeated four-qubit layer left partly unmerged by Qiskit's highest preset optimization level, given the pre-lowered circuit, needs hundreds to thousands of times the logical T states of our compiler's output, a ratio that grows with the number of repetitions, while TKET's PauliSimp, given the same input, rebuilds the phase structure and needs about half as many. Recoverability is thus a quantity that a software stack can budget for: where the interface allows, randomize and lower late, and keep phase semantics until the coefficients have been merged.

REPRODUCIBILITY

For scale, the headline campaigns comprise 592 traced reference-compiler runs; a 1,080-cell matched-representation design and a separate 1,728-cell external campaign containing TKET PauliSimp and PyZX full-reduce, with completed metrics accepted only when supported by the stated symbolic or dense certificate; 288 complete fixed-total-error fault-tolerant estimates under a common algorithmic, synthesis, and logical-failure budget; and 21,060 natural-workload and negative-control cells crossing a six-factor ablation, in which 5 structured families satisfy the predeclared all-size/all-seed criterion and the two negative controls show 0 systematic regressions. Explicitly labelled historical diagnostics retain their original artifact provenance.

The canonical registry is `data/manifest.yaml`; the normalized schema is `schema/experiment.schema.json`. Its complete key is

`experiment_id, instance_id, representation_id, method_id, seed, backend_hash, tool_version, artifact_commit, config_hash.`

The normalized Parquet index contains 24,748 unique records and retains 48 predicate-error rows outside quality means. The data audit checks CSV/Parquet mirrors, conflicting numeric cells, duplicate TeX labels, source hashes, row counts, orphaned frozen files, and historical versus canonical campaign disposition. It finds no duplicate primary keys, registered-file omissions, conflicting mirror cells, hash drift, or unregistered frozen artifacts among the committed canonical sources.

All measurement macros, tables, and figures used by the W3–W9 sections are generated from registered frozen sources by `scripts/build_all_tables.py` and `scripts/build_all_figures.py`; the operation and unique source row for each printed number are recorded in `data/provenance_map.yaml` and tested by `tests/test_paper_number_provenance.py`. The correctness language, symbolic checker, and mutation/dense audit are in `certificates/`; the QRE schemas and two physical profiles are in `qre/`. The approximately 700-MB raw W3–W8 run tree, including manifests, logs, certificates, seeds, and package records, is versioned in the separate private `OIerYangJZ/paper1-data` archive at the revision recorded in `data/manifest.yaml`; its absence is reported but does not invalidate the committed frozen-data rebuild.

The one-command entry point is `reproducibility/run_submission.sh`. From a fresh clone it creates the locked project environment when needed, reruns frozen-data audits, regenerates all registered tables and figures, executes the full test suite, builds the canonical, arXiv, Quantum-submission, and supplement PDFs, and rejects unresolved references. This procedure reproduces the submitted analysis from immutable frozen datasets; it does not rerun the complete measurement campaigns. The clean-room runner copies a minimal artifact to an independent temporary directory, creates a fresh environment bound to the locked project packages, reruns 9 headline-theorem tests, compiles and densely certifies a matched QAOA/Ising instance through the authors and TKET paths, and exercises 17 deterministic synthesis pairs in a fixed-total-error QRE subset. This is working-directory isolation, not a claim of host-binary independence. Complete audit and limitation records are `DATA_AUDIT_REPORT.md`, `CLEAN_ROOM_REPORT.md`, `QRE_REPORT.md`, and `NATURAL_WORKLOAD_REPORT.md`.

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Generative AI tools, including large language model assistants such as Claude (Anthropic), were used to assist with manuscript drafting and revision, code and repository maintenance, literature discovery, and consistency checks. The authors directed their use through task-specific prompts and independently reviewed the generated text, citations, code, and numerical claims against the cited literature, source files, executable tests, and the symbolic or dense certificates reported in this work. AI output was not treated as a scientific source or as independent proof verification. The authors take full responsibility for the manuscript, proofs, code, data, figures, citations, and conclusions.

AUTHOR CONTRIBUTIONS

J.Y. developed the theory and proofs, implemented the compiler instrumentation and experimental pipeline, and performed the majority of the measurements and analyses. Y.L. supervised the research, reviewed and revised the manuscript, and provided project administration and funding support. X.-H.D. independently validated the theoretical derivations, numerical results, and computational artifacts. All authors wrote the manuscript together, discussed the results, and approved the final version.

DATA AVAILABILITY

The frozen datasets, analysis and figure scripts, the compiler snapshot used in the experiments, and the build that regenerates every table, figure, and reported number are openly available at `https://github.com/OIerYangJZ/representation-as-resource`. The approximately 700-MB raw run tree described in the Reproducibility section is held in a separate archive and is available from the corresponding authors upon reasonable request.

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Appendix A: Supplementary Theory and Full Proofs

1. Operational Gap Notation

For reference, the experimental gap can be written without adding another formal theorem hypothesis. Let:

- $L_B(C)$ be exact lowering into target basis B,
- κ be a conservative structural cost,
- $\mathcal{T}_K(C)$ be the bounded set of semantic candidates produced by the semantic lift under budget K,
- $\mathfrak{A}$ be a configured post-lowering pipeline family used for an experiment.

Define the K-representation gap of C by

$$\mathrm{Gap}^K_{\mathrm{rep}}(C) := \inf_{A\in\mathfrak{A}} \kappa(A(L_B(C))) - \inf_{T\in\mathcal{T}_K(C)} \kappa(L_B(T)).$$

Define the operational recoverability gap by

$$\mathrm{Gap}^K_{\mathrm{rec}}(C) := \kappa(C^\star_{\mathrm{flat}}) - \kappa(C^\star_{\mathrm{term}}),$$

where $C^\star_{\mathrm{flat}}$ is the best candidate recovered from the flat basis-lowered representation and $C^\star_{\mathrm{term}}$ is the best candidate recovered from the bounded semantic term representation.

The meaning is simple: semantics may survive lowering, while cheap recoverability of the opportunity may not. This notation is operational and experimental; the formal

lower bounds in the main text are the budget-explicit output and cut-transcript theorems for the declared finite-window stream model.

Definition 27 (Aggregate recovery)**.** *Consider the Fourier-layer family* $C_{m,r} = H_\Lambda D_m(\boldsymbol{\theta})^r H_\Lambda$ *with* $D_m(\boldsymbol{\theta}) = \prod_{j=1}^m e^{-i\theta_j P_j/2}$. *A compiler performs* aggregation-equivalent recovery *on this family if, for each non-removable term* P_j*, its successful output can be associated with an* $O(1)$*-size record, subcircuit, or algebraic object whose coefficient is equivalent modulo* 2π *to the aggregate* $r\theta_j$*. The object may be explicit, as in a phase-polynomial or commuting-diagonal IR, or implicit, as in a specialized rewrite, lookup, ZX transformation, or synthesis routine that computes the same aggregate diagonal action without storing all* r *lowered occurrences separately.*

Remark 18 (Operational recovery channels)**.** *For the Fourier-layer family, a pipeline that recovers the constant fixed-width output can be classified by the channel through which it carries the aggregate coefficient information. Examples include:*

1. *store enough persistent summary information to carry the aggregate;*
2. *lift enough lowered volume into a richer IR to reconstruct the aggregate;*
3. *use nonlocal provenance information from before basis lowering that already contains the aggregate;*
4. *reconstruct a global semantic representation equivalent to a phase-polynomial, ZX, stabilizer-phase, or commuting-diagonal Hamiltonian object.*

Consequently, a successful future flat-looking implementation would not by itself refute the main theorem. It would instead classify the implementation by identifying where the aggregate information crosses the finite-state boundary, or by showing that the formal transducer model is missing a cheaper channel. The fixed-width theorem lower-bounds final output length for constant-budget finite-window transducers on a certified noncolliding family. A pipeline that achieves the semantic constant output must still represent the aggregate coefficients that distinguish the noncolliding cases. The listed channels are implementation-level ways to carry that information. This is a classification statement rather than an optimality claim, and the list is not exhaustive.

Additional Structural Families

The artifact also targets phase-ladder, mirrored, and conjugation patterns that appear in QPE/QFT-style workloads and Grover-style workloads. These families are not used here as additional theorem instances. They are best read as structural stress families: the semantic controller preserves repeated phase-heavy ladders, mirrored shells, and conjugation shells long enough to compare projected costs before materializing the selected candidate.

For phase-ladder workloads, the relevant empirical question is whether bounded semantic terms preserve a repeated commuting-diagonal opportunity that is harder to recover after early lowering. For mirrored and conjugation workloads, the relevant empirical question is whether the artifact recognizes repeated shell structure and avoids materializing every candidate before selection. The Grover and mirrored-shell experiments in Appendix C are therefore empirical positive cases and robustness checks, not support for a separate theorem-level family separation.

2. Self-Contained Rational CP Period Certificate

The canonical four-qubit rational witness has diagonal block

$$\begin{aligned} D &= \prod_{j=0}^{3} RZ_j(\theta_j) \prod_{0\le a<b\le 3} CP_{ab}(\theta_{ab}),\\ \boldsymbol{\theta}_{RZ}/\pi &= (1/7, 1/8, 1/9, 1/10),\\ (\theta_{01},\theta_{02},\theta_{03})/\pi &= (1/12, 1/13, 1/14),\\ (\theta_{12},\theta_{13},\theta_{23})/\pi &= (1/14, 1/15, 1/16). \end{aligned}$$

For a reduced rational angle $\theta/\pi = p/q$, both $RZ(\theta)$ and $CP(\theta)$ have projective single-term order

$$\operatorname{ord}(p/q) = \frac{2q}{\gcd(|p|, 2q)},$$

because the relevant relative phase is p/q modulo 2. The complete single-term certificate is

term	support	θ/π	T_j
RZ_0	0	1/7	14
RZ_1	1	1/8	16
RZ_2	2	1/9	18
RZ_3	3	1/10	20
CP_{01}	01	1/12	24
CP_{02}	02	1/13	26
CP_{03}	03	1/14	28
CP_{12}	12	1/14	28
CP_{13}	13	1/15	30
CP_{23}	23	1/16	32

Hence

$$\begin{aligned} T_{\text{terms}} &= \operatorname{lcm}(14, 16, 18, 20, 24, 26, 28, 28, 30, 32)\\ &= 2^5 \cdot 3^2 \cdot 5 \cdot 7 \cdot 13 = 131040, \end{aligned}$$

which proves that D^{131040} is a global phase. To certify minimality for the product rather than merely obtain this common upper bound, subtract the phase of 0000. For $z = z_0z_1z_2z_3$, the exact relative phase is

$$\begin{aligned} w(z) = {} & z_0/7 + z_1/8 + z_2/9 + z_3/10\\ & + z_0z_1/12 + z_0z_2/13 + z_0z_3/14\\ & + z_1z_2/14 + z_1z_3/15 + z_2z_3/16 \pmod 2. \end{aligned}$$

In particular,

$$w(1011) = \frac{37007}{65520},$$
$$\operatorname{ord}(w(1011)) = \frac{2 \cdot 65520}{\gcd(37007, 2 \cdot 65520)} = 131040.$$

Thus no smaller positive power of D is a global phase, and $T = 131040$ exactly. It follows that $D^1, \ldots, D^R$ are pairwise distinct up to global phase for every $R \leq T$, since $D^i \simeq D^j$ requires $T \mid j - i$.

The machine certificate uses only Python `fractions.Fraction` and prints all 16 relative numerators, denominators, and orders. The frozen integer output contains

`term_order_lcm=131040`,
`relative_order_lcm=131040`,
`minimality_state_1011_order=131040`,
`exact_global_phase_period_T=131040`.

The reproducibility files and hashes are listed below; each digest is split at its 32-hex midpoint only for typesetting.

- verifier: `research/rational_witness/verify_period.py`, SHA256 `c7c90f26e00e515383e94e4bbd707574 eb3aabf94ad71c3cd5e2cdb8d5db7e76`;
- integer output: `research/rational_witness/period_certificate.txt`, SHA256 `375295d776f21ea2a82ee4f2bab5205d 25bfef85ee864900baacfbd80be9bd6e`;
- manifest: `research/rational_witness/SHA256SUMS`.

From the repaired-manuscript root, the byte-for-byte check is

```
cd research/rational_witness
python3 verify_period.py \
-check period_certificate.txt
```

3. Full Proofs for the Transducer Lower Bound

This section supplies the full finite-state proof layer underlying Corollaries 1 to 3. The main text states the shorter version; the appendix keeps the pumpability data explicit. The following statements restate the main-text theorem and corollaries with full hypotheses and complete proofs; they are not additional independent results.

a. Pumpable Pair Families

Persistent store can depend on r, so the induction tracks both a store value and a tape word. In this subsection, D in pumpability data denotes a period modulus, not the diagonal witness layer in $C_r = H_\Lambda D^r H_\Lambda^{-1}$. The transducer model is the one of Remark 5. The next definition records pumpability with explicit residue data.

Definition 28 (Pumpable pair family)**.** *Let Γ be a finite alphabet and P a finite set of store values. A family*

$$X(r) = (\beta(r), W(r)) \in P \times \Gamma^*, \qquad r \in \mathbb{N},$$

is pumpable *if there exist integers $h \geq 0$ and $D \geq 1$ such that for every residue $\rho \in \{0, \ldots, D-1\}$ there are*

$$\beta_\rho \in P, \qquad A_\rho, B_\rho, C_\rho \in \Gamma^*,$$

with

$$X(h + \rho + kD) = (\beta_\rho, A_\rho B_\rho^k C_\rho)$$

for every $k \geq 0$. The word B_ρ is the pump word on residue ρ. The residue is live *if $B_\rho \neq \epsilon$ and* collapsed *if $B_\rho = \epsilon$.*

Lemma 11 (Input pumpability)**.** *For the fixed-width witness $T_0(r) = \Pi M^r \Sigma$, the pair family consisting of the initial persistent store and $T_0(r)$ is pumpable with $h = 0$, $D = 1$, pump word M, and fixed store value.*

Proof. Take $A_0 = \Pi$, $B_0 = M$, and $C_0 = \Sigma$. □

b. One-Pass Propagation

Fix a precision bound q and a charged alphabet $\Gamma = \Gamma_B(q)$. Under this bound, a pass of $\mathcal{A}$ is an ordinary deterministic subsequential transducer over a finite alphabet whose transitions do not read head positions (Remark 5). Its finite state consists of the ephemeral state, the persistent store, and the w-token window. The end-of-pass persistent store is one component of the final finite state.

Lemma 12 (One-pass pumpability propagation)**.** *Let $X(r) = (\beta(r), W(r))$ be a pumpable pair family over a finite input alphabet. Let τ be one deterministic finite-window, finite-burst pass whose output alphabet and store set are finite. Let*

$$\tau(\beta(r), W(r)) = (\beta'(r), W'(r))$$

be the resulting final store and output tape. Then $X'(r) = (\beta'(r), W'(r))$ is pumpable.

Proof. Fix pumpability data $h, D, \{\beta_\rho, A_\rho, B_\rho, C_\rho\}_\rho$ for X. We first treat one residue ρ, then recombine all residues into r-indexed pumpability data for X'.

For a fixed residue ρ, write

$$W_k = AB^kC, \qquad \beta_k = \beta,$$

where $k \geq 0$ and A, B, C, β abbreviate the residue data. Absorb the ephemeral state, persistent store, and window into the pass's finite state set S. Starting from the state

determined by β, process A. This emits a fixed word O_A and reaches a state $s_A \in S$.

Let $f : S \to S$ be the state map obtained by reading the block B, and let $o(s) \in (\Gamma')^*$ be the output emitted while reading B starting in state s. The sequence

$$s_A,\ f(s_A),\ f^2(s_A), \dots$$

in the finite set S is ultimately periodic. Thus there exist $u \geq 0$ and $\ell \geq 1$ with $u + \ell \leq |S| + 1$ such that

$$f^{t+\ell}(s_A) = f^t(s_A) \qquad \text{for all } t \geq u.$$

For $k = u + v\ell + \eta$ with $0 \leq \eta < \ell$, the output while reading B^k is

$$\begin{aligned}&\big(o(s_A)o(f(s_A)) \cdots o(f^{u-1}(s_A))\big)\\ &\quad \cdot \Big(o(f^u(s_A)) \cdots o(f^{u+\ell-1}(s_A))\Big)^v\\ &\quad \cdot \big(o(f^u(s_A)) \cdots o(f^{u+\eta-1}(s_A))\big).\end{aligned}$$

After this, the pass reads the fixed suffix C, emits a word depending only on the state $f^{u+\eta}(s_A)$, and ends in a persistent store value depending only on the same state.

Restore the residue subscripts. For each original residue $\rho \in \{0, \dots, D-1\}$, let $u_\rho \geq 0$ and $\ell_\rho \geq 1$ be the preperiod and cycle length obtained above, let f_ρ, s_{A_ρ}, and O_{A_ρ} be the corresponding state map, post-prefix state, and prefix output. For $0 \leq \eta < \ell_\rho$, let $F_{\rho,\eta}$ be the complete word emitted while reading the fixed suffix C_ρ from state $f_\rho^{u_\rho+\eta}(s_{A_\rho})$, including the end-of-tape finalizer output, and define

$$\begin{aligned}P_\rho &:= o\big(s_{A_\rho}\big)o\big(f_\rho(s_{A_\rho})\big) \cdots o\big(f_\rho^{u_\rho-1}(s_{A_\rho})\big),\\ Q_\rho &:= o\big(f_\rho^{u_\rho}(s_{A_\rho})\big) \cdots o\big(f_\rho^{u_\rho+\ell_\rho-1}(s_{A_\rho})\big),\\ R_{\rho,\eta} &:= o\big(f_\rho^{u_\rho}(s_{A_\rho})\big) \cdots o\big(f_\rho^{u_\rho+\eta-1}(s_{A_\rho})\big) \cdot F_{\rho,\eta}.\end{aligned}$$

Let $\beta'_{\rho,\eta}$ be the end-of-pass persistent store value reached from that state. Define

$$\begin{aligned}u^* &:= \max_\rho u_\rho, \qquad \ell^* := \operatorname{lcm}_\rho \ell_\rho,\\ h' &:= h + Du^*, \quad D' := D\ell^*.\end{aligned}$$

Each new residue $\rho' \in \{0, \dots, D' - 1\}$ decomposes uniquely as

$$\rho' = \rho + D\tilde{\eta}, \qquad \rho \in \{0, \dots, D-1\}, \quad \tilde{\eta} \in \{0, \dots, \ell^* - 1\}.$$

For $r = h' + \rho' + k'D'$ with $k' \geq 0$, the old representation $r = h + \rho + kD$ has

$$k = u^* + \tilde{\eta} + k'\ell^*.$$

Since $u^* \geq u_\rho$, we have $k \geq u_\rho$ for every $k' \geq 0$; and since $\ell_\rho \mid \ell^*$, the residue of $k - u_\rho$ modulo ℓ_ρ does not depend on k'. Set

$$\eta := (u^* + \tilde{\eta} - u_\rho) \bmod \ell_\rho, \qquad v_0 := \left\lfloor \frac{u^* + \tilde{\eta} - u_\rho}{\ell_\rho} \right\rfloor \geq 0,$$

so that

$$k = u_\rho + (v_0 + k'\ell^*/\ell_\rho)\ell_\rho + \eta.$$

By the output decomposition above, for every $k' \geq 0$,

$$X'(h' + \rho' + k'D') = \big(\beta'_{\rho,\eta},\ A'_{\rho'}(B'_{\rho'})^{k'}C'_{\rho'}\big),$$

where

$$A'_{\rho'} := O_{A_\rho}P_\rho Q_\rho^{v_0}, \qquad B'_{\rho'} := Q_\rho^{\ell^*/\ell_\rho}, \qquad C'_{\rho'} := R_{\rho,\eta}.$$

This is exactly pumpability of X' with data (h', D') and residue data $\{\beta'_{\rho,\eta}, A'_{\rho'}, B'_{\rho'}, C'_{\rho'}\}_{\rho'}$. □

Corollary 9 (Per-tape induction, for proof)**.** *Assume $\mathcal{A}$ has constant budget, so all executions use one finite alphabet $\Gamma_B(q_*)$. For every pass index $i = 0, 1, \dots, p$, the pair family consisting of the persistent store after pass i and the tape $T_i(r)$ is pumpable.*

Proof. The base case is Lemma 11. Apply Lemma 12 successively to passes $1, \dots, p$. □

Remark 19 (Unary scratch tapes)**.** *If pass 1 writes one marker per input period, then $T_1(r) = \texttt{marker}^r$ is pumpable with live pump word* `marker`*. Lemma 12 applies to every later pass. A later constant-budget pass may output one marker every c input markers, permute markers, or add a fixed periodic decoration, but it remains pumpable. It cannot turn* `marker`^{r} *into a family of $O(1)$ many correct aggregate circuits for all noncolliding r; if the final pump collapses, the final output repeats on a residue class.*

c. Collapse Versus Final-Output Length

Lemma 13 (Pumpable outputs are bounded-or-linear on each residue)**.** *Let $W(r)$ be a pumpable word family with data $(h, D, A_\rho, B_\rho, C_\rho)$. For $r = h + \rho + kD$:*

(a) if $B_\rho = \epsilon$, then $W(h + \rho + kD)$ is independent of k;

(b) if $B_\rho \neq \epsilon$, then

$$|W(h + \rho + kD)| \geq k.$$

Consequently, on a live residue,

$$|W(r)| \geq \frac{r - h - D + 1}{D}$$

for all $r \geq h + D$ in that residue.

Proof. Both statements are immediate from $W(h + \rho + kD) = A_\rho B_\rho^k C_\rho$. If $B_\rho \neq \epsilon$, then $|B_\rho| \geq 1$, so $|A_\rho B_\rho^k C_\rho| \geq k$. Since $r = h + \rho + kD$ and $0 \leq \rho < D$, $k \geq (r - h - D + 1)/D$. □

Lemma 14 (Correct noncolliding final outputs cannot collapse)**.** *Assume finite-range noncollision (Assumption 1) with $N(R) = R$ on the interval $1 \le r \le R$. Suppose the final output family $T_p(r)$ is pumpable with data $(h, D, A_\rho, B_\rho, C_\rho)$ and $\mathcal{A}$ is correct on every $r \in \{1, \dots, R\}$. Without loss of generality take $h \ge 1$: if $h = 0$, replace h by $h + D$; this preserves pumpability after discarding the finite prefix of indices $k = 0$ on each residue and absorbing one copy of B_ρ into A_ρ. Then every residue class appearing twice in the tail $\{h, h+1, \dots, R\}$ is live. In particular, for every $r \in \{h + D, \dots, R\}$,*

$$|T_p(r)| \ge \frac{r - h - D + 1}{D}.$$

Proof. Let $r = h + \rho + kD$ with $k \ge 1$ and $r \le R$. Then $r - D = h + \rho + (k-1)D$ also lies in the tail, and by the convention $h \ge 1$ it satisfies $r - D \ge h \ge 1$, so both r and $r - D$ lie in the correctness domain $\{1, \dots, R\}$. If $B_\rho = \epsilon$, Lemma 13 gives

$$T_p(r) = T_p(r - D)$$

as literal circuit strings. Hence the two outputs have the same unitary. Correctness at both r and $r - D$ would imply

$$U(\mathcal{L}_B(C_r)) = U(\mathcal{L}_B(C_{r-D}))$$

up to global phase, which by exactness of the lowering means $D^r \equiv D^{r-D}$ up to global phase, contradicting Assumption 1 with $N(R) = R$. Therefore $B_\rho \ne \epsilon$, and Lemma 13 gives the stated length bound. □

d. Final-Output Theorem and Corollaries

Theorem 6 (p-pass final-output lower bound, restated with full hypotheses)**.** *Fix a fixed-width diagonal witness family $C_r = H_\Lambda D^r H_\Lambda^{-1}$ with exact lowering $\mathcal{L}_B(C_r) = \Pi M^r \Sigma$, and assume unbounded noncollision (Assumption 2): the powers D^r are pairwise distinct up to global phase for all $r \ge 1$, so $N(R) = R$ on every range. Let $\mathcal{A}$ be a uniform deterministic p-pass finite-window, finite-burst transducer with constant budget that is correct on the input $\mathcal{L}_B(C_r) = \Pi M^r \Sigma$ for every $r \ge 1$. Here uniform means one fixed deterministic transducer, with the same passes, transition rules, finite state, window, burst bound, and store bound for all repetition counts r.*

Then there exist constants $h \ge 1$ and $D \ge 1$, depending on $\mathcal{A}$ and the fixed witness but not on r, such that for every $r \ge h + D$,

$$|T_p(r)| \ge \frac{r - h - D + 1}{D} = \Omega_{\mathcal{A}}(r).$$

Equivalently, no constant-budget transducer in this model that is correct on the whole noncolliding fixed-width family can emit $o(r)$-length final outputs on that family.

Proof. By constant budget, all executions use a finite charged token alphabet. Corollary 9 gives pumpability of the final pair family and hence of $T_p(r)$, with data (h, D) depending only on $\mathcal{A}$ and the witness. By the convention of Lemma 14, take $h \ge 1$, so every comparison point $r - D \ge h \ge 1$ used below lies in the correctness domain.

Fix any $r \ge h + D$ and any $R \ge r$. By Assumption 2, $N(R) = R$, and $\mathcal{A}$ is correct on every $r' \in \{1, \dots, R\}$. Thus Lemma 14 applies and gives the displayed bound at r. In particular every final residue is live: if any final residue collapsed, two sufficiently large inputs in that residue would receive the same final output string, contradicting correctness and noncollision. □

a. Correctness quantifier. Correctness in Theorem 6 is required at every $r \ge 1$; the conclusion then holds at every $r \ge h + D$, without exception. This global quantifier is not decorative: the collapse argument in Lemma 14 compares the outputs at r and $r - D$ and needs correctness at both points. Correctness at isolated points is not sufficient; if $\mathcal{A}$ were correct at r but not at $r - D$, identical final strings at those two inputs would yield no contradiction.

Corollary 10 (Finite-range tail form, restated with full hypotheses)**.** *Let $\mathcal{A}$ be a constant-budget transducer, and let (h, D) be pumpability data for its final pair family as provided by Corollary 9. Suppose finite-range noncollision (Assumption 1) holds with $N(R) = R$ on the range $1 \le r \le R$ and $\mathcal{A}$ is correct on every $r \in \{1, \dots, R\}$. Then*

$$|T_p(r)| \ge \frac{r - h - D + 1}{D} \qquad \text{for every } h + D \le r \le R.$$

The constants (h, D) are determined by $\mathcal{A}$ and the witness family alone; in particular they do not depend on R. No statement is made for transducers outside the constant-budget regime: there, pumpability data in the sense of Definition 28 need not exist, and any finite-range inequality would require explicit control of pumping constants not supplied by this theorem.

Proof. Constant budget puts all executions over one finite charged alphabet, so Corollary 9 supplies pumpability data (h, D) for the final tape. The construction of this data uses only the fixed transducer, its finite alphabets and state sets, and the fixed witness words Π, M, Σ; it is obtained before any finite endpoint R is chosen. Hence (h, D) are independent of R. Lemma 14 then applies verbatim on the range $1 \le r \le R$. □

Corollary 11 (Bit-level output counting toll, restated for proof)**.** *Assume finite-range noncollision supplies $N(R)$ pairwise distinct target unitaries, up to global phase, on a range. Suppose a deterministic exact compiler emits final outputs $T_p(r)$. Then*

$$\max_{1 \le r \le R} B_{\mathrm{self}}(T_p(r)) \ge \lceil \log_2 N(R) \rceil .$$

The same inequality holds for a fixed family-relative code when one public dictionary is shared by the whole range.

For the rational-angle CP witness, only the displayed per-range inequality is claimed on fixed ranges $R \le T = 131040$.

Proof. A circuit string determines its unitary, so a deterministic exact compiler that is correct on two inputs whose target unitaries are distinct up to global phase must emit two distinct output strings. The noncollision hypothesis therefore forces at least $N(R)$ pairwise distinct codewords. If their prefix-free lengths are $\ell_1, \dots, \ell_{N(R)}$, Kraft's inequality gives $\sum_i 2^{-\ell_i} \le 1$, so $N(R)2^{-\max_i \ell_i} \le 1$. Hence the largest length is at least $\lceil \log_2 N(R) \rceil$. The argument applies independently to either fixed code and does not convert token count or gate count into bits. □

4. Bookkeeping for the Masked-Share Cut

This subsection records the restart property used by the masked-share tradeoff (Theorem 2). It also makes explicit why a materialized update log, phase table, graph, RAM/cache layout, old-input spool, dynamic-pass state, or random seed is part of the cut message rather than an uncharged reconstruction channel.

Lemma 15 (Restart from the serialized cut)**.** *Fix $\mathcal{A}$ as in Definition 9, receiving the masked-share stream once (Theorem 2). At the inter-round cut c_{MS}, the five fields in* (4)*, together with any prefix already committed to the final write-once output, determine the continuation of that run for every supplied second-round continuation v. For a randomized run this statement is conditional on the serialized seed/tape and cursor.*

Proof. The control field restores the exact transition, all head positions, dynamic pass/termination schedule, and random seed/tape cursor. The store and window fields restore RAM, stack, hash/cache metadata, nonparameter machine data, and ordered live token structures. The parameter field restores every omitted coefficient by its location tag. The IR field restores every other live intermediate tape, DAG, table, addressed record, and any consumed input block retained for rescanning. These disjoint fields therefore reconstruct the complete machine configuration. Conditional on the serialized random tape, the receiver can append the supplied second round and execute the same dynamically chosen continuation. The separately transmitted final-output prefix is prepended to the suffix produced after restart. No additional state, old-input file, or public-family side channel is used. The single-delivery hypothesis is needed: if the source redelivered the stream for a later pass, the continuation would reread the first round, which the receiver does not hold. □

Appendix B: Artifact Engineering

The implementation described in this appendix is a bounded recognizer and candidate controller, not a general maximal-commuting-region extractor for arbitrary DAGs. Its semantic nodes are limited to explicit supported Pauli/phase records, local phase-ladder or Fourier spans, mirrored/self-inverse composite blocks, and repeated-run or preset semantic references. The word "region" below means a contiguous span accepted by that configured grammar under its scan order. Soundness of a selected rewrite is supplied by the independent certificate and conservative fallback, not by a claim of global maximality.

1. Candidate Control and Bounded Pipeline

The semantic path is advisory, not authoritative: semantic terms are lowered late, semantic subterms are compiled once and reused, and a semantic candidate loses to a flat reference whenever it does not clearly improve the conservative structural cost. The term caches, compiled-subterm caches, and repeated-run block caches in the implementation follow from treating the semantic term rather than the materialized circuit as the unit of comparison. The whole method is the following bounded pipeline.

1. **Input.** A high-level circuit C, target basis B, and iteration budget K.

2. Compute stable signatures for instructions and candidate blocks in C.

3. Run bounded preprocessing $\mathcal{S}_K(C)$: repeated-prefix and repeated-run detection and reuse; adjacent block–inverse(block) cancellation; commutative inverse cancellation; semantic lifting into instruction-span, phase-ladder, and mirrored/self-inverse nodes where applicable; early stop if one full round does not improve the selected cost.

4. Construct the candidate set

$$\mathcal{C}(C) = \{\mathcal{L}_B(C), \mathcal{P}_{\mathrm{ucc}}(\mathcal{L}_B(C)), \mathcal{P}_{\mathrm{ucc}}(\mathcal{L}_B(\mathcal{S}_K(C))), \mathcal{R}(C)\},$$

   where $\mathcal{R}(C)$ is an optional semantic or preset-reference candidate for large repeated structures, itself produced from the semantic term algebra (Fourier-layer lowering, mirrored/self-inverse lowering, or a bounded repeated-run rebuild).

5. If the circuit factors as $C = P \cdot B^r \cdot S$, form the bounded block-candidate set $\mathcal{B}(B) = \{B_1, \dots, B_q\}$ and select $i^\star = \arg\min_i \kappa_{\mathrm{proj}}(P, B_i, r, S)$ on the projected full-circuit cost, rebuilding only $B_{i^\star}$. This is what lets the repeated-run path avoid materializing several giant candidates merely to compare them.

6. Evaluate all candidates under the conservative structural cost κ, with fixed lexicographic tie-breaking, and return $\widehat{C} = \arg\min_{X \in \mathcal{C}(C)} \kappa(X)$.

The role of candidate control is not to dominate all downstream flows. It is to prevent avoidable structured-circuit regressions by preserving the best available output among a small family of semantically equivalent candidates.

2. Complexity Discussion

The method is intentionally bounded and does not attempt global optimization. Let $n := |C|$ be the number of high-level instructions. Suppose one outer iteration consists of r bounded scans, and scan j has cost $T_j(n)$. Then the total runtime of preprocessing satisfies

$$T_{\mathrm{pre}}(n) \in O\left(K \cdot \sum_{j=1}^{r} T_j(n)\right).$$

The dominant scans are near-linear in n up to implementation-dependent hashing, block-comparison, and bookkeeping costs, so practical behavior stays near-linear up to a bounded constant on repeat-heavy workloads. A large-circuit fast path handles repeat-dominant cases, source-level short-circuiting avoids downstream work on Qiskit-native all-to-all composites, and backend-aware portfolio expansion is restricted to cases where instability matters. The method is therefore a bounded systems heuristic, not a globally optimizing pass.

Proposition 14 (Expected regime of benefit)**.** *The preprocessing stage is most likely to help on circuits whose useful structure is already explicit at the high-level instruction stream or in the bounded semantic lift, especially exact inverse blocks, repeated blocks, phase-ladder layers, mirrored/self-inverse shells, and dominant repeated prefixes. It is not expected to uniformly improve families whose performance is dominated by later synthesis or routing decisions that are not captured by the preprocessing signatures or semantic nodes.*

Justification. This follows from the construction of the pass. Every preprocessing rule operates on high-level structural regularities detected through stable signatures, exact inverse relations, or the bounded semantic lift into semantic terms such as Fourier layers and mirrored/self-inverse shells. Therefore the pass can only exploit structure that survives in this representation. Conversely, if output quality is dominated by later basis-specific synthesis or hardware-routing effects not visible at this stage, the preprocessing pass may improve little or only preserve parity with stronger downstream baselines. The representation-aware compilation principle is therefore selective rather than universal: it claims that some opportunities are materially easier to preserve and compare before lowering, not that all useful optimization should be moved to the semantic layer. □

3. Integration Into UCC

The semantic artifact changes the UCC flow in six places:

1. it performs structural simplification before basis lowering,
2. it lifts suitable subcircuits into a bounded semantic term algebra before candidate generation,
3. it compares candidate outputs conservatively to avoid structured-circuit regressions,
4. it uses source-level short-circuiting on suitable all-to-all Qiskit-native cases,
5. it restricts backend-aware portfolio expansion to cases where extra probing is justified,
6. it memoizes bounded semantic references and repeated-run block selections for reuse.

The organizing distinction is that some circuit growth is unavoidable basis-translation overhead while some is structural regression caused by losing recoverable semantic evidence before candidate selection.

4. Method and Provenance Map

Every frozen numerical row is addressed by the complete primary key

(`experiment`, `instance`, `representation`, `method`, `seed`, `backend_hash`, `tool_version`, `artifact_commit`, `config_hash`).

The backend and configuration hashes are SHA256 digests of canonical JSON (sorted keys and no insignificant whitespace). A row collision on all nine fields is rejected. In particular, the reorder-probe hardware rows and the canonical hardware-aware rows have different experiment and configuration keys, and therefore different reader labels even when the underlying MQT instance name is the same. The frozen source is `research/experiment_registry/frozen_hardware_rows.json`; `scripts/generate_keyed_tables.py` validates key uniqueness and generates the three affected table fragments plus `expanded_primary_keys.json`. The generated labels expose the campaign, instance, seed, and the first eight configuration-hash digits. Hand-edited numeric rows are not accepted by this path.

The canonical validation data are grouped by experiment family in the artifact. The external-baseline method map immediately below uses the disclosed `timeout_s` = 600 second wall-clock budget per method per instance; it does not override the distinct `RP23`, `HW24`, or `REAL24` campaign manifests. The table maps paper-facing names to JSON keys, tool/pass sequences, artifact flags, and

source classes. The older real-panel UCC default row is included only to define that frozen-control label in the main text and Appendix C; it is not one of the canonical methods used in the 600 s external-baseline comparison.

The source classes in Table XI expand as follows: Here "main" denotes membership in the canonical comparison campaign and does not necessarily imply inclusion in the two headline tables.

- main: canonical 600 s external-baseline comparison.
- abl: Fourier-layer IR ablation.
- res: resource-consequence check.
- corr: large-instance correctness extension.
- irr: irrational independent-Pauli unbounded witness suite.
- status: provenance-only configuration excluded from numerical quality comparison when its predicate fails.
- real: official real-instance frozen panel.
- appx: older frozen-control appendix panels.

The new executable protocol and provenance are in `research/round6_major_revision/`: the witness declaration and exact-rank verifier, `unbounded_witness_results.json`, `rational_cp_round6_results.json`, `public_phase_folding_results.json`, `tket_paulisimp_results.json`, per-cell logs, and the environment snapshot. After global-phase alignment, the implemented check computes the maximum elementwise error ϵ and requires $\epsilon \le \mathrm{atol} + \mathrm{rtol}\max_{i,j}|U_{ij}|$. Thus $\mathrm{atol} = \mathrm{rtol} = 10^{-9}$ gives an effective upper threshold of approximately 2×10^{-9} because unitary entries have magnitude at most one. Recorded maximum errors increase with repetition count r, while every row marked completed and valid satisfies this criterion. The irrational-witness PyZX 10k result passes at approximately 1.05×10^{-9}; the 20k result at approximately 2.68×10^{-9} is a correctness failure. Timeout, predicate-error, unsupported-conversion, and correctness-failure statuses are kept distinct. The staq provenance records public commit `a2acd39` (with the full hash in the artifact) and the generic double-precision OpenQASM patch used for continuous-angle round trips.

5. Certificate, Cut-Trace, and QRE Artifacts

The exact canonicalizer and symbolic checker are `certificates/canonicalize.py` and `certificates/symbolic_checker.py`. The frozen audit `data/frozen/certificate_audit.json` contains 12 accepted/rejected dense cross-check cases over widths one through six and 56 adversarial mutations. The checker and dense evaluator use separate implementations: the former propagates complete signed Clifford tableaus and exact rational-π Pauli coefficients, while the latter constructs full numerical matrices and a projective distance. Only `completed_valid` records carrying an independent certificate enter numerical means; unsupported, predicate-error, and numerical-inconclusive records remain status-only.

The canonical W3 theorem-model reference compilers are under `compiler/reference_streaming/` and are driven by `scripts/run_tradeoff_sweep.py`. After every input update, their version-one event schema records `committed_output_bytes`, the five cut fields from (4), pass index, RSS, elapsed nanoseconds, artifact paths, and digests. The verifier reopens and decodes every field slice and checks the committed-prefix hashes. The complete 592-run campaign is stored under `data/runs/w3-tradeoff-20260802-local/`; its registered summary and freeze manifest are `data/frozen/tradeoff_summary.csv` and `data/frozen/tradeoff_freeze_manifest.json`. The older `research/cut_trace/` bundle was produced by an earlier serializer and lacks the per-event field files, so its 1,320 events and 24 summaries are historical only and are not accepted by the version-one verifier. The checked three-cut codec fixture and its manifests remain in the project-root `instrumentation/example_trace/`.

The W7 runner and allocator are the project-root `scripts/run_qre_campaign.py` and `scripts/allocate_synthesis_error.py`; their total-error schema, two QEC profiles, and factory grid are under `qre/`. The 288-row Parquet/CSV, full angle records, analysis, figures, and manifests are under `data/frozen/`, `data/runs/w7-fixed-total-error-20260803-local/`, and `figures/`. The earlier single-profile runner and its six-row result are retained as historical artifacts under `research/fixed_budget_qre/` but are not the W7 main result. The grid synthesizer is built from public staq commit `a2acd39e60ed0e5f1978fa530df7e1e0c7cedf7a`; the disclosed artifact change only replaces the nondeterministic candidate-search seed with 20260801. The runner omits staq's `-time` fast path because that path bypasses its internal detail/check branch; it instead measures subprocess time externally and requires `Check flag = 1` for every unique angle. The QRE is the explicit surface-code model family stated in the main text, not a vendor service or an independent architecture implementation.

6. Immutable Campaign Hierarchy

The machine-readable index `research/campaign_manifests/campaign_index.json` defines two base profiles and five noninterchangeable campaigns. `RP23` is the pass-order probe, `HW24` is the canonical line-backend campaign, `REAL24` is the official real-instance anti-regression panel, `MR25` is the earlier 525-cell matched-representation matrix, and `NF25` is the earlier 1,120-cell natural-kernel factorial ablation. Each prefix is unique and must appear in a reader-facing label before the source instance name.

Paper name	JSON key	Tool/pass sequence	Artifact flags	Budget	Source classes
semantic UCC (Fourier-layer IR enabled)	semantic_ucc	ucc.compile to {cx, rx, ry, rz, h} with Fourier-layer IR enabled	UCC_DISABLE_FOURIER_LAYER_IR unset	600 s	main, abl, res, corr
upstream UCC v0.4.12 pre-semantic default	baseline_ucc	ucc.compile loaded from the legacy baseline repository in research/compare_real_instances.py; target basis {cx, rx, ry, rz, h}	legacy baseline repo; semantic candidate controls absent	240 s in real-panel runner	real, appx
artifact UCC (Fourier-layer IR disabled)	no_fourier_ucc	same artifact compile path with Fourier-layer IR disabled	UCC_DISABLE_FOURIER_LAYER_IR=1	600 s	abl, res
qiskit opt3	qiskit_opt3	qiskit.transpile(...,optimization_level=3,layout_method="trivial",routing_method="none")	none	600 s	main, abl, res
PyZX configured bridge	pyzx_configured	Qiskit QASM2 to PyZX, to_basic_gates, basic_optimization, Qiskit target-basis cleanup	none	600 s	main
PyZX full_reduce	pyzx_full_reduce	Qiskit QASM2 to PyZX graph, full_reduce, circuit extraction, Qiskit target-basis cleanup	none	600 s	main
TKET FullPeephole	tket_fullpeephole	Qiskit-to-TKET bridge when available, otherwise QASM fallback, then FullPeepholeOptimise	none	600 s	main
TKET PauliSimp (unrebased configuration)	tket_paulisimp_unrebased	direct Qiskit–TKET bridge, then PauliSimp; retained only to record the predicate error	none	600 s	status
TKET PauliSimp (rebased)	tket_paulisimp_rebased	direct bridge, DecomposeBoxes, AutoRebase({CX, RZ, Rx}), RemoveRedundancies, PauliSimp, RemoveRedundancies, target-basis cleanup	generic identical preamble	600 s	main, irr
TKET GuidedPauliSimp	tket_guided_paulisimp	same TKET bridge/fallback sequence, then GuidedPauliSimp	none	600 s	main
staq rotation folding	staq_rotation_folding	Qiskit opt0 target-basis QASM2, public staq fold_rotations, QASM2 parse, target-basis cleanup	no family labels; generic double-precision parser/serializer patch disclosed	600 s	main, irr
phase-polynomial reference	phase_poly_reference	recognize the HD^rH witness, aggregate commuting rz/cp coefficients, then lower through Qiskit opt0	none	600 s	main, abl, res, corr

Table XI. Method/provenance map for the canonical validation protocol. The direct Qiskit-to-TKET bridge was available in the validation environment; the QASM fallback is documented for reproducibility when that bridge is unavailable.

The matched-representation benchmark and the natural-workload ablation reported in the main text are the later campaigns `w4-matched-representation-20260802` and `w8-natural-factorial-20260803-local`, registered in `data/manifest.yaml` with receipts `data/frozen/matched_representation_freeze_manifest.json` and `data/frozen/natural_workload_analysis.json`; `MR25` and `NF25` are retained as historical diagnostics. The per-bundle `SHA256SUMS` files under `PaperDraft/research/` seal each bundle at its freeze; five of them no longer match the current files (generator scripts and, for the claim-evidence bundle, its rows), which were revised after the freeze, and the seals are kept unchanged as historical records.

Every campaign manifest records or explicitly marks unavailable the backend topology, native gate set, seed, timeout policy, memory cap, tool versions, artifact/freeze commit, bridge sequence, certificate policy, immutable config hash, and frozen dataset hash. Historical campaigns did not enforce a memory cap or independent certificate; their manifests say `not_enforced_historical` and "diagnostic only" rather than imputing a modern policy. In particular, `HW24` uses 120 s for QPE/QAOA and 240 s for Grover, whereas `RP23` uses 600 s. Those rows may share an MQT source-instance string but cannot share a short label or be averaged.

The configuration hierarchy is single-parent: a campaign inherits only its named base profile and then supplies explicit overrides. Backend and config hashes use canonical JSON. The frozen source snapshots `frozen_hardware_aware_rows.json` and `frozen_real_panel_rows.json` live beside the manifests, so later changes to a runner cannot silently change a displayed row.

7. Claim–Evidence Matrix

Every headline statement in the abstract, introduction, resource section, and conclusion is registered with one primary theorem, proposition, table, figure, or artifact test and an explicit scope. The machine-readable source is `research/claim_evidence/claim_evidence.json`; the compact reader map is Table XII.

Appendix C: Supplementary Experiments

In this appendix, rows labeled "baseline UCC" are frozen-control rows: they denote the upstream UCC v0.4.12 pre-semantic default path defined in the provenance map of Appendix B, not a separate semantic method.

1. Exact-Semantics Witness Data

The following tables contain the full uniform-600 s comparison data for the exact-semantics witnesses discussed in Corollary 1 of the main text.

2. Fourier Witness Diagnostics

The tables in this subsection support the main-text Fourier witness without serving as additional theorem instances. The canonical theorem-linked CP witness is the finite-range case in Table XIII; the generalized topology/angle rows are robustness diagnostics, and the external stress rows use the budgets recorded in their frozen source files.

ID	claim location	primary evidence	authorized scope
C01	abstract/introduction	Definition 9 and Eq. (4)	formal model only
C02	abstract/conclusion	Theorem 1	declared family-relative and self-contained codes
C03	abstract/conclusion	Theorem 2	two-round masked-share stream
C04	introduction/discussion	Proposition 8	ordered-share family-relative codec
C05	validation	Fig. 4	1,080 frozen cells; 24 cells per representation–compiler pair
C06	validation	Fig. 5	592 formal-model reference-compiler runs
C07	validation/workloads	Table X and Fig. 12	9 families; 2^6 factors; 3 scales and 3 seeds
C08	resource consequences	Table VIII	one disclosed surface-code model
C09	compiler	Proposition 12	declared symbolic domain only
C10	validation	Fig. 4 and Section VIII B	native and unified configurations; bridge loss reported separately
C11	application workloads	Table IX	REAL24 campaign
C12	conclusion	Table IV	evidence-class separation
C13	abstract/conclusion	Theorem 5	round-major forward scans; all 2p-1 crossings charged
C14	abstract/discussion	Proposition 10	family-relative p-block codec; compact-output corner for every r; full curve for r=2

Table XII. Claim–evidence matrix. The full JSON also contains the exact claim text and evidence class. A claim outside the displayed scope is not made by this paper.

Requested	Actual	semantic UCC	`qiskit opt3`	PyZX `full_reduce`	TKET PauliSimp (rebased)	staq rotation folding	TKET GuidedPauliSimp
4,000	3,998	42/24/12	9588/5593/4788	107/59/31	52/29/19	9188/4996/4788	13574/6392/4788
10,000	9,998	42/24/12	23988/13993/11988	timeout	62/36/28	22988/12496/11988	33974/15992/11988
20,000	19,998	42/24/12	47988/27993/23988	timeout	59/27/20	45988/24996/23988	67974/31992/23988
50,000	49,998	42/24/12	119988/69993/59988	timeout	73/44/35	114988/62496/59988	169974/79992/59988
100,000	99,998	42/24/12	239988/139993/119988	timeout	66/37/28	229988/124996/119988	339974/159992/119988

Table XIII. Uniform-600 s comparison on the rational CP finite-range witness. Completed cells report gates/depth/cx; other entries are status outcomes and are not scored as structural wins. The family is certified only for $R \leq T = 131040$, with per-transducer tail constants. No external row is a formal theorem instance.

3. Controlled Fixed-Basis Results

a. Repeated $\mathrm{QFT} + \mathrm{QFT}^{-1}$ *(100k gates)*

b. Repeated $\mathrm{QFT} + \mathrm{QFT}$ *control (100k gates)*

These finite controls show that, for this configured family and the tested pipelines, a large structural opportunity remains after controlling for the final target basis. They are not a separate theorem-level witness.

c. Systematic $\mathrm{QFT} + \mathrm{QFT}^{-1}$ *external scaling*

The inverse-QFT scaling control is not the primary theorem witness, because `qiskit opt3` also finds the zero-gate circuit at smaller sizes. Its role is to test scalability when both paths return the same zero-gate output. As shown in Table XXII, the semantic artifact returns the zero-gate circuit at all tested sizes, is faster than `qiskit opt3` wherever `qiskit opt3` completes, and avoids the `qiskit opt3` timeouts at 50k and 100k. As an additional control, it also matches `qiskit_commutative_inverse` in output quality and is faster at every tested size.

4. Why Simple Pass Reordering Is Not Enough

To test whether the observed gains are mainly explained by a missing pass or pass-ordering issue inside `UCCDefaults`, we ran a controlled comparison between baseline UCC, several minimal `UCCDefaults`-style alternatives, and the semantic artifact. The minimal alternatives were `defaults_preinverse_only`, `defaults_reordered_only`, and `defaults_minimal_bundle`; the all-to-all cases were normalized to the same final target basis $B = [\mathtt{cx}, \mathtt{rx}, \mathtt{ry}, \mathtt{rz}, \mathtt{h}]$, and the backend-aware cases used fixed transpiler seed 12345. This is a separate reorder-probe experiment, so its runtime numbers should be read independently from the frozen mainline tables below. Its hardware rows have reader labels `RP23-qpeexact-s12345-c89463b96` and `RP23-qaoa-s12345-c89463b96`; the canonical hardware campaign below instead uses the `HW24-...-c0f32747f` labels. The differing labels are different full experiment keys, not repeated measurements of one row.

Two qualifications are not visible in Table XXIII itself. On `RP23-qaoa-s12345-c89463b96` the minimal bundle already matches the semantic artifact, so the reorder-only explanation is adequate there. On `RP23-qpeexact-s12345-c89463b96` it is not: the minimal bundle gives 1,893 gates, depth 576, `cx` 1,311, against 1,593, depth 420, `cx` 772 for the semantic artifact.

5. Public Benchmark Suite Results

To move beyond official library examples, we added two public benchmark sources: `MQT Bench` and `SupermarQ`.

a. MQT Bench. Representative results are:

- `mqt_qpeexact_32`: `qiskit opt3` = 1,891 gates, baseline UCC = 5,494, semantic UCC = 1,891;

Requested	Actual	semantic UCC	`qiskit opt3`	PyZX `full_reduce`	TKET PauliSimp (rebased)	staq rotation folding	TKET GuidedPauliSimp
4,000	3,998	27/18/6	4799/2801/2394	38/19/8	36/14/6	3402/2201/2394	8786/4000/2394
10,000	9,998	27/18/6	12000/7001/5994	48/33/12	35/15/8	8502/5501/5994	21986/10000/5994
20,000	19,998	27/18/6	24000/14001/11994	correctness failure	38/16/8	17002/11001/11994	43986/20000/11994
50,000	49,998	27/18/6	60000/35001/29994	timeout	36/15/8	42502/27501/29994	109986/50000/29994
100,000	99,998	27/18/6	120000/70001/59994	timeout	37/16/8	85002/55001/59994	219986/100000/59994

Table XIV. Uniform-600 s comparison on the irrational independent-Pauli witness. Completed cells report gates/depth/cx; other entries are status outcomes and are not scored as structural wins. The exact rank and irrational-angle certificate, rather than this numerical table, instantiate Corollary 1.

Topology	Angles	Points	`qiskit` ok/TO	Semantic completed	Semantic output signatures
chain CP	3	12	9 / 3	all 12	23–27 gates, 17–18 depth, 6 `cx`
ring CP	3	12	9 / 3	all 12	32 gates, 23 depth, 8 `cx`
sparse CP 0.5	3	12	6 / 6	all 12	25–27 gates, 11–12 depth, 6 `cx`
full-pair CP	2	6	4 / 2	all 6	42 gates, 24 depth, 12 `cx`
Total points	–	42	28 / 14	all rows	repetition-independent at fixed topology/angle

Table XV. Generalized fixed-width Fourier witness suite. Each point is a paired run of semantic UCC and `qiskit opt3` on an HD^rH circuit at $n = 4$, with requested sizes up to 50k gates. The rows are filtered from the recorded 284-row generalized suite by the $n = 4$ topology/angle/size slice. Recorded `qiskit opt3` timeout cells in that source use legacy 60 s or 90 s budgets, so the table is a robustness diagnostic rather than a uniform-600 s comparison.

Width (n)	Checks	Output Gates	Depth	CX Count	Seed Variance
4	15 / 15	42	24	12	0
5	15 / 15	65	32	20	0
6	15 / 15	93	40	30	0

Table XVI. Correctness and seed-robustness certificate for the full-pair Fourier witness. The correctness experiment checks 3 widths × 3 requested sizes × 5 angle seeds using Qiskit `Operator.equiv`; all 45 configurations are equivalent to the original circuit up to global phase. The seed-robustness experiment contains 90 rows including `qiskit opt3`; semantic UCC has zero output gate-count variance across the five tested nonresonant angle seeds at each width.

- `mqt_qpeinexact_24`: `qiskit opt3` = 1,206 gates, baseline UCC = 3,682, semantic UCC = 1,206;
- `mqt_ae_8`: `qiskit opt3` = 207 gates, baseline UCC = 542, semantic UCC = 207;
- `mqt_draper_qft_adder_32`: `qiskit opt3` = 1,630 gates, baseline UCC = 5,278, semantic UCC = 1,630;
- `mqt_qaoa_32`: `qiskit opt3` = 1,422 gates, baseline UCC = 6,310, semantic UCC = 1,422;
- `mqt_grover_20`: `qiskit opt3` = 3,848,810 gates in 38.043 s, baseline UCC times out at > 120 s, and semantic UCC reaches the same output quality in 39.123 s.

Thus, on `MQT Bench`, the semantic artifact repeatedly recovers `qiskit opt3`-level output quality and clearly improves over baseline UCC. The added `qpeinexact`, `ae`, and `draper_qft_adder` cases are especially relevant because they extend the phase-ladder/Fourier-layer story beyond one exact-QPE benchmark into inexact QPE, amplitude estimation, and QFT-based arithmetic. Runtime still remains weaker on the hardest Grover-style case.

Additional mirrored/conjugation positive cases make the mirrored-shell heuristic more concrete. On the official `grover_real` instance, baseline UCC returns 287,047 gates while the semantic artifact and `qiskit opt3` both return 79,029. On `MQT Bench` Grover instances, the same pattern persists across scale: `mqt_grover_8` gives 18,992 → 5,170, `mqt_grover_12` gives 259,797 → 71,706, and `mqt_grover_16` gives 2,112,285 → 581,098, where in each case the semantic artifact records the same gate count as `qiskit opt3` while clearly improving on baseline UCC. The large fixed-basis mirrored Grover benchmark shows the same parity pattern: baseline UCC gives 4,122,006 gates, while the semantic artifact and `qiskit opt3` both give 1,158,011. These cases make the mirrored/conjugation line less dependent on the routed timeout-recovery interpretation of `hw_mqt_grover_20`.

A dedicated mirrored/conjugation scaling sweep makes this support more systematic. On public `MQT Bench` Grover instances, the semantic artifact records the same gate count as `qiskit opt3` at sizes 8, 12, and 16, reducing baseline UCC from 18,992 to 5,170 gates, from 259,797 to 71,706 gates, and from 2,112,285 to 581,098 gates, respectively. On the fixed-basis `grover_mirrored` repeated-shell scaling sweep, the same parity pattern holds at 10k, 20k, 50k, and 100k input gates: baseline UCC gives 412,206, 824,406, 2,061,006, and 4,122,006 output gates, while the semantic artifact and `qiskit opt3` give 115,811, 231,611, 579,011, and 1,158,011 gates. Thus this second structural family targeted by the artifact has empirical positive cases in both public Grover scaling and large repeated-shell fixed-basis scaling.

b. SupermarQ. Representative results are:

- `supermarq_mermin_bell_8`: baseline UCC = 386 gates, depth 135; semantic UCC = 76 gates, depth 44; `qiskit opt3` = 76 gates, depth 44;
- `supermarq_qaoa_vanilla_12`: baseline UCC = 947

Method	Requested	Actual	Status	Gates/depth/cx	Certificate
semantic UCC (Fourier-layer IR enabled)	100,000	99,998	completed	42/24/12	`Operator.equiv=True`
phase-polynomial reference	100,000	99,998	completed	42/24/12	`Operator.equiv=True`

Table XVII. Large-instance correctness extension on the canonical $n = 4$, $r = 9999$ Fourier witness. The check is a finite-instance Qiskit `Operator.equiv` certificate for the two completed compressed outputs only; it does not certify the generalized topology/angle suite or prove an algebraic identity for all r.

Width	Requested	semantic UCC	`qiskit opt3`	PyZX bridge	TKET FullPeephole
5	4,000	65	11,711	14,640	30,369
5	10,000	65	29,315	36,640	timeout
6	4,000	93	12,108	15,321	33,881
6	10,000	93	30,417	38,487	timeout

Table XVIII. Representative external stress tests on the width-axis full-pair Fourier witness. Entries report output gates after the configured bridge pipeline and final target-basis compilation. PyZX completes all four runs but does not recover the bounded semantic forms 65 and 93; TKET completes the 4k runs with larger outputs and times out at 10k. The recorded budget in this stress file is 300 s.

gates, depth 225; semantic UCC = 222 gates, depth 80; `qiskit opt3` = 222 gates, depth 80;

- `supermarq_hamiltonian_sim_8`: baseline UCC = 71 gates, depth 36; semantic UCC = 51 gates, depth 24; `qiskit opt3` = 51 gates, depth 24.

This second public benchmark source provides additional finite anti-regression checks on nontrivial families and also exposes a limitation: the semantic artifact is not uniformly better on every benchmark family.

6. Hardware-Aware Results

We added a backend-aware comparison using a fixed 20-qubit bidirectional line backend with native operations `u`, `sx`, `p`, `cx`, `measure`, and `id`. The benchmark source is `MQT Bench`. The reader-facing rows are `HW24-qpeexact-s12345-c0f32747f`, `HW24-qaoa-s12345-c0f32747f`, and `HW24-grover-s12345-c0f32747f`; the full source instance remains part of each registry primary key.

a. HW24-qpeexact-s12345-c0f32747f. This is a clean fixed-seed parity case: the semantic artifact matches `qiskit opt3` in total gates, depth, and `cx` count, while still remaining much better than baseline UCC on all structural metrics. The tradeoff is runtime rather than output quality.

b. HW24-qaoa-s12345-c0f32747f. This is the clearest hardware-aware external-baseline improvement in the paper: the semantic artifact improves on `qiskit opt3` in total gates, depth, and `cx` count, while also improving on baseline UCC in total gates, depth, and `cx` count.

c. HW24-grover-s12345-c0f32747f. `qiskit opt3` and baseline UCC still time out under this setting, but the semantic artifact returns a result in 1.999 s via a stronger mirrored/self-inverse repeated-run backend fallback. The resulting circuit is still extremely large, but the selector lowers total gates, depth, and `cx` simultaneously (to 6,467,857 total gates, depth 4,094,196, and `cx` count 3,045,048), so we treat this as a robustness result rather than a main competitive routed-quality claim.

7. Stability and Seed Sensitivity

We measured repeated-run stability on the frozen artifact in two ways:

1. five repeated runs for the official real instances and the canonical hardware-aware seed,
2. a five-seed sweep for the backend-aware `MQT Bench` cases.

The main conclusions are:

- the official real-instance outputs are deterministic across repeated runs,
- the hardware-aware outputs are deterministic for each fixed seed,
- `hw_mqt_qaoa_20` remains better than `qiskit opt3` in total gates, depth, and `cx` count across all five tested seeds 0, 1, 42, 12345, and 54321,
- `hw_mqt_qpeexact_20` is now a fixed-seed parity case rather than a strict improvement.

Thus, the hardware-aware claim is a stable workload-family result across a small fixed seed portfolio, rather than a single favorable seed.

8. Scaling and Ablation

The main scaling trends are summarized in Fig. 14. The size sweep covers 4k, 10k, 20k, 50k, and 100k inputs

Topology	semantic UCC	`qiskit opt3`	PyZX bridge	TKET FullPeephole
chain CP	27	6,851 / 17,136	10,838 / 27,121	20,555 / timeout
ring CP	32	7,996 / 19,996	11,984 / 29,984	23,987 / 59,987
sparse CP 0.5	27	6,851 / 17,136	10,838 / 27,121	timeout / timeout
full-pair CP	42	10,783 / 26,983	13,574 / 33,974	timeout / timeout

Table XIX. Topology-axis external stress test at $n = 4$ with nonresonant seed 0. Each external-baseline cell reports output gates at requested sizes 4k / 10k. The deterministic sparse-CP draw uses the same number of CP edges as chain CP and is graph-isomorphic to a chain at $n = 4$, which explains the matching `qiskit opt3` and PyZX gate totals for those two rows. The recorded budget in this stress file is 300 s.

Pipeline	Representation tested	Observed behavior in the disclosed suites	Interpretation
semantic UCC (Fourier-layer IR enabled)	pre-basis Fourier-layer term	completed 5/5; 42 gates, depth 24, 12 `cx` at every tested scale	semantic side
phase-polynomial reference	same Qiskit input; only commuting-diagonal coefficient aggregation	completed 5/5; 42 gates, depth 24, 12 `cx` at 4k–100k	authors' protocol-specific constructive control
`qiskit opt3`	preset target-basis pipeline	completed 5/5; grows from 9,588 gates at 4k to 239,988 gates at 100k	strong completed baseline, but not bounded
PyZX configured bridge	Qiskit–QASM–PyZX basic optimization–Qiskit	completed 4/5; grows through 169,974 gates at 50k and times out at 100k	configured bridge does not recover bounded form
PyZX `full_reduce`	QASM to PyZX graph; `full_reduce`; extract and lower	rational witness: completes only at 4k with 107 gates, depth 59, 31 `cx`; timeout at 10k–100k. Irrational witness: completes and validates at 4k and 10k with 38 and 48 gates; correctness failure at 20k; timeout at 50k and 100k	strong small-scale simplification; correctness failure remains distinct from timeout and error
TKET FullPeephole	direct Qiskit–TKET bridge to FullPeepholeOptimise	timeout on all five canonical sizes under the 600 s budget	no completed bounded recovery in this protocol
TKET PauliSimp (unrebased configuration)	direct Qiskit–TKET bridge to PauliSimp	predicate error on all five sizes for both witness families	provenance-only configuration; not a structural loss
TKET PauliSimp (rebased)	generic DecomposeBoxes, AutoRebase to {CX,Rz,Rx}, RemoveRedundancies, PauliSimp, RemoveRedundancies	completes and validates at all five scales: 52–73 gates on the rational witness and 35–38 on the irrational witness	generic aggregate reconstruction crosses the formal boundary
staq rotation folding	generic target-basis QASM2 through public `fold_rotations`	completes and validates at all five scales, but grows from 9,188 to 229,988 gates on the rational witness and from 3,402 to 85,002 on the irrational witness	completed public baseline; growing output is not an impossibility result
TKET GuidedPauliSimp	direct Qiskit–TKET bridge to GuidedPauliSimp	completed 5/5 on each witness; returns 13,574 to 339,974 gates on the rational witness, exactly the gate count of its lowered input, and 8,786 to 219,986 on the irrational witness	no optimization on flat input: the pass acts only inside `CircBox` regions
artifact UCC (Fourier-layer IR disabled)	same artifact with Fourier-layer semantic path disabled	in ablation/resource runs, completes at 4k, 10k, and 20k with 9,588, 23,988, and 47,988 gates; times out at 50k and 100k	semantic-path causality check; completed rows materialize large outputs

Table XX. External-pipeline status matrix and constructive controls for the rational CP and irrational independent-Pauli witnesses. Completed outputs and status outcomes are separated. The rebased and unrebased PauliSimp configurations are distinct, and no row is treated as a formal transducer instance.

	$\mathrm{QFT} + \mathrm{QFT}^{-1}$		$\mathrm{QFT} + \mathrm{QFT}$	
Method	Output Gates	Output Depth	Output Gates	Output Depth
translation only	400,000	142,500	400,000	140,000
baseline UCC	968,784	276,266	1,207,504	317,501
semantic UCC	0	0	347,500	125,000

Table XXI. Controlled fixed-basis results for the repeated inverse and non-inverse QFT controls at 100k input gates.

for QFT-inverse, QPE-style, QAOA-ring, and mirrored-Grover families; the QAOA-ring block uses 20 qubits. The most important observations are stable quality behavior across scale and improving repeated-family runtime once the semantic representation and dispatch controller are enabled. The method is most effective on exact or highly repeated structure: `qft_inverse` remains maximally strong across all tested sizes, `qpe_style` remains a strong positive fixed-basis case, `qaoa_ring` consistently avoids UCC inflation, and the mirrored-Grover family continues to match Qiskit quality at 100k while remaining more expensive. Direct spot checks on the semantic artifact give `qpe_style` = 82,200 gates at 50k in 2.504 s and 164,400 gates at 100k in 4.636 s, reflecting the phase-ladder plus repeated-run controller improvements.

The non-monotonic semantic runtime between the 20k and 50k points is a dispatch effect visible in the implementation, not an inferred hardware effect. The 20,000-gate full-preset limit `_MAX_FULL_PRESET_REFERENCE_SIZE` allows the 19,998-gate input to enter the full preset-

Input gates	qiskit opt3		qiskit_commutative_inverse		semantic UCC
	Gates	Runtime	Gates	Runtime	Runtime
4,000	0	1.052 s	0	0.082 s	0.063 s
10,000	0	6.090 s	0	0.182 s	0.124 s
20,000	0	24.596 s	0	0.341 s	0.279 s
50,000	timeout	> 90 s	0	0.979 s	0.829 s
100,000	timeout	> 90 s	0	2.271 s	1.651 s

Table XXII. External scaling on repeated $\mathrm{QFT} + \mathrm{QFT}^{-1}$. The semantic UCC artifact returns zero gates at all tested sizes; the table reports runtime because output quality is identical to the zero-gate reference wherever the references complete. Timeout cells in this source use the recorded 90 s budget.

Case	Method	Gates	Depth	CX count	Runtime
phase_estimation_real	baseline UCC	471,819	328,271	58,491	75.538 s
	defaults_preinverse_only	485,780	332,924	82,925	80.045 s
	defaults_reordered_only	485,780	332,924	82,925	80.005 s
	defaults_minimal_bundle	485,780	332,924	82,925	80.084 s
	semantic UCC	166,805	119,204	58,491	1.989 s
	qiskit opt3	166,805	119,204	58,491	1.254 s
qaoa_ring_100k	baseline UCC	300,055	163,055	40,000	2.431 s
	defaults_preinverse_only	434,104	217,962	40,000	11.352 s
	defaults_reordered_only	445,216	217,962	40,000	10.120 s
	defaults_minimal_bundle	434,104	217,962	40,000	10.818 s
	semantic UCC	100,000	62,000	40,000	3.131 s
	qiskit opt3	100,000	62,000	40,000	0.307 s
RP23-qpeexact s12345-c89463b96	baseline UCC	2,077	626	1,262	0.126 s
	defaults_preinverse_only	1,893	576	1,311	0.292 s
	defaults_reordered_only	1,900	582	1,312	0.289 s
	defaults_minimal_bundle	1,893	576	1,311	0.287 s
	semantic UCC	1,593	420	772	0.296 s
	qiskit opt3	1,572	407	812	0.033 s
RP23-qaoa s12345-c89463b96	baseline UCC	2,088	522	1,439	0.098 s
	defaults_preinverse_only	2,050	487	1,458	0.270 s
	defaults_reordered_only	2,050	487	1,458	0.269 s
	defaults_minimal_bundle	2,050	487	1,458	0.271 s
	semantic UCC	2,050	487	1,458	0.278 s
	qiskit opt3	2,091	532	1,530	0.048 s

Table XXIII. Controlled comparison of baseline UCC, minimal UCCDefaults-style alternatives, the semantic artifact, and qiskit opt3. Small default-pipeline fixes explain part of the improvement on some cases, but they do not account for the main gains on the real and structured workloads. The overlapping phase_estimation_real rows use the per-method output cx counts from the recorded real-instance data. The keyed hardware rows are generated from the frozen registry; their RP23 labels distinguish this reorder-probe configuration from the HW24 campaign below.

(a) HW24-qpeexact-s12345-c0f32747f

Method	Output Gates	Output Depth	CX Count	Runtime
qiskit opt3	1,572	407	812	0.056 s
baseline UCC	1,976	648	1,350	0.174 s
semantic UCC	1,572	407	812	0.235 s

(b) HW24-qaoa-s12345-c0f32747f

Method	Output Gates	Output Depth	CX Count	Runtime
qiskit opt3	2,091	532	1,530	0.078 s
baseline UCC	2,057	513	1,503	0.141 s
semantic UCC	2,050	487	1,458	0.112 s

Table XXIV. Canonical hardware-aware results for the QPE and QAOA reader labels, generated from the frozen keyed dataset.

reference comparison, whereas the 49,998-gate input exceeds that threshold and uses the cheaper repeated-prefix/hierarchical route. The corresponding canonical witness runtimes are 28.314 s and 20.934 s. This threshold explains the local drop; it is not evidence of asymptotically decreasing compilation time.

The ablation picture is narrow. The dominant practical component is conservative candidate control on recoverability-sensitive families, most clearly on the 10,000-input-gate qaoa_ring row, where the full configuration returns 10,000 gates while the no-candidate-selection variant returns 30,055. By contrast, qft_inverse, qft_control, qpe_style, and grover_mirrored are less sensitive to the removal of individual structural rules. This reinforces the paper's current framing: the main contribution is a bounded semantic representation and dispatch controller that preserve recoverability before flat materialization, rather than a single local identity.

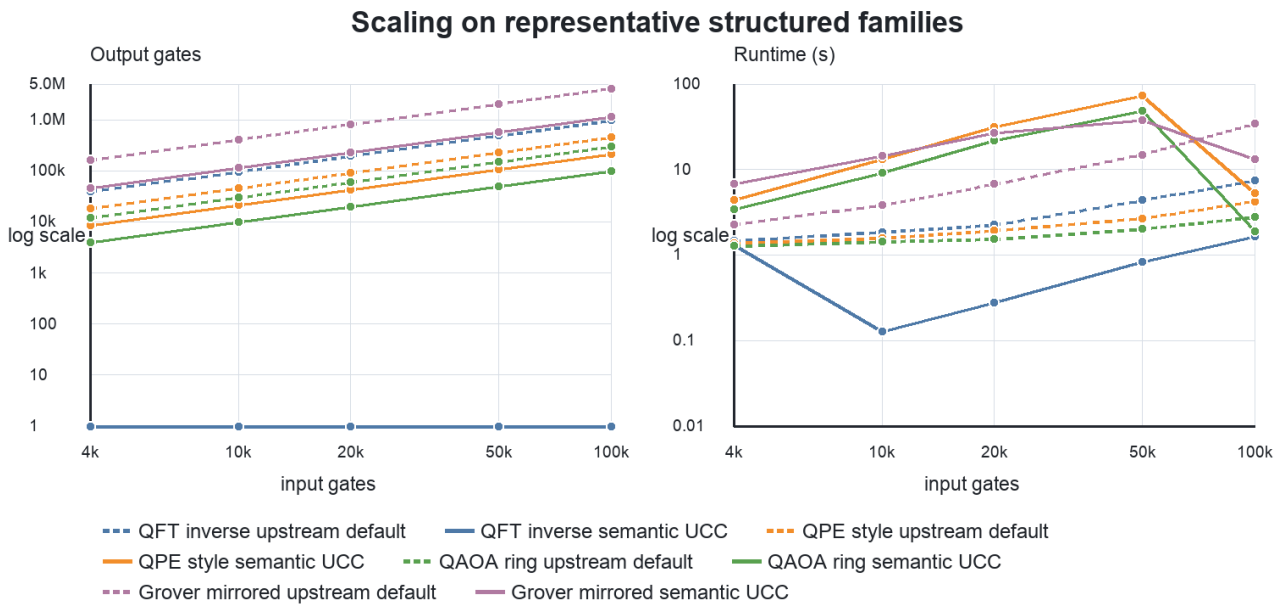


Figure 14. Scaling behavior on representative structured families. Left: output gate count versus input gate count for upstream default (baseline UCC) and the semantic UCC artifact. Right: runtime versus input gate count. The semantic path preserves clear quality advantages across scale, while runtime behavior remains family-dependent.